

UNIVERSITY OF Waterloo



NANOTECHNOLOGY ENGINEERING
FOURTH YEAR DESIGN PROJECT FINAL REPORT

SunDOG

Luminescent Downshifting for Optimized Generation

Presented to Dr. Hany Aziz

2026-04-06

NE_2026_20

Ryan Miller
20973259
Ryan Miller

Ryan Becze
20958526
Ryan Becze

Alexandre Dufresne-Nappert
20948586
Alexandre Dufresne-Nappert

Nathan Frey
20957309
Nathan Frey

Consultant: Dr. Siva Sivoththaman

1 Executive Summary

Solar power is one of the largest growing energy markets. Especially in Canada, and in provinces without hydro power, solar is the key to making a green energy transition. At the same time, solar panels are an expensive, long-term investment. With advances in solar technology every year, older panels cannot compete with newer models in terms of power conversion. Additionally, small drops in efficiency over a panel's lifetime leads to early recycling, which imposes a significant environmental burden. This project aims to increase the efficiency of existing solar panels in a low cost and accessible way.

This project proposes to fabricate a polymer film embedded with luminescent downshifting QDs. We aim to target high energy blue and UV light that cannot be effectively converted in silicon solar cells. QDs in the film will absorb UV light and reemit lower energy orange-infrared light that can be better absorbed by silicon panels. This film must increase the power conversion efficiency of the panel without causing detrimental effects and must do so with a long lifespan. Additionally, the film should be easy to apply, should be resilient to the environment around it, and should not have negative ecological factors.

2 Acknowledgements

We would like to express sincere appreciation to our consultant Dr. Siva Sivoththaman for providing such valuable technical guidance and direction throughout our project. We'd also like to recognize Dr. Hrilina Ghosh, Zhiqiao Yang and Yizun Wang for their help in the lab. Their contributions were instrumental in enabling the successful fabrication and testing of our prototype.

We'd also like to thank Dr. Hany Aziz for his oversight and coordination of the capstone program, Jenn Coggan for her technical insight and logistical help, Dr. Ahmad Ghavami for his help in the lab, Dr. Sebastian Schulz for his design recommendations, and Wendy Gauthier for her logistical help. Finally, we'd like to acknowledge Trusscore, especially Dr. Chad Smithson and Dr. Sam Yang, for supplying materials, tools and technical advice.

3 Glossary of Terms and Acronyms

A: Absorptance
AR: Anti-Reflective
ARC: Anti-Reflective Coating
FYDP: Fourth Year Design Project
GRIN: Gradient Refractive Index
I-V: Current-Voltage
JIT: Just-In-Time
NIL: Nanoimprint Lithography
PCE: Power Conversion Efficiency
PDMS: Poly(dimethyl siloxane)
PET: Poly(ethylene Terephthalate)
PLQY: Photoluminescent Quantum Yield
PMMA: Poly(methyl methacrylate)
PV: Photovoltaic
QD: Quantum Dot
R: Reflectance
RCWA: Rigorous Coupled Wave Analysis
T: Transmittance
TE: Transverse Electric
TIR: Total Internal Reflection
TM: Transverse Magnetic
UV: Ultra-Violet
Vis: Visual Light

4 Table Of Contents

1	Executive Summary	ii
2	Acknowledgements	iii
3	Glossary of Terms and Acronyms	iv
4	Table Of Contents	v
5	Table of Figures	viii
6	Table of Tables	xiii
7	Introduction	14
7.1	Theory	15
7.2	Design Introduction	15
7.3	Customer Requirements	16
7.3.1	Primary Requirements	16
7.3.2	Secondary Requirements	17
8	Discussion	19
8.1	Design Choices	19
8.1.1	Quantum Dots	19
8.1.2	Anti-Reflective Surface	22
8.1.3	Film Material	23
8.1.4	Adhesive-backed Substrate Layer	25
8.1.5	Manufacturing Technique	25
8.2	Design Verification	27
8.2.1	RCWA	27
8.2.2	Photovoltaic Power Conversion Efficiency Simulations	32
8.2.3	Cost Analysis	33
8.3	Prototype Construction	33

8.4	Test Results	34
9	Deviations/Shortfalls.....	38
10	Summary	42
11	References.....	46
Appendix A	Design Specifications.....	A-1
A.1	Primary Specifications	A-1
A.2	Secondary.....	A-2
Appendix B	Verification Plan for the Conceptual Design	B-1
B.1	Surface Energy Calculations.....	B-1
B.2	Solcore Power Conversion Efficiency Simulations.....	B-2
B.3	RCWA Simulations	B-4
B.4	Raytracing Simulations.....	B-7
B.5	Raytracing Simulation Configurations – Python Engine.....	B-8
B.6	Raytracing Simulation Configurations – C++ Engine.....	B-11
B.7	Appendix B References	B-26
Appendix C	Test Plan for Constructed Design Prototype.....	C-1
C.1	Photoluminescent Quantum Yield (*links to ‘Increased PCE’).....	C-2
C.2	QD Absorbance and Emission (*links to ‘Invisibility’).....	C-2
C.3	QD Particle Size and Distribution (*links to ‘Increased PCE’).....	C-2
C.4	Film Transparency (*links to ‘Invisibility’)	C-3
C.5	Print Feasibility (*links to ‘Cost Effectiveness’).....	C-3
C.6	Anti-Reflectivity (*links to ‘Anti-reflectivity’).....	C-3
C.7	Device Flexibility (*links to ‘Ease of Application’).....	C-4
C.8	Final Stage PCE Characterisation (*links to ‘Increased PCE’).....	C-4
C.9	Long Term Weathering Resistivity. (*links to ‘Lifespan’)	C-5

C.10	AFM (*links to ‘Anti-reflectivity’)	C-6
Appendix D Prototype Construction Blueprint D-1		
D.1	Appendix D References	D-4
Appendix E Verification Data.....E-1		
E.1	RCWA Verification Data Results	E-1
E.2	Raytracing Verification Data Results	E-5
E.3	Cost-Savings Analysis	E-12
E.4	Appendix E References	E-14
Appendix F Prototype Test/Measurement Data.....F-1		
F.1	Measurement Conditions	F-1
F.2	Extracted Electrical Parameters	F-1
F.3	Reflectance Data Summary	F-3
F.4	Requirements Summary	F-5
Appendix G FYDP Resources..... G-1		
G.1	Budget	G-1
G.2	Lab Equipment Usage	G-2
G.3	Activity Log	G-3

5 Table of Figures

Fig. 1: SunDOG Device Stack On-Top of a Solar Cell	19
Fig. 2: Band Diagrams of QD Types [4].....	21
Fig. 3: Diagram of Antireflective Textured Surface [9]	23
Figure B-1: Downshifted Solar Spectrum.....	B-3
Figure B-2: Comparison of RCWA Cone Structure Cross-Sections	B-6
Figure B-3: Python Raytracing Scene 1 testing the Waveguide approximation of a Moth-Eye pattern	B-10
Figure B-4: Python Raytracing Scene 2 testing the Sinusoidal geometric interpretation of a Moth-Eye pattern	B-10
Figure B-5: Extrapolated Empirical Moth-Eye Refractive index profile from the Broadband and Crack-Free Antireflection Coatings by Self-Assembled Moth-Eye Patterns paper	B-13
Figure B-6: Max Capture Angle Test Configuration.....	B-15
Figure B-7: TIR of Sinusoidal geometric approximation of Moth-Eye layer Test Configuration	B-15
Figure B-8 : TIR of Moth-Eye Waveguide approximation Test Configuration using Circular QDs (Unit Cell)	B-16
Figure B-9: TIR of Moth-Eye Waveguide approximation Test Configuration using Circular QDs (Large Scale).....	B-16
Figure B-10: TIR of Moth-Eye Waveguide approximation Test Configuration using Cone QDs (Unit Cell)	B-17

Figure B-11: TIR of Moth-Eye Waveguide approximation Test Configuration using Cone QDs (Large Scale)	B-17
Figure B-12: Real Life Device Test Configuration with Circular Quantum Dots at Incident Angle of 0 and 26°	B-18
Figure B-13: Solcore Raytracing Simulation 1 Configuration – Transmittance Wavelength Sweep	B-22
Figure B-14: Solcore Raytracing Simulation 2 Configuration – Transmittance Wavelength Sweep with Perturbances	B-22
Figure B-15: Solcore Raytracing Simulation 3 Configuration – Transmittance Wavelength Sweep with Wavy layers and Perturbances	B-23
Figure B-16: Solcore Raytracing Simulation 4 Configuration – TIR Wavelength Sweep	B-23
Figure B-17: Solcore Raytracing Simulation 5 Configuration – TIR Wavelength Sweep with Perturbances	B-24
Figure B-18: Solcore Raytracing Simulation 6 Configuration – TIR Wavelength Sweep with Wavy layers and Perturbances	B-24
Figure B-19: Solcore Raytracing Simulation 7 Configuration – Control Wavelength Sweep .	B-25
Figure D-1: Entire Prototype Fabrication Process Demonstration;	D-3
Figure D-2: Final Prototype Assembled Onto a 1" Solar Panel	D-3
Figure E-1: Comparison of the Transmittance values across a spectral sweep from 300-1000 nm for all RCWA Simulations	E-1
Figure E-2: Comparison of the Reflectance values across a spectral sweep from 300-1000 nm for all RCWA Simulations	E-1

Figure E-3: Comparison of the Absorbance values across a spectral sweep from 300-1000 nm for all RCWA Simulations	E-2
Figure E-4: Average Transmittance values across all RCWA Moth-Eye Cone structures	E-2
Figure E-5: Average Reflectance values across all RCWA Moth-Eye Cone structures	E-3
Figure E-6: Average Absorbance values across all RCWA Moth-Eye Cone structures	E-3
Figure E-7: TRA values across the 4 Completed RCWA tests for the hillLike Moth-Eye cone structure.....	E-4
Figure E-8: TRA values across the 4 Completed RCWA tests for the linear 250 Moth-Eye cone structure.....	E-4
Figure E-9: TRA values across the 4 Completed RCWA tests for the linear 300 Moth-Eye cone structure.....	E-4
Figure E-10: TRA values across the 4 Completed RCWA tests for the realistic shape Moth-Eye cone structure	E-5
Figure E-11: Comparison of Transmitted power through a Waveguide interpretation of a Moth-Eye AR layer using Empirically Approximated and Linear GRIN	E-5
Figure E-12: TIR Comparison of Linear and Empirically Approximated GRIN using a Cone QD (Large Scale).....	E-6
Figure E-13: TIR Comparison of Linear and Empirically Approximated GRIN using Cone QD (Unit Cell).....	E-6
Figure E-14: TIR Comparison of Linear and Empirically Approximated GRIN using a Circular QD (Large Scale)	E-7
Figure E-15: TIR Comparison of Linear and Empirically Approximated GRIN using Circular QDs (Unit Cell).....	E-7

Figure E-16: TIR power absorbed using a Circular and Cone QD's using a Sinusoidal geometric representation of a Moth-Eye pattern	E-8
Figure E-17: Transmitted Power through a Waveguide approximation of the Film Device Stack and Circular QD's with an Incident Angle of 0 and 26°	E-8
Figure E-18: TRA AR spectral responses from C++ Raytracing Moth-Eye layer (Simulation 1)E-9	
Figure E-19: TRA AR spectral responses from C++ Raytracing Moth-Eye layer (Simulation 2)E-9	
Figure E-20: TRA AR spectral responses from C++ Raytracing Moth-Eye layer (Simulation 3)E-9	
Figure E-21: TRA TIR spectral responses from C++ Raytracing Moth-Eye layer (Simulation 4)	E-10
Figure E-22: TRA TIR spectral responses from C++ Raytracing Moth-Eye layer (Simulation 5)	E-10
Figure E-23: TRA TIR spectral responses from C++ Raytracing Moth-Eye layer (Simulation 6)	E-10
Figure E-24: TRA AR spectral responses from C++ Raytracing Single PDMS layer (Simulation 7)	E-11
Figure F-1: I-V Testing with a Control Solar Panel and One with Device Stack with Different Illumination Conditions	F-2
Figure F-2: PLE of Quantum Dots. Emission measured with excitation $\lambda=380\text{nm}$. Absorption from UV-Vis.....	F-2

Figure F-3: Measured Reflectance with Incoming Light at 8° Under /h Conditions of a Blue-ray Imprinted AR PDMS Surface Film Vs. A Flat PDMS Sample	F-3
Figure F-4: Measured Reflectance with Incoming Light at 8° Under /h Conditions of Plain Solar Panel Vs. One With the Device Stack	F-4

6 Table of Tables

Tab. 1: Table of Specified Primary and Secondary Requirements for SunDOG	18
Tab. 2: Requirements and Associated Test with Success Ranking	37
Table B-1: Solvent Surface Energy Comparison.....	B-2
Table B-2: Test Configurations used for each nano-cone structure	B-6
Table B-3: Raytracing Simulation Benchmark parameter sweeps for FYDPSims module	B-14
Table B-4: Exploratory Parameter sweep bounds for SolcoreSims Module	B-20
Table B-5: Final Parameters for SolcoreSims Module Test Configurations	B-20
Table C-1: Summary of Primary and Secondary Requirements.....	C-1
Table E-1: Annual Cost-Savings with 15% Efficient Solar Panel System.....	E-12
Table E-2: Annual Cost-Savings with 16% Efficient Solar Panel System.....	E-13
Table F-1 Electrical Parameters Extracted from I-V Testing to get Power Increase.....	F-1
Table F-2: Table of Reflectance Data Summarized.....	F-3
Table F-3: Copy of Level of Success for Each Requirement	F-5
Table G-1: NE_2026_20 Final Budget	G-1
Table G-2: NE_2026_20 Lab Equipment Usage	G-2
Table G-3 Lab Equipment Usage per Group Member with Associated Usage Time	G-3

7 Introduction

Solar power is the champion of renewable energy. Canada accumulated 7.64 TWh of solar energy produced in 2023, all while being the 8th largest producer of clean electricity, much of which being due to its high hydropower production [1]. Solar offers an increasingly promising alternative, especially in provinces without hydropower such as in the prairies. Solar panels are measured in terms of their power conversion efficiency (PCE), which is a measure of the power of output electrons over the power of incident photons

$$PCE = \frac{P_{out}}{P_{in}} \quad (1)$$

Modern commercial silicon-based solar panels achieve a PCE of roughly 20-22%, while research cells have achieved PCEs of up to 27% for crystalline silicon cells, 36% for tandem cells and 48% for multijunction cells.

Solar panels are a significant long-term investment. Residential solar panel arrays cost between \$2.42-\$3.05 per Watt, or over \$25,000 for a 7.5 kW system [2]. Although the solar panel generates revenue in the form of power savings, it doesn't return a profit until after 10 or 15 years. SunDOG hopes to provide a solution to consumers, both residential and institutional, that have already committed to solar power. These consumers may be locked into relatively lower yielding panels from years ago. Additionally, a small drop in panel efficiency over its lifetime may lead consumers to recycle their panels early. This leads to an oversized environmental impact due to the high costs of making and recycling panels that are normally offset by their long lifespan. Our solution aims to increase their yield to keep up with the continual advancements in the field at a lower cost and simpler process than replacing their solar panels.

7.1 Theory

The spectral response (SR) of a solar panel is related to its external quantum efficiency (EQE), where the EQE is the ratio of collected carriers and incident photons of a certain frequency

$$EQE(\lambda) = \frac{\# \text{ carriers}}{\# \text{ photons with wavelength } \lambda} \quad (2)$$

While spectral response is the ratio of current collected relative to incident power

$$SR(\lambda) = \frac{q\lambda}{hc} \cdot EQE(\lambda) \quad (3)$$

Solar panels show low spectral response at both short and long wavelengths. Wavelengths longer than the bandgap of the silicon (1.12 eV) cannot be absorbed by solar panels and are out of the scope of a top-applied film. Short wavelengths can be absorbed efficiently by silicon but do not show a spectral response due to their absorption depth. Absorption is proportional to the inverse of a photon's energy, with k being the extinction coefficient of the material

$$\alpha = \frac{4\pi k}{\lambda} \propto E_{\text{photon}} \quad (4)$$

High energy photons are absorbed at the top layer of a solar panel where there is a low carrier collection probability due to high surface recombination rates and defects. Medium wavelengths in the orange to near infrared (NIR) range have an ideal response.

7.2 Design Introduction

Semiconducting nanocrystals, or quantum dots (QDs), are nanoparticles with specific quantized energies and bandgaps that can be tuned with varying size. QDs show strong and controlled absorbance in the ultra-violet (UV) range and good reemission in a targeted visible wavelength, which is called downshifting. QDs are enveloped between a durable PDMS top layer and a PET substrate with adhesive backing. Additionally, a geometric anti-reflective layer is embossed into

the PDMS to further increase the possible efficiency. The design is chosen to mirror a scalable roll-to-roll process.

7.3 Customer Requirements

7.3.1 Primary Requirements

Increase PCE by 1%. The core performance metric of this design is an absolute improvement of one percentage point in power conversion efficiency. The increase in power savings must offset the cost of the film over some moderate time frame. Increasing PCE by 1% (absolute, i.e. from 20% to 21%) would be a substantial enough selling point.

Invisible. The film must not interfere with the existing solar panel's efficiency. Silicon absorbs across the UV-IR range, especially between 400 nm and 1100 nm. The new film must have less than 1% absorbance, on average, in this spectra range, with an absorbance of 0% in the 600 nm to 800 nm range where silicon has the highest absorption. The film should also have relatively low absorption (<10%) in the 250 nm to 400 nm range. Low scattering is also implied in this requirement. Optical haze and reabsorption of fluoresced photons that would redirect or attenuate incident photons should also be avoided.

10 Year Lifespan. Solar panels have lifespans between 20 and 40 years. As such, any additional product must have a long lifespan to align with consumer expectations. Due to the nature of the film material, extremely long lifespans are not realistic.

Anti-reflectivity. The film should have low reflectance to maximize photon flux into the solar panel. A typical solar panel addresses reflectance by creating a pyramidal surface which allows any reflected light another chance to come incident on the glass. Adding a film may reduce the effectiveness of this technique, so the film should also add some amount of anti-

reflectiveness. A reduction in reflectance by 3% absolute would lead to a net positive impact on the solar panel's reflectance.

7.3.2 Secondary Requirements

Ease of application. The product must be designed for straightforward installation by end users without the need for specialized tools or professional assistance. To meet this requirement, the film should be flexible, lightweight and easy to adhere to the panel surface.

Hydrophobic. The film must present a hydrophobic surface to minimize water retention, reduce surface fouling, and preserve optical transmission over time. This can be quantified as the film achieving a water contact angle $\geq 100^\circ$, which would allow water to bead and roll off.

Cost-effectiveness/Scalability. The film must deliver performance improvements at a cost point that is acceptable for residential, institutional, and utility-scale customers. For homeowners, this typically means that the return on electricity over the film's lifetime should be lower than the film's cost. A similar calculation can be done for solar farms. A scalable process would allow for much cheaper manufacturing.

Winterproof. Canadian winters can reach temperatures as low as -40°C . A film applied to a solar panel must match the thermal properties of the solar panel's glass to avoid tension under extreme heat or cold. The film should be designed to have a thermal expansion coefficient within 10% of tempered glass or be elastic enough to not crack under stress.

Tab. 1: Table of Specified Primary and Secondary Requirements for SunDOG

Requirement	Primary/Secondary	Functional/ Non-Functional	Minimum Performance Specification
PCE Increase by +1%	Primary	Functional	PCE +1%
Transmit 99% of light	Primary	Functional	99% Transmittance (400 nm -1100 nm)
10-year lifespan	Primary	Non-Functional	10 years
Lower reflectance by 3%	Primary	Functional	-3% Reflectance
Easy to apply	Secondary	Non-Functional	Flexible
Hydrophobic surface	Secondary	Functional	$\geq 100^\circ$ water-film contact angle
Cost efficient/ scalable process	Secondary	Non-Functional	Material cost < energy savings
Winterproof	Secondary	Functional	$\leq 10\%$ mismatch of thermal expansion coefficients

8 Discussion

SunDOG's design consists of four components that are layered into a device stack. This device stack can then be applied to a solar panel. A model of the SunDOG device stack can be seen in **Error! Reference source not found.** consisting of a solar cell base (dark purple), an adhesive-backed PET substrate (white), a layer of QDs (dark teal), an encapsulating PDMS layer (light blue), and a textured AR surface (gray). The base solar cell is independent of the device stack and is intended to represent existing panels that have been applied to the roofs of buildings. The device was constructed on a PET substrate with an adhesive backing for ease of application. A layer of QDs was applied directly to the substrate and then encapsulated in a PDMS layer to ensure stability and prevent leakage to the environment. The top surface of the PDMS was textured with an AR pattern to ensure that minimal light was prevented from reaching the device.

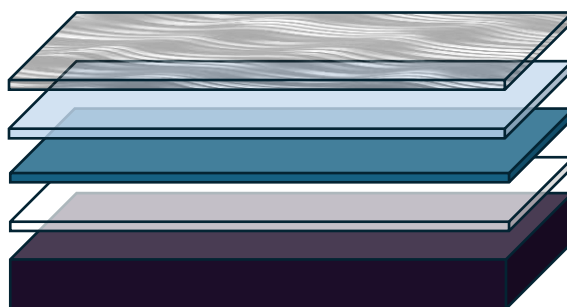


Fig. 1: SunDOG Device Stack On-Top of a Solar Cell

8.1 Design Choices

8.1.1 Quantum Dots

QDs are semiconductor nanocrystals that are often functionalized to improve stability and solubility. Their optoelectronic properties are heavily dependent on their size and material

composition, meaning they can be engineered for specific applications. When the radius of a QD is smaller than the exciton Bohr radius of the bulk semiconductor material, the exciton gets delocalized over the entire dot and becomes confined (Equation 5).

$$E_n = \frac{\hbar^2 \pi^2 n^2}{2m^* R^2} \quad (5)$$

Since the energy levels in a QD are inversely related to its size, smaller quantum dots have wider bandgaps than larger ones and thus emit higher energy light. In general, QDs cannot be made to have smaller bandgaps than their bulk semiconductor, as shown in the *Brus Equation* (Equation 6) [3, pp. 298–304].

$$E_{g,QD} = E_{g,bulk} + \frac{\hbar^2 \pi^2}{2m_r^* R^2} - \frac{1.8q_e^2}{4\pi\epsilon_r\epsilon_0 R} \quad (6)$$

In these core QDs, surface defects often lead to premature exciton quenching, severely reducing their photoluminescent quantum yield (PLQY). A second semiconducting material can be grown on the QD as a shell which can passivate the surface defects and provide a surface that can be functionalized without affecting the QD's optical properties; however, the shell can cause additional problems. If the crystal structures of the two semiconducting materials have a large mismatch in lattice parameter, it can cause physical stress to the core of the QD, leading to nonradiative recombination.

There are several “types” of core/shell QDs, depending on the relative size and position of the shell bandgap compared to that of the core. Type I QDs have a shell with a wider bandgap with both a deeper conduction band and a shallower valence band relative to vacuum which ensures the excitons are confined to the core (Fig. 2). This greatly improves the PLQY of the QDs and is ideal for downshifting applications [4].

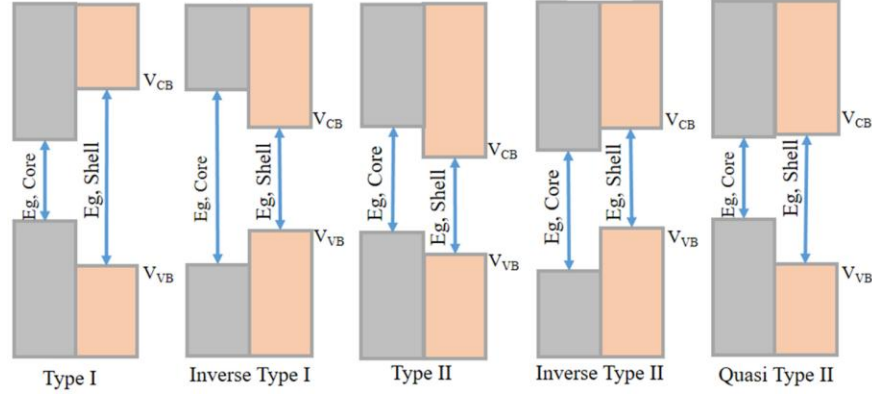


Fig. 2: Band Diagrams of QD Types [4]

There were several requirements that the QDs chosen for SunDOG had to meet. Firstly, high PLQY, Type I QDs were essential due to the nature of the project: a low PLQY would mean that light was being effectively prevented from reaching the solar cell due to a lack of reemission, lowering the overall efficiency of the device. Their emission peak must fall within the orange-red section of the visible spectrum since this is in the optimal range for silicon PV absorption. They had to be stable when encased in the PDMS layer, and they had to be relatively safe for the people and the environment in case of any leakage. The first QDs considered were CdSe/ZnS due to their high PLQY, ease of synthesis, and familiarity. However, cadmium is a heavy metal, and it is extremely toxic to people and the environment. It is listed as a “Schedule 1 Toxic Substance” by the Government of Canada, meaning its use would be heavily regulated and increase production costs [5]. Similarly, lead-based QDs were not chosen due to environmental and health concerns. Two environmentally friendly QDs were identified as viable options: copper-based QDs (CuInS_2) and indium-based QDs (InP). Both options have a high PLQY and are non-toxic; however, InP dots were ultimately chosen because the ones available to purchase had a higher PLQY in the target spectral region. The chosen QDs were InP/ZnSe/ZnS which have a core/shell/shell structure designed to improve stability and PLQY by reducing the lattice mismatch between the layers

(atypical core/shell QDs would be InP/ZnS with a lattice mismatch of 7.84% whereas as the core/shell/shell structure has a mismatch of ~4% between each layer) [6]. The QDs were purchased from *NanoOptical Materials Inc.* with a listed emission peak of 580 nm, a peak width of 62 nm, and a PLQY of >90% [7].

8.1.2 Anti-Reflective Surface

The top layer of the SunDOG device is an anti-reflective surface. This component is essential for ensuring that the majority of the light that hits the device is directed downwards towards the solar panel rather than reflected away and lost, along with keeping emitted light from the QDs inside the device. There are two common methods to achieve this: an antireflective coating or a nano-textured surface pattern. Antireflective coatings are created by adding a thin layer of a dielectric material to the surface of the device. By controlling the thickness of the coating, reflections from the top surface and the dielectric-device interface destructively interfere, reducing the amount of light reflecting away from the device [8, pp. 54–55]. This method is ideal for targeted wavelengths but is generally not applicable to arbitrary wavelengths [9]. Additionally, the method requires nanoscale precision in height, which is not suitable for a print process.

An alternative approach is a refractive index gradient. At each interface, light is refracted further downwards into the device. To control a gradient, several layers of modified refractive index materials are often used. Alternatively, nano-scaled domes can be used, called a moth-eye pattern. When the dome feature is smaller than the wavelength of the incident light, the textured surface is seen as one continuous layer with a refractive index proportional to the ratio between air and the surface material. Fig. 3 shows such a patterning and the effective refractive index through the surface. [9]. An additional benefit to using a textured surface is being able to maintain the physical properties of the surface material.

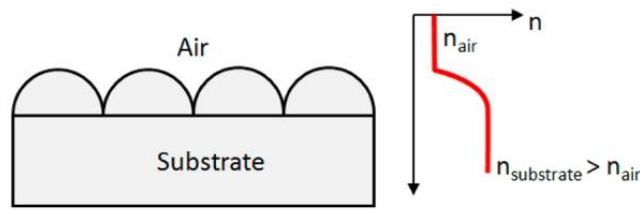


Fig. 3: Diagram of Antireflective Textured Surface [9]

It was determined that the textured surface would be ideal for the SunDOG device due to the necessity of capturing a broad spectrum of light and the desire to maintain the hydrophobic and weather-resistant properties of the PDMS film. Several methods were explored for texturing the surface of the film. The first method involved creating a uniform pore structure via a breath-method evaporation of a polymer dissolved in chloroform. This method creates uniform pores with a 250 nm diameter which is essential for capturing wavelengths in the UV to visible range [9]. These pores would then be used as a mold to cure PDMS in the desired dome. The polymer used for this method is complex and requires a multi-step synthesis to create which proved too difficult and time-consuming to complete within the constraints of the Capstone Design Project.

Blu-ray discs have a feature size of roughly 320 nm, which can be accessed by peeling off the back coating of the disc. This feature size is larger than desired and will not provide anti-reflective benefits to light with wavelengths of 350nm and lower, but should provide a suitable reduction in reflectance across most of the solar spectrum [10]. The Blu-ray discs can be used in a nanoimprint lithography (NIL) process to transfer the moth-eye pattern to PDMS.

8.1.3 Film Material

The bulk of the device stack is the clear encapsulation layer. Poly(dimethyl siloxane) (PDMS) was chosen due to good optical and physical properties, including its resistance to weathering. Other

common polymers, such as poly(ethylene terephthalate) (PET) and poly(methyl methacrylate) (PMMA) were considered but discarded for various reasons.

PDMS demonstrates exceptional flexibility, allowing the finished film to be rolled up, which would aid the ease of use for an end customer. Additionally, this high elasticity means the surface should withstand impacts, such as from hail, better. PMMA is intrinsically brittle and would not be easily rolled, and PET has a higher degree of plastic deformation than PDMS. Across a range of temperatures, PDMS maintains its desired rubbery state due to a low glass transition temperature ($T_g \approx -120^\circ\text{C}$), far lower than the glass transition temperatures of PMMA ($T_g \approx 105^\circ\text{C}$) and PET ($T_g \approx 70^\circ\text{C}$), enabling the same performance over the winter as during the summer.

Additionally, PDMS has a more interesting chemistry. The Si-O backbone in PDMS is much more stable than the C-C backbone of most common transparent polymers. Si-O bonds have a dissociation energy of 460 kJ/mol, compared to 360 kJ/mol for C-C bonds. This stronger backbone means that under sustained solar irradiation, PDMS will not show yellowing, compared to carbon backbone polymers which typically show yellowing after only a year. Yellowing both embrittles the polymer and reduces its transparency. The nonpolar methyl groups give PDMS a strong hydrophobic surface characteristic which lowers the surface energy. Water has a much higher surface energy than PDMS, and to show hydrophobicity, the resultant contact angle should be higher than 100° . Due to this hydrophobic nature, PDMS has low swelling and won't absorb water, which could quench the quantum dots. The film will also exhibit a self-cleaning 'lotus effect' where water beads and rolls off the surface, carrying away any surface contaminants.

Finally, PDMS is a heat cured polymer. Sylgard 184 is chosen as the type of PDMS due to its good properties and use in other opto-electronic applications. This PDMS is made up of two

parts, the first being a viscous resin with vinylated end groups and the second having resin with hydride end groups and a platinum catalyst. After mixing, a hydrosilylation reaction occurs to crosslink the two components which can be accelerated under heat. Due to this controllable curing, PDMS can be cured into any mold shape, enabling the patterning of the antireflective layer with NIL.

8.1.4 Adhesive-backed Substrate Layer

The substrate layer is essential for manufacturing the device while minimizing defects. Additionally, choosing an adhesive-backed material allows for ease of application of the device to the solar cell. The chosen material was required to balance UV resistance for outdoor applications and be tough and flexible for easy application. The chosen material also had to be easily available to maintain low production costs. Poly(ethylene terephthalate) (PET) was identified as an ideal candidate due its low cost, common use in outdoor applications, good physical properties, and high transparency in the visible range [11], [12], [13]. Although PET does not have the best UV resistance out of all available plastics, this was determined to not be an issue because the majority of the incoming UV light would be absorbed by the SunDOG film. The composition of the adhesive backing was also determined to be unimportant due to the thinness of the layer. For these reasons, 127 μm thick PET with a pre-applied adhesive backing layer was purchased from McMaster-Carr and used as the substrate for the SunDOG device.

8.1.5 Manufacturing Technique

The design is chosen to be compatible with roll-to-roll processing. Scalable print techniques allow for much lower manufacturing costs, enabling an affordable end cost. A QD ink can be coated onto a PET substrate using slot-die coating and dried. To simplify steps during the prototyping, a

100% toluene ink is used. Toluene has a surface energy of 29 mN/m while PET has a surface energy of 45 mN/m, with a contact angle of 65° (good, but not excellent wettability). As discussed in Deviations/Shortfalls, the ink showed poor wettability and would not be suitable for large scale printing.

Additionally, solvents can be classified by their greenness score [14], a measure of a solvent's health, safety and environmental impact, with high greenness scores being required in some large-scale print facilities. The ink can be further modified with thickening agent to increase viscosity, leading to a more even, thick film, or with surfactants, lowering the surface energy further and ensuring better coating. Finally, a binary or ternary solvent system can be considered. By mixing several solvents with different drying times, good solubility and surface coverage can be achieved in systems that don't have one solvent ideal for both.

In a large-scale print operation, further modifications can be made to the PET substrate to improve adhesion and wettability. Corona treatment creates dangling hydroxy functional groups on the plastic surface, increasing the polar surface energy and creating a more hydrophilic surface.

As described in Appendix B, anisole and ethoxybenzene both offer a compelling alternative to toluene. Both have high greenness scores (above 7.0) and are chemically similar to toluene, with the only difference being the replacement of toluene's methyl end group with a methoxy or ethoxy end group. The added oxygen provides a polar component to the solvent. While toluene should be a compatible solvent with PET, toluene does not benefit from corona surface functionalization because it lacks a polar component. As such, the adhesion and wettability of a toluene ink will not show any improvements after treatment. Contrarily, an anisole or ethoxybenzene based ink would be able to interact with the surface functionalization and improve their film qualities. Anisole and ethoxybenzene both have a higher boiling point compared to toluene (154°C, 170°C vs 111°C)

which could impact the film's drying ability. A binary system composed of anisole or ethoxybenzene and a low boiling point solvent like ethyl acetate ($\text{bp} = 77^\circ\text{C}$, $R_a(\text{anisole}) = 4.1$, $G = 6.7$) could be made so that a good film would be produced with a bulk of the ink evaporating quickly after deposition.

A layer of PDMS can be coated on top of the dried QDs using the same slot-die technique. The viscosity of PDMS can be controlled by mixing in a small amount of toluene or other volatile solvent to reduce viscosity, and/or by heating the polymer to allow some initial crosslinking to increase. A NIL mold of the moth-eye pattern can be incorporated in line to cure the PDMS into the antireflective pattern while maintaining the continuous process. This reusability allows the cost of the mold to be amortized across more product, allowing a more expensive and precise mold to be made.

8.2 Design Verification

To verify the design met the performance and functional requirements set forth in Design Choices, a multi-scale verification framework bridging photonic behavior to the design efficiency was developed. This consists of validating the device stack across 3 regimes: nano-scale electromagnetic response, macro-scale light propagation and total photovoltaic PCE. Additionally, a device cost-savings analysis was done to ensure its economic viability.

8.2.1 RCWA

Rigorous Coupled Wave Analysis (RCWA) was used to confirm the AR capabilities of the periodic moth-eye nanostructure, providing a rigorous electromagnetic baseline for the devices' surface performance at the nanoscale. By comparing the results against a custom-developed raytracing engine, a high degree of correlation was found between the two methods, confirming

the ray-based approximations accurately capture the device primary optical characteristics. Consequently, macro-scale simulations and extensive parameter sweeps were performed using the raytracing engine to provide computational scalability necessary for modelling bulk angular and internal reflection dependencies. This enables a robust, statistically significant predictive estimation of the devices capabilities under realistic operating conditions, the results of which were used in integration tests to predict the final PCE of the PV device.

Using the synthesized data from these design verification tests, provides a robust proof-of-concept, demonstrating the moth-eye AR layer and the integrated device stack satisfy the projects technical requirements. A detailed account of the execution plan and the subsequent results are provided in the following sections. Furthermore, comprehensive simulation configurations and raw data plots are documented in Verification Plan for the Conceptual Design and Verification Data respectively.

Upon analysis, it is observed that Test 3 did not complete across all simulation categories. A detailed explanation of the theoretical framework governing RCWA and the specific parameters used for the simulations is provided RCWA Simulations. The failure of Test 3 is attributed to the significant computational overhead of approximating the structure using 200 uniform layers and a Harmonic Order (N) of 5, which exceeded the reasonable execution time frame.

However, the remaining tests, whose results are documented in RCWA Verification Data Results, indicate all geometric categories performed nearly identically. The average transmittance and reflectance of the simulations remained within narrow margin across the 300-1000 nm spectrum. Each test and geometry exhibited peak transmittance within the 450-500 nm range and a corresponding minimum in the 650-750 nm region as shown in Figure E-1 - Figure E-10. The

reflectance maintained a reciprocal relationship, while the absorbance of each test remained constant and effectively negligible.

Furthermore, a complete azimuthal independent response is observed as evidenced from the results of Test 1 and Test 2. A noticeable trend is observed regarding the base PDMS layer thickness. As the thickness is increased from 0 nm to the ideal specification of 171 450 nm, the transmittance and reflectance curves exhibited increased noise. Conversely, the absorbance signal demonstrated increased stability as thickness increased. Averaging the results, the “realistic” cone structure emerged as the optimal configuration, demonstrating the best performance in the 450-500 nm spectral range, while maintaining median performance in the infrared region. These findings provide a robust baseline, confirming the moth-eye pattern satisfies optical requirements for the final device integration.

Further simulations were run with raytracing, a robust simulation method that models light transport by tracing paths from a virtual camera through a pixel plane into a 3D environment [15]. The approach utilizes a backward traversal where rays are emitted from an eye and tracked as they interact with scene geometry until they lose sufficient power, exit the simulation volume or reach a predefined number of bounces [16]. Beyond its origins in the film and gaming industry, raytracing serves as a powerful tool for scientific analysis. By calculating geometric ray intersections and applying fundamental optical laws, properties of complex devices with reactive components can be accurately approximated. This methodology is uniquely suited for simulating the complete device stack including the moth-eye pattern, the PDMS film volume and the re-emission from QD's.

A series of eleven test categories were executed to verify the optical performance of the device and provide a comparative benchmark against RCWA results. A detailed explanation of the

computational framework governing the raytracing engine and the specific parameters used for these simulation configurations is provided in Raytracing Simulations. Initial verification focused on characterizing the total internal reflection within the wavy moth-eye geometry and validating a waveguide approximation of the AR layer. This approximation sampled uniform layer refractive indices based on an established empirical curve defining the effective refractive index as a function of height [9].

Subsequent simulations evaluated QD integration, utilizing a 550 nm solar source to characterize the transmittance and reflectance efficiency of the complete device stack. The final seven simulation categories were dedicated to generating the spectral and angular response curves required for PCE modelling with Solcore. These tests included extensive sweeps from 300 – 1000 nm across varying incident angles and layer counts. To account for moth-eye formation defects, structural perturbations were introduced by applying Gaussian-sampled angular shifts to the surface normals and modelling layer interfaces with sinusoidal defects. These stochastic methods provide a predictive account of how structural irregularities affect global transmittance and internal reflection, ensuring the final PCE estimation reflects realistic device behavior. Complete execution parameters and sweep configurations for these tests are documented in Raytracing Simulation Configurations – C++ Engine.

Initial results across the first four verification categories demonstrate the impact of discretization and geometry on the optical performance of the AR layer. In the Maximum Capture Angle tests, a direct correlation is observed between the number of layers and the resulting transmittance. As the number of layers increase, the structure showed improved AR performance, though a sharp drop-off in transmittance occurred as the incident angle approached 50° as seen in Figure E-11. Similarly, the TIR simulations yielded promising results, showing that a high

percentage of light remains trapped within the device stack regardless of the QD representation. As the number of layers increased, the internal reflection efficiency approached 100% establishing a robust waveguide effect as seen in Figure E-12. Notably, the implementation of the simulation's parameters, such as the moth-eye pattern's gradient refractive index (GRIN), geometry and the QD interpretation have large effects on the films presumed performance. Interestingly, the linear GRIN performed the best in the Transmission regime, while the empirically approximated GRIN performed best in the TIR regime, suggesting the AR layer will be great at trapping light emitted by QDs, but may struggle to capture incident light. The effects of poor interpretation and simulation parameters can be observed in Figure E-12 - Figure E-16, showing the criticality of accurately configuring the device and tuning the devices parameters. A specific trend is observed regarding QD density across all simulations: increasing the number of QDs after a threshold resulted in marginal reduction in TIR. This is attributed to the increased probability of isotropic re-emission in unfavourable directions, suggesting that the concentration of QDs must be carefully tuned to balance absorption and light trapping. Additionally, the methodology used to represent the refractive index gradient at different height fractions proved critical to simulation accuracy. Using a linear shift in refractive index across layers produced unfavourable results in the TIR regime compared to the empirically approximated curves from the paper "Broadband and Crack-Free Antireflection Coatings by Self Assembled Moth Eye Patterns", while the inverse is true for the transmittance regime [9]. Furthermore, the limitations of ray-based approximations at the nanoscale were evidenced when modelling the Moth-Eye as sinusoidal structures, as visualized in Figure B-7. This configuration failed to capture light in TIR scenarios, evidenced in Figure E-16, confirming the waveguide representation is superior model for macro-scale propagation analysis.

Consequently, the final seven simulation categories utilized a mathematically derived fill-factor refractive index function, which incorporated the Sellmeier equation to generate accurate spectral response curves for Solcore modelling. Extensive parameter sweeps consolidated the predicted film performance, showing a peak transmittance of 97.5% at 0° incidence, which gradually decreased to 93.5% at a 60° incident angle as shown in Figure E-18. Introduction of perturbation played a minor role in reducing the transmittance at higher angles of incidence, evidenced by Figure E-19, while representing the layers as sinusoidal structures introduced sudden spectral noise observed in Figure E-20. In the case of the TIR characterization, simulations 4 through 6 were conducted to evaluate the trapping efficiency of device stack. While simulation 4 established a baseline, the addition of normal perturbances and sinusoidal geometry introduced significant stochastic noise as seen in Figure E-21 - Figure E-23. Despite this noise, the plots confirm a high degree of internal reflection is maintained across the 300-1000 nm range, even under non-ideal structural conditions. Comparatively, the single-layer PDMS simulation without a moth-eye structure reported a consistent 2-3% reduction in transmittance across all angles, validating the functional necessity of the gradient AR interface as evidenced in Figure E-24.

8.2.2 Photovoltaic Power Conversion Efficiency Simulations

The downshifting effect of quantum dots can be quantified by computing the I-V characteristic of a simulated solar cell under various spectral shapes. A full breakdown of the simulation steps is described in Appendix B. At low PLQY, across all absorbance values, there is a direct decrease in PCE. This is expected as the quantum dots absorb light without compensating for it. Contrarily, at high PLQY, there is an increase at all absorbance levels, with more absorbance leading to a larger increase. This validates the theory directly of high energy light being inefficient in solar energy and that high energy photons would have a larger impact on the sample. While the absorption

factor used in the calculation should in theory be usable to determine the needed thickness of the quantum dot film, in practice, the factor does not directly align with a specific concentration or thickness. As such, the model's main role is to validate the existence of an increase in PCE with the SunDOG stack.

8.2.3 Cost Analysis

The device cost-savings analysis was done using a hypothetical 5 kW capacity solar panel system with 15% efficiency panels set in Kitchener, Ontario. To compare cost savings, the SunDOG device was simulated by increasing the panel efficiency to 16%, without changing any other parameters. This resulted in additional annual cost savings of \$58.78, or \$587.77 over the 10-year lifetime of the device. An analysis was also done to compare SunDOG production costs to the generated cost-savings. It was found that significant improvements to the manufacturing process, including in-house QD synthesis, would be required to turn SunDOG into a viable product. The full cost analysis break-down can be found in Appendix E.

8.3 Prototype Construction

The final prototype consisted of films sequentially layered, designed with the backbone of roll-to-roll scalability for easy application onto a solar panel surface. The device stack included a solar cell base, and adhesive-backed PET substrate, a coated QD layer, all encapsulated within the bulk PDMS layer with a moth-eye patterned surface. Further details within this section will be discussed in Appendix D.

Procedurally, QDs in toluene were first drop-cast onto the PET substrate and allowed to dry, forming the active optical layer. Separately, the base component of Sylgard 184 was mixed with 10 wt% toluene using a vortex stirrer for 10 minutes to ensure homogeneous mixing. The cure component of the PDMS was then added and the solution was mixed again. The uncured

PDMS was then drop-cast onto a sacrificial Teenage Mutant Ninja Turtles Blu-ray disc. This assembly was placed on a hot plate at 120 °C for roughly 10 to 15 minutes, allowing the PDMS to cure fully. Once cured, the textured PDMS layer was peeled off the Blu-ray and sandwiched with the QD-coated PET substrate to form the complete film. The final film was then applied to a small solar panel for prototype testing. Additional tape was added on the edges to prevent the layers from separating during handling.

8.4 Test Results

The performance of the SunDOG prototype was evaluated through a series of experimental tests designed to assess its optical and electrical impact on PV systems. The complete dataset is provided in Appendix F, while the key findings and representative results are discussed in this section.

To evaluate the effect of the film on PV efficiency, I-V testing was conducted on baseline and coated solar panels under controlled white light and UV illumination separately. This allowed for the extraction of key parameters such as open-circuit voltage, short-circuit current, and maximum power output. The coated samples showed a slight decrease in maximum power output of approximately 3-4% under white light illumination when compared to the uncoated control. However, an increase in power output by 17% was observed under UV illumination, indicating that the film interacts favorably within the targeted spectral region.

The small reduction in performance under white light conditions is almost entirely a result of the fabrication and application inconsistencies, such as non-uniform air pockets within the film and potential surface defects, which could introduce additional optical losses through scattering or partial absorption. Importantly, these results do not invalidate the main design theory but rather highlight the sensitivity to the system regarding manufacturing quality and process control.

Additionally, testing limitations, including the lack of a calibrated full-spectrum light source, prevented accurate calculation of absolute PCE. Overall, while the prototype did not meet the intended performance improvement under all conditions, the observed spectral-dependent behavior and UV enhancement support the feasibility of the concept, with performance expected to improve through refined fabrication techniques.

Optical performance of the film was evaluated primarily through reflectance measurements, from which transmittance behavior was inferred using the related $T = 1 - R - A$. Direct absorbance measurements were not to be conducted, but assumed to be negligible due to PDMS' known optical transparency in the visible range, allowing transmittance to be approximated primarily as the complement of reflectance.

At the material level, reflectance testing of standalone PDMS films showed that applying the anti-reflective (AR) surface texture reduced reflectance, confirming that the surface pattern effectively modified optical interactions. However, when the film was applied to a solar panel, system-level measurements indicated an overall reflectance of approximately ~13%, which limited transmittance and prevented achieving the 99% transmission target.

Despite this, a relative decrease in reflectance of about 2% was still observed compared to the uncoated panel, indicating that the AR effect was partially maintained after integration. An increase in measured reflectance at shorter wavelengths was also observed. This behavior is likely due to photon absorption followed by re-emission, which would be detected as reflected light by the spectrometer, which does not have a monochromator. These results indicated that while the AR structure was effective at the material level, system-level performance was affected by fabrication quality and interactions with the underlying solar panel surface. Further optimization is required to achieve the desired transmittance targets.

Overall, the optical testing results indicate that the SunDOG film partially satisfies the design requirements related to reflectance and transmittance. The anti-reflective (AR) surface successfully reduced reflectance at the material level and achieved an approximate 2% reduction when applied to a solar panel, demonstrating that the underlying design concept is valid. However, the overall reflectance remained higher than desired ($\sim 13\%$), preventing the system from achieving the target of 99% light transmission. As a result, the transmittance requirement is considered only partially met. The difference between material-level and system-level performance is also primarily due to fabrication inconsistencies and integration effects with the existing solar panel surface. Therefore, while the results confirm the feasibility and effectiveness of the AR approach, further optimization in manufacturing and application processes is required to fully meet the specified performance targets.

Several design requirements, including long-term lifespan, hydrophobicity, and environmental durability for winter conditions, were not experimentally validated within the scope and timeline of this project. These characteristics were instead evaluated based on established material properties and literature-supported behavior of the selected components. The PDMS substrate is known for optical stability and resistance to yellowing under prolonged UV exposure, supporting an expected operational lifespan on the order of multiple years, potentially approaching 10 years.

The PDMS surface is naturally hydrophobic. Additionally, due to the added roughness from the moth-eye surface patterning, this hydrophobicity is accentuated, promoting water repellence. This surface structure, combined with the flexibility and chemical stability of PDMS, suggests resistance to environmental factors such as moisture, temperature variation, and icing, supporting suitability for outdoor and winter conditions.

However, these performance characteristics were not directly tested under controlled environmental conditions due to time constraints and therefore remained theoretical within the context of this project. As a result, these requirements were classified as not validated, with further work required to experimentally confirm long-term durability, hydrophobic performance, and environmental resilience under real-world operating conditions.

Tab. 2: Requirements and Associated Test with Success Ranking

Requirement	Test Conducted	Level of Success
PCE Increase by +1%	IV-Testing	Semi-successful
Transmit 99% of light	UV-Vis	Semi-successful
10-year lifespan	Not tested	N/A
Lower reflectance by 3%	UV-Vis	Semi-successful
Easy to apply	Not tested	N/A
Hydrophobic surface	Not tested	N/A
Cost efficient/scalable process	Calculated cost vs power generated per area, Designed to be scalable	Semi-successful
Winterproof	Not tested	N/A

9 Deviations/Shortfalls

Although the final prototype demonstrated partial success during validation of the main design concept, several deviations from the original design and shortfalls from the set requirements were observed. The most significant limitation was not being able to achieve the intended system-level performance once the film was placed onto a solar panel. While some individual components of the design performed as intended on their own, specifically the UV-responsive enhancement from the QDs and the reduced reflectance on textured PDMS samples, the gains did not fully compensate for the other errors incorporated into the final prototype. Instead, the final prototype experienced optical and fabrication-related losses that limited its ability to meet the target specifications for increased power conversion efficiency, invisibility, and anti-reflectivity.

A major deviation from the original design occurred in the layer architecture. The initial design proposed embedding the QDs directly within the PDMS matrix to form a single, integrated layer. After the successful characterization of individual components, QDs were introduced to the Sylgard 184 base for prototype construction. Following the standard PDMS curing procedure, the curing agent was then added to the base mixture containing the QDs and deposited to form a film. Upon addition of the QDs to the PDMS mixture, it was found that the film was no longer able to cure. Further research has found that the oleylamine ligands that were used to stabilize the QDs favourably bind to the platinum catalyst in the PDMS curing agent, severely inhibiting reactivity [17]. Potential solutions include incorporating the QDs in a separate polymer first (with the same refractive index as PDMS to minimize refraction) or performing a ligand exchange to ones that would not inhibit the platinum. Available resources did not allow for exploring these options, so a separate workaround had to be used instead: the QDs in toluene were drop-cast directly onto the PET substrate and then a separate PDMS layer was added on top.

This modified approach introduced its own challenges regarding the interface between the QD layer and PET substrate. While toluene and PET should have favourable surface energies, very poor wettability was observed, indicating there was some surface modification on the PET. Leftover residue from the manufacturing process could drastically reduce the surface energy of the plastic meaning toluene would no longer fully wet the surface. Modifications such as corona treatment were attempted without success, due to toluene not have a polar surface energy component. To proceed with the final prototype, the ink was applied to the PET and constantly spread with a plastic pipette to ensure the entire film was covered. The result was an inconsistent film that should still have most of the same properties as a uniform film.

Although this revised fabrication method allowed the prototype to be completed, it introduced an additional interface into the system and increased the likelihood of air gaps, non-uniform contact, and optical mismatch between layers. These factors, combined with the non-uniform quantum dot distribution, contributed to scattering and reflectance losses that reduced overall device effectiveness.

The initially chosen method to create the antireflective surface patterning for the device proved to have a much more complex and difficult synthesis than anticipated. It was determined that the polymer synthesis could not be completed to the required purity and yield with the available time and resources, so a new method had to be used. The chosen method, Blu-ray NIL, resulted in feature sizes that were suboptimal for capturing light in the target range, reducing the overall effectiveness of the AR surface. This was unfortunately unavoidable due to the cost involved with more advanced and precise lithography techniques which would have produced a more ideal structure.

Additional shortfalls were observed in the intrinsic optical performance of the quantum dots. The measured PLQY was approximately 64%, significantly lower than the manufacturer's advertised value of greater than 90% for the material supplied by NanoOptical Materials. This reduction in PLQY indicated less efficient photon conversion, which directly limited the effectiveness of the intended downshifting mechanism. Potential causes include degradation during handling or storage, solvent interactions, and differences between supplier characterization conditions and those used in this work.

At the system level, these fabrication and material limitations impacted the optical requirements of the design. The film did not fully meet the invisibility requirement, as non-uniform deposition, interfacial effects, and re-emission introduced scattering and optical losses within the active spectral range of the solar cell. Similarly, while the AR PDMS structure was effective at the material level, the assembled system did not achieve the targeted reduction in reflectance due to integration-related losses and surface imperfections.

Several secondary requirements also remained incomplete rather than conclusively failed. Long-term lifespan, hydrophobicity, and winterproofing were not experimentally validated within the available project timeline and budget, therefore, being instead assessed based on expected material behavior. Cost-effectiveness was not achieved at the prototype scale due to low absolute energy gains relative to fabrication cost, although the selected processes remained compatible with scalable manufacturing techniques.

An additional limitation was also encountered in the ability to directly measure the primary performance metric of the system. The intended method for evaluating PCE relied on the use of a calibrated solar simulator operating under AM1.5G standard test conditions. However, this crucial device would experience a system failure relating to the light source that could not be replaced for

at least a few months, all during the final stages of the project. As a result, direct measurement of the PCE could not be performed. Instead, performance assessment would be conducted through I-V measurements under available UV and white light sources, from which relative changes in power output could be measured but where input power remained unknown. While this approach provided insight into the separate performances between the coated and uncoated samples, it did not allow for validation of the absolute PCE improvement specified in Appendix A. This limitation further constrained the ability to get a true reading on the primary requirement being met.

Collectively, these deviations and shortfalls all contribute to the inability of the prototype to meet the primary requirement of achieving a 1% absolute increase in PCE. Electrical testing showed improvements under the target UV conditions however experienced a slight reduction in performance under white light. Although absolute PCE could not be calculated directly, relative changes in maximum power output were used instead to see the effects on the PCE, being directly proportional to each other. The small observed decrease in power under broadband white light therefore indicated a corresponding decrease in effective PCE, confirming that the prototype did not achieve the intended performance improvement. This was somewhat expected, as the downshifting mechanism would not interact with non-UV light. Regardless, without a combined testing method, the results are inconclusive. The combined effects of fabrication inconsistencies, optical losses, non-uniform QD distribution, and reduced PLQY ultimately prevent the realization of the intended system-level performance.

10 Summary

This project aimed to develop a low-cost, scalable film capable of improving the performance of existing silicon-based solar panels through luminescent downshifting and optical surface engineering. The primary objective was to achieve a 1% absolute increase in PCE while maintaining optical transparency, reducing reflectance, and ensuring long-term durability and ease of application. The motivation for this work was rooted in the growing need to improve the efficiency of already-installed solar panels without requiring a full system replacement, therefore extending their usable lifetime and reducing environmental impact associated with premature disposal, all while avoiding inline manufacturing troubles related to the production of solar panels.

The core design approach involved the integration of multiple functional layers into a flexible polymer film that could easily be applied directly to a solar panel. The device consisted of an adhesive-backed PET substrate, a QD layer for luminescent downshifting, and a PDMS encapsulation layer with a textured AR surface. The QDs were selected to absorb high-energy UV photons and re-emit light in the orange-red spectral region, where silicon solar cells exhibit higher spectral response. This downshifting process was intended to increase the effective utilization of incident solar energy. Along with that, the textured PDMS surface was designed to reduce reflectance by creating a gradient refractive index profile, improving light coupling into the device, and preventing remission from the QDs to escape. The overall system was developed with compatibility for roll-to-roll manufacturing, supporting scalability and cost-effectiveness as key design considerations.

To validate the design, both simulation and experimental approaches were conducted. RCWA and raytracing simulations were used to model optical behavior across the device stack, confirming that the anti-reflective structure and layered system could achieve high transmittance

and low reflectance under ideal conditions. These simulations predicted strong optical performance, including transmittance values approaching 97.5% under normal incidence. However, experimental validation revealed discrepancies between predicted and realized performance due to fabrication limitations and system-level integration effects.

Prototype fabrication followed a layered construction approach, where QDs were drop-cast onto a PET substrate and combined with a separately cured PDMS layer containing the AR surface. While this method enabled successful construction within project constraints, it introduced multiple fabrication challenges. Surface energy mismatch between the QD solution and PET led to non-uniform deposition, producing inconsistent QD distribution. Additionally, attempts to integrate QDs directly into PDMS were unsuccessful due to interference with polymer curing, forcing a redesign of the layer structure. These procedures introduced optical losses through scattering, interfacial mismatch, and non-uniformity, limiting system performance.

Experimental testing focused on evaluating both electrical and optical performance. I–V measurements under UV and white light illumination showed that the device behaved differently across the different spectral regions. Under UV illumination, the coated panel demonstrated an increase in maximum power from approximately 0.18 mW to 0.21 mW, corresponding to a relative improvement of approximately 16–17%. This confirmed that the quantum dot downshifting mechanism was functioning as intended within the targeted spectral range. However, under white light illumination, the maximum power decreased from approximately 7.74 mW to 7.48 mW, representing a reduction of approximately 3–4%. Since PCE is directly proportional to power output under fixed illumination conditions, this decrease indicated a corresponding reduction in effective PCE at the system level.

Optical testing further showed these findings. Reflectance measurements of normal PDMS films demonstrated a reduction in reflectance when the AR texture was applied, validating the effectiveness of the surface patterning approach. However, when placed onto a solar panel, the system resulted in an overall reflectance of approximately 13%, which prevented achieving the target of 99% transmittance. Although a relative reduction in reflectance of approximately 2% was still observed compared to the uncoated panel, fabrication defects, interfacial effects, and scattering limited overall performance. Additionally with the full device stack, reflectance increases at shorter wavelengths suggested that re-emission from the QDs have not been reflected into the device stack towards the solar panel, suggesting improvements should be directed towards the AR fabrication process.

Several design requirements were only partially met or not validated. The primary requirement of achieving a 1% absolute increase in PCE was not satisfied due to the combined effects of fabrication inconsistencies, optical losses, and reduced QD performance. The invisibility requirement was also not fully met, as non-uniform deposition and scattering introduced unwanted optical losses. Anti-reflectivity showed partial success, with improvements observed at the singular material level but limited gains at the system level. Secondary requirements such as hydrophobicity, long-term lifespan, and winter durability were not experimentally tested within the project timeline and were instead evaluated based on expected material properties. Cost-effectiveness was not achieved at the prototype scale, although the process remains scalable and compatible with industrial manufacturing methods.

An additional limitation in the evaluation of the prototype was the inability to directly measure absolute PCE due to failure of the solar simulator required for standardized testing. As a result, assessing the performance of the device stack relied on relative power measurements

obtained from I–V curves. While this approach provided meaningful comparative insights, it limited the ability to fully validate compliance with the primary design specification under standard conditions.

Despite these limitations, the results of this project proved that the main design concept is valid. The observed UV enhancement confirms the usage of QDs for spectral modifications, and the AR surface showed decent improvements at the material level. The differences between simulated and experimental performance highlight the importance of fabrication quality and process control for turning theoretical designs into functional devices.

Future work should focus on improving fabrication consistency and material integration. This includes developing alternative deposition techniques such as slot-die or spin coating to eliminate non-uniform QD distribution and optimizing the solvent system to better match substrate surface energy. Simply exploring methods to successfully embed QDs within PDMS without inhibiting curing would prevent the surface energy mismatching all together. Improvements in AR pattern fidelity and layer adhesion would further reduce optical losses. Additionally, access to calibrated solar simulation equipment is important for future accurate PCE characterization. Long-term testing should also be implemented to validate durability, hydrophobicity, and environmental resistance under real-world conditions. Finally, further optimization of QD properties, including maintaining high PLQY during processing, will be critical in maximizing device performance.

In conclusion, while the prototype did not meet all primary performance requirements, it successfully demonstrated the feasibility of a luminescent downshifting film as a method for enhancing solar panel performance. The project provides a strong foundation for future development, with clear pathways for improving both fabrication processes and system-level integration to achieve the desired performance targets.

11 References

- [1] “Topic: Electricity in Canada,” Statista. Accessed: Apr. 03, 2026. [Online]. Available: <https://www.statista.com/topics/11724/electricity-in-canada/>
- [2] R. Urban, “Cost of Solar Power In Canada 2024,” energyhub.org. Accessed: Apr. 03, 2026. [Online]. Available: <https://www.energyhub.org/cost-solar-power-canada/>
- [3] George W. Hanson, “Chapter 9: Models of Semiconductor Quantum Wells, Quantum Wires, and Quantum Dots,” in *Fundamentals of Nanoelectronics*, 1st ed., pp. 280–316.
- [4] A. Sahu and D. Kumar, “Core-shell quantum dots: A review on classification, materials, application, and theoretical modeling,” *J. Alloys Compd.*, vol. 924, p. 166508, Nov. 2022, doi: 10.1016/j.jallcom.2022.166508.
- [5] E. and C. C. Canada, “Toxic substances list: schedule 1.” Accessed: Apr. 04, 2026. [Online]. Available: <https://www.canada.ca/en/environment-climate-change/services/canadian-environmental-protection-act-registry/substances-list/toxic/schedule-1.html>
- [6] John R. Rumble, ed., “PROPERTIES OF SEMICONDUCTORS,” in *CRC Handbook of Chemistry and Physics*, 106th ed., Boca Raton, FL: CRC Press/Taylor & Francis, 2025.
- [7] “InP580-OS Specifications and Spectra - NanoOptical Materials.” Accessed: Apr. 04, 2026. [Online]. Available: <https://nomcorp.com/inp580-os-specifications-and-spectra/>
- [8] “Optoelectronics & Photonics - Principles & Practices.”
- [9] F. Galeotti, F. Trespidi, G. Timò, and M. Pasini, “Broadband and Crack-Free Antireflection Coatings by Self-Assembled Moth Eye Patterns,” *ACS Appl. Mater. Interfaces*, vol. 6, no. 8, pp. 5827–5834, Apr. 2014, doi: 10.1021/am500687f.
- [10] A. J. Smith, C. Wang, D. Guo, C. Sun, and J. Huang, “Repurposing Blu-ray movie discs as quasi-random nanoimprinting templates for photon management,” *Nat. Commun.*, vol. 5, no. 1, p. 5517, Nov. 2014, doi: 10.1038/ncomms6517.
- [11] C. E. Carraher Jr., *Seymour/Carraher’s Polymer Chemistry*, 0 ed. CRC Press, 2007. doi: 10.1201/9781420051032.
- [12] “UV light Resistance & Properties: Polymer Properties.” Accessed: Apr. 05, 2026. [Online]. Available: <https://www.specialchem.com/plastics/guide/uv-light-resistance>
- [13] A. Szczurek *et al.*, “Polyethylene terephthalate (PET) optical properties deterioration induced by temperature and protective effect of organically modified SiO₂–TiO₂ coating,” *Mater. Chem. Phys.*, vol. 306, p. 128016, Sep. 2023, doi: 10.1016/j.matchemphys.2023.128016.
- [14] C. Larsen *et al.*, “A tool for identifying green solvents for printed electronics,” *Nat. Commun.*, vol. 12, no. 1, p. 4510, Jul. 2021, doi: 10.1038/s41467-021-24761-x.
- [15] “Ray Tracing,” NVIDIA Developer. Accessed: Apr. 03, 2026. [Online]. Available: <https://developer.nvidia.com/discover/ray-tracing>
- [16] “Ray tracing (graphics),” *Wikipedia*. Mar. 26, 2026. Accessed: Apr. 03, 2026. [Online]. Available: [https://en.wikipedia.org/w/index.php?title=Ray_tracing_\(graphics\)&oldid=1345468903](https://en.wikipedia.org/w/index.php?title=Ray_tracing_(graphics)&oldid=1345468903)
- [17] Q. Zhao, H. Li, X. Zhang, S. Yu, S. Wang, and G. Sun, “Platinum in-situ catalytic oleylamine combustion removal process for carbon supported platinum nanoparticles,” *J. Energy Chem.*, vol. 41, pp. 120–125, Feb. 2020, doi: 10.1016/j.jechem.2019.05.006.

Appendix A Design Specifications

A.1 Primary Specifications

Increase PCE by 1%. The core performance metric of this design is an absolute improvement of one percentage point in power conversion efficiency. The increase in power savings must offset the cost of the film over some moderate time frame. Increasing PCE by 1% (absolute, i.e. from 20% to 21%) would be a substantial enough selling point. This would be measured under standardized test conditions on test solar cells and can be measured from current-voltage measurements before and after the film is applied. The increased revenue from the solar panels will offset the cost of the film and the increase in long-term return will be substantial enough to be used as a selling point to consumers.

Invisible. The film must not interfere with the existing solar panel's efficiency. Silicon absorbs across the UV-IR range, especially between 400 nm and 1100 nm. The new film must have less than 1% absorbance, on average, in this spectra range, with an absorbance of 0% in the 600 nm to 800 nm range where silicon has the highest absorption. The film should also have relatively low absorption (<10%) in the 250 nm to 400 nm range. Low scattering is also implied in this requirement. Optical haze and reabsorption of fluoresced photons that would redirect or attenuate incident photons should also be avoided. The overall objective is to ensure that, aside from the intended QD down conversion, the film remains optically transparent to the active absorption band of silicon and does not impose penalties on baseline device performance. This requirement can be verified by UV-Vis-NIR spectroscopy on the various layers of the design

10 Year Lifespan. Solar panels have lifespans between 20 and 40 years. As such, any additional product must have a long lifespan to align with consumer expectations. Due to the nature

of the film material, extremely long lifespans are not realistic. We aim for a lifespan of 10 years, tested under accelerated aging with inspiration from standards such as IEC 61215, IEC 61646. Achieving this benchmark will allow the product to be integrated into existing photovoltaic systems without compromising long-term reliability, while still offering a cost-effective and technically realistic path to performance enhancement.

Anti-reflectivity. The film should have low reflectance to maximize photon flux into the solar panel. A typical solar panel addresses reflectance by creating a jagged [find better word**] surface which allows any reflected light another chance to come incident on the glass. Adding a film may reduce the effectiveness of this technique, so the film should also add some amount of anti-reflectiveness. A reduction in reflectance by 3% absolute would lead to a net positive impact on the solar panel's reflectance. Meeting this specification ensures that the film does not impose optical penalties relative to standard PV encapsulants and contributes positively to overall module efficiency by increasing the light going into silicon.

A.2 Secondary

Ease of application. The product must be designed for straightforward installation by end users without the need for *specialized tools or professional assistance*. *Consumer adoption is strongly influenced by the convenience of integration into existing systems, particularly in the residential market segment. To meet this requirement, the film should be flexible, lightweight and easy to adhere to the panel surface.*

Hydrophobic. The film must present a hydrophobic surface to minimize water retention, reduce surface fouling, and preserve optical transmission over time. This can be quantified as the film achieving a water contact angle $\geq 100^\circ$, which would allow water to bead and roll off.

Cost-effectiveness/Scalability. The film must deliver performance improvements at a cost point that is acceptable for residential, institutional, and utility-scale customers. For homeowners, this typically means that the return on electricity over the film's lifetime should be lower than the film's cost. A similar calculation can be done for solar farms. A scalable process would allow for much cheaper manufacturing. For solar farms and institutions, the benchmark is stricter: the cost must align with or improve upon the levelized cost of electricity (LCOE) associated with existing PV modules. Materials and processes will be evaluated and priced at scale to justify this requirement. Since solar panels are long-lived capital investments, any aftermarket product must demonstrate a favorable cost-to-benefit ratio to justify adoption.

Winterproof. Canadian winters can reach temperatures as low as -40°C . A film applied to a solar panel must match the thermal properties of the solar panel's glass to avoid tension under extreme heat or cold. The film should be designed to have a thermal expansion coefficient within 10% of tempered glass or be elastic enough to not crack under stress.

Appendix B Verification Plan for the Conceptual Design

B.1 Surface Energy Calculations

To show hydrophobicity of the final film, the water (1) – PDMS (2) contact angle should be $>100^\circ$.

$$\gamma_1(H_2O) = 72.8 \text{ dynes/cm}, \quad \gamma_2(PDMS) = 19.8 \text{ dynes/cm}$$

$$H_2O: \gamma_1^d = 22.1 \text{ dynes/cm}, \quad \gamma_1^p = 50.7 \text{ dynes/cm}$$

$$PDMS: \gamma_2^d = 19.0 \text{ dynes/cm}, \quad \gamma_2^p = 0.8 \text{ dynes/cm}$$

$$\gamma_{12} = \gamma_1 + \gamma_2 - \frac{4\gamma_1^d\gamma_2^d}{\gamma_1^d + \gamma_2^d} - \frac{4\gamma_1^p\gamma_2^p}{\gamma_1^p + \gamma_2^p} \quad (B.1)$$

$$\gamma_{12} = 72.8 + 19.8 - \frac{4 \cdot 22.1 \cdot 19.0}{22.1 + 19.0} - \frac{4 \cdot 50.7 \cdot 0.8}{50.7 + 0.8} = 48.58 \text{ dynes/cm}$$

$$\gamma_2 = \gamma_{12} + \gamma_1 \cos \theta$$

$$\theta = \arccos\left(\frac{\gamma_2 - \gamma_{12}}{\gamma_1}\right) = \arccos\left(\frac{19.8 - 48.58}{72.8}\right) = 113.29^\circ$$

Surface energy components can be taken from the squares of their respective Hansen solubility parameters (HSP) and then normalized with the known total surface energy. The Green Solvent Selection tool helpfully provides the three HSP, the total surface energy and the greenness score for most common solvents. [B-0]

$$\gamma^d \propto \delta_D^2, \quad \gamma^p \propto (\delta_P^2 + \delta_H^2)$$

$$A = \frac{\gamma}{\gamma^d + \gamma^p} \quad (B.2)$$

From this, we can compute the dispersive and polar surface energy components for a range of common solvents. Additionally, we can compare the greenness scores of each, and their distance from toluene in HSP space

$$R_a^2 = 4(\delta_{D,1} - \delta_{D,2})^2 + (\delta_{P,1} - \delta_{P,2})^2 + (\delta_{H,1} - \delta_{H,2})^2 \quad (B.3)$$

With the same intuition as ‘like dissolves like’, solvents that are close in HSP space should dissolve similar solutes. **Error! Reference source not found.** describes the computed γ_d and γ_p values, and their greenness and distance.

Table B-1: Solvent Surface Energy Comparison

Solvent	Surface Energy γ (dynes/cm)	δ_D	δ_P	δ_H	γ_d	γ_p	G (Greenness)	Ra
Toluene	28.5	18.0	1.4	2.0	28.0	0.5	6.0	0.0
Anisole	34.1	17.8	4.4	6.9	28.1	6.0	7.4	5.8
D-Limonene	26.0	17.2	1.8	4.3	24.2	1.8	5.5	2.8
Ethoxybenzene	32.7	18.4	4.5	4.0	29.5	3.2	7.2	3.8
Cyclohexanone	35.0	17.8	8.4	5.1	26.8	8.2	7.2	7.7
Ethyl acetate	24.0	15.8	5.3	7.2	18.2	5.8	6.7	7.9
Butyl acetate	23.0	15.8	3.7	6.3	18.9	4.1	7.5	6.7
Water	72.0	15.5	16.0	42.3	7.6	64.4	7.3	43.2

Solvents with greenness scores above 7 are considered as suitable inks in our system to meet sustainability goals. Ethoxybenzene and anisole are both chemically similar to toluene, but with an ethoxy and methoxy group respectively to replace toluene’s methyl group, creating a polar component. These solvents can then interact with the polar component of corona treated PET, which toluene could not.

B.2 Solcore Power Conversion Efficiency Simulations

The simulation starts with the AM1.5G standard solar spectrum. This spectrum is passed into a downshifting function, meant to emulate the effects of quantum dots. This function is fitted with experimental parameters and produces a new spectral distribution, as shown in Figure F-2. The emission also compensates for the PLQY of the dots and has a factor to account for light being emitted upwards away from the solar cell. This data is then fed into the solcore solver, a python

module. Solcore predicts the full IV curve, of which the important parameters for PCE can be extracted, those being short circuit current (J_{sc} , taken at $V=0$), open circuit voltage (V_{oc} , taken at $I=0$), maximum power (P_{max} , taken at $\max(I \cdot V)$) and fill factor (FF, the ratio $P_{max}/J_{sc}V_{oc}$).

$$PCE = \frac{FF \cdot J_{sc} \cdot V_{oc}}{P_{in}} \quad (B.4)$$

PCE is then plotted across several absorbance values, representing different concentrations of quantum dots, and PLQY values of the dots. Higher absorbance means a larger fraction of the target light is absorbed, while higher PLQY means more of the absorbed light is added back to the spectrum at and around the emission wavelength.

The simulated solar panel is an P-N amorphous silicon junction with emitter $N_d = 10^{18} \text{cm}^{-3}$, $d = 0.4 \mu\text{m}$, $S_p = 10^4 \text{cm} \cdot \text{s}^{-1}$ and base $N_a = 10^{16} \text{cm}^{-3}$, $d = 200 \mu\text{m}$, $S_n = 10^2 \text{cm} \cdot \text{s}^{-1}$. Voltage is swept from -1.2 V to 0V and TMM is used as the optics method inside the cell.

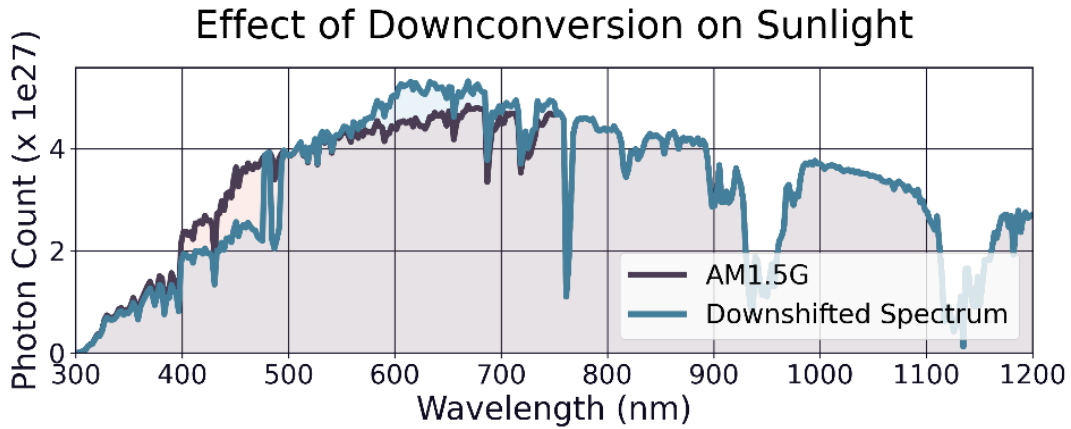


Figure B-1: Downshifted Solar Spectrum

B.3 RCWA Simulations

RCWA, also known as Fourier Modal Method, is a computational electromagnetic technique used to solve light scattering in periodic dielectric structure [B-2], [B-3]. It functions by discretizing a structure into a series of uniform layers along the direction of light propagation. In complex, non-planar geometries like the Moth-Eye nanostructure, a staircase approximation is used to divide the profile into many thin slices. Within each layer, the Maxwell's equations are solved analytically in the Fourier domain by representing the device geometry and electromagnetic fields as finite sums of spatial harmonics [B-2], [B-3]. The global solution is obtained by matching boundary conditions in each interface through scattering matrices, these propagate the electromagnetic modes across the device stacks. This approach provides a high-fidelity estimation of surface level reflectance and device transmittance, establishing the baseline electromagnetic performance required to validate the AR layer.

The implementation of the RCWA framework utilized the `RigorousCoupledWaveAnalysis.jl` Julia library to evaluate the electromagnetic response and establish the baseline performance of the Moth-Eye AR layer [B-1]. To model the non-planar geometries of the nanocones, the periodic structures were discretized into a staircase approximation of either 80 or 200 uniform thin slices along the light propagation direction. Within each layer, Maxwell's equations are solved analytically in the Fourier domain, and boundary conditions are matched at each interface using scattering matrices to propagate electromagnetic modes across the device stack [B-2], [B-3]. Four distinct geometric profiles were evaluated to characterize the impact of morphology on optical performance: linear radius cutoffs at 250 nm and 300 nm, a sinusoidal hill-like structure, and a smoothened "realistic" equivalent. The device stack was subjected to a spectral sweep from 300 nm to 1000 nm across 141 points to validate effective AR operation across the ultraviolet and visible spectrum. These configurations were dispatched to

a remote server for parallel execution, requiring between 12 to 36 hours of processing time depending on the specific layer count and harmonic orders utilized. Detailed simulation parameters and structural comparisons are documented in the RCWA Simulations section and Figure B-2 respectively.

To characterize the impact of the surface morphology, four distinct geometric radius functions were implemented to define the pillar profile: linear radius cutoffs at 250 nm and 300 nm, a sinusoidal “hill-like” structure and a “realistic” smoothened cone. These profiles are depicted in Figure B-2 below. The simulation parameters were configured to establish a baseline for spectral performance from 300 to 1000 nm. The specific configuration for each test run across all structures are detailed in Table B-2.

The materials were modeled using a constant permittivity for the PDMS ($\epsilon_r = n^2 \approx 1.41^2$) and air ($\epsilon_r = 1$) while the substrate used to represent the solar panels glass was modeled using the built-in fused silica material. For each sampled wavelength, the solver evaluated the TE and TM polarizations, averaging their results to determine the Reflectance (R), Transmittance (T) and Absorbance (A). These simulations provide high-fidelity electromagnetic baseline required to validate subsequent macro-scale raytracing approximations.

The execution of the RCWA scripts was managed by transferring the 20 distinct Julia files to a remote server, into a structured directory to ensure ease of access during the data extraction process. Each script is run in parallel on the server utilizing all its cores to maximize throughput, requiring between 12 to 36 hours depending on the number of layers. In all cases, Test 3 did not complete since a Harmonic Order N of 5 required excessive computational power to complete. The scripts were started in parallel using a Bash script, which executed and ran the simulations as Linux background processes. Upon completion of the sweeps, each simulation generated a

standardized CSV output file containing the transmittance, reflectance and absorbance spectral responses. These datasets were subsequently aggregated, analysed and plotted to define a baseline to compare against Raytracing, these plots can be found in RCWA Verification Data Results.

Cross-Sectional Comparison of Cone Structures

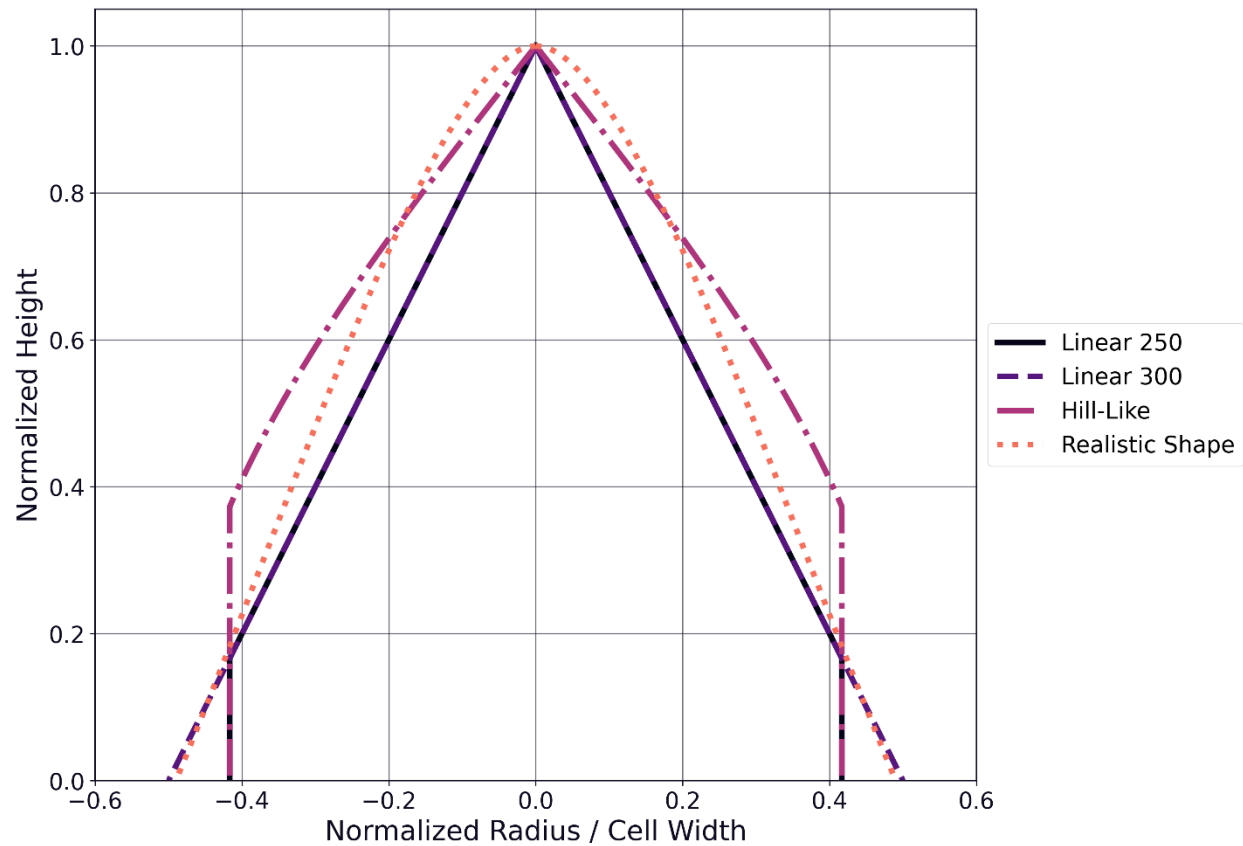


Figure B-2: Comparison of RCWA Cone Structure Cross-Sections

Table B-2: Test Configurations used for each nano-cone structure

Test	Azimuth Angle (α)	Harmonic Order (N)	Number of Layers	PDMS Thickness (nm)
1	0	3	200	0
2	45.0	3	200	0
3	0	5	200	0
4	0	3	80	100
5	0	3	80	171 450

B.4 Raytracing Simulations

Raytracing is a computational technique simulating light transport by tracing paths from a virtual camera or ray emitter through a viewing pane and into a 2D or 3D environment. The “backward” traversal is utilized for computational efficiency, calculating geometric ray intersections to model physical optical phenomena [B-4]. A ray is defined parametrically using an origin vector $\vec{s}$ and a normalized direction vector $\vec{d}$ such that any point $\vec{x}$ along the ray at a distance t is expressed as shown in Equation B.5 [B-5].

$$\vec{x} = \vec{s} + t\vec{d} \quad (B.5)$$

Finding the intersection between a ray and a geometric boundary requires substituting the ray equation into the shape’s definition to solve for the smallest positive distance t . Upon intersection, fundamental optical laws, including Snell’s Law (Equation B.6) and the Fresnel equations (Equations B.7 – B.10) are applied to determine the fraction of power reflected and transmitted. The algorithm recursively generates new rays for these components, tracking their energy until they exit the system, exceed a maximum bounce limit, or fall below a predefined power threshold [B-5].

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2) \quad (B.6)$$

$$t_s = \frac{2n_1 \cos(\theta_1)}{n_1 \cos(\theta_1) + n_2 \cos(\theta_2)} \quad (B.7)$$

$$r_s = \frac{n_1 \cos(\theta_1) - n_2 \cos(\theta_2)}{n_1 \cos(\theta_1) + n_2 \cos(\theta_2)} \quad (B.8)$$

$$t_p = \frac{2n_1 \cos(\theta_1)}{n_1 \cos(\theta_2) + n_2 \cos(\theta_1)} \quad (B.9)$$

$$r_p = \frac{n_1 \cos(\theta_2) - n_2 \cos(\theta_1)}{n_1 \cos(\theta_2) + n_2 \cos(\theta_1)} \quad (B.10)$$

This pipeline enables custom sampling, allowing for the characterization of the film under real-world operating conditions through large-scale parameter sweeps. To maintain computational scalability during these extensive simulations, accelerating structures such as BVH are utilized to focus light path computations within a defined view frame [B-7]. By establishing that the device stack can be effectively approximated through this method, the raytracing engine facilitates the optimization of the film and anti-reflective layer, leading to a precise final prediction of the device's optical capabilities.

B.5 Raytracing Simulation Configurations – Python Engine

To evaluate the device's optical behavior, an initial 2D raytracing engine was developed in Python. The architecture discretized environmental boundaries into line segments defined by a starting node $\vec{A}$ and an ending node $\vec{B}$ and an associated refractive index for the enclosed medium. To model the Moth-Eye AR interface, a parametrized wave object generated a sinusoidal profile discretized into the segments governed by Equation B.11, where A is the amplitude, B dictates the spatial frequency, C is the phase offset and D is the vertical offset. Alternative representations of the AR structure is a discretized stack of segments with a GRIN transitioning from air to the films bulk PDMS index. To optimize the intersection mathematics, the Numba library's JIT compilation was implemented to accelerate array operations and cross-product calculations required to identify the intersection distance t between a ray and a segment [B-6].

$$y = A \sin(B(x - C)) + D \quad (B.11)$$

Initial verification utilized scenes, such as the ones visualized in Figure B-3 and Figure B-6, to evaluate power capture and core optical logic. These tests configured a virtual environment with a bounding perimeter of mirrors, and a Moth-Eye AR layer approximation or geometric interpretation. The approximation consisted of 4 stacked waveguide layers with progressively larger refractive indices to simulate the GRIN transition. A “target” is placed at the bottom of the environment, which is designed to absorb incident rays and sum their cumulative power. An array of incident rays is propagated from a point source, and the engine iterates through every geometric segment to calculate the minimum intersection distance, evaluate the Fresnel coefficients and dispatch the resulting child rays. While these test scenes validated the libraries core functionality and the viability of utilizing raytracing, the Python implementation relied on a brute-force intersection approach without hierarchical spatial acceleration structure. Consequently, the computational complexity scaled poorly as the structural discretization of the Moth-Eye layer increased. The engine proved too slow to compute the millions of ray paths required for the large-scale spectral and angular parameter sweeps, establishing the necessity to migrate the core optical computation logic into a high-performance, compiled C++ architecture.

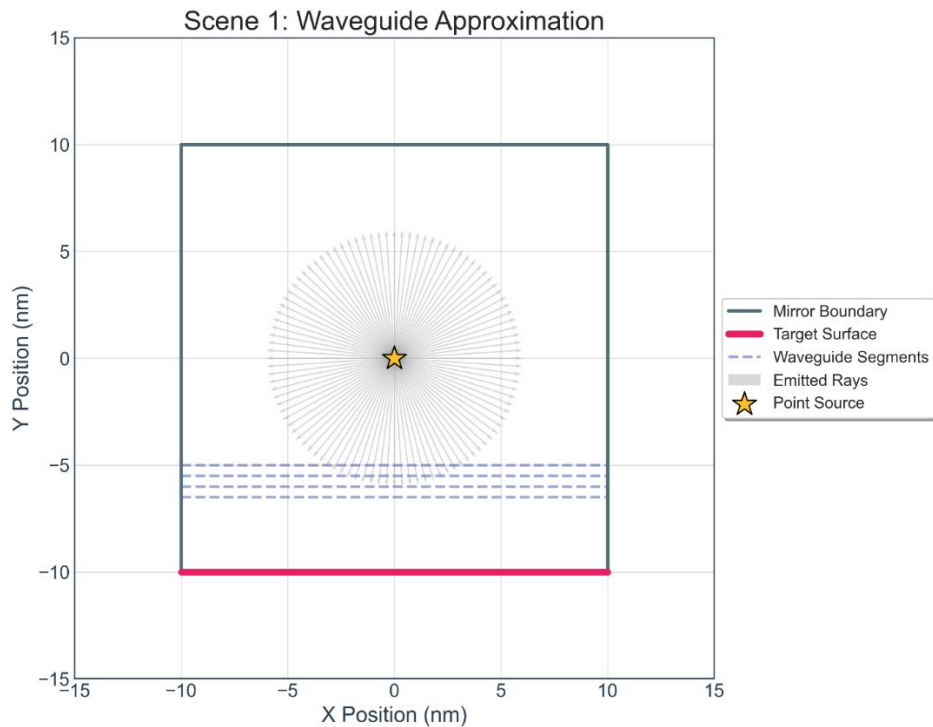


Figure B-3: Python Raytracing Scene 1 testing the Waveguide approximation of a Moth-Eye pattern

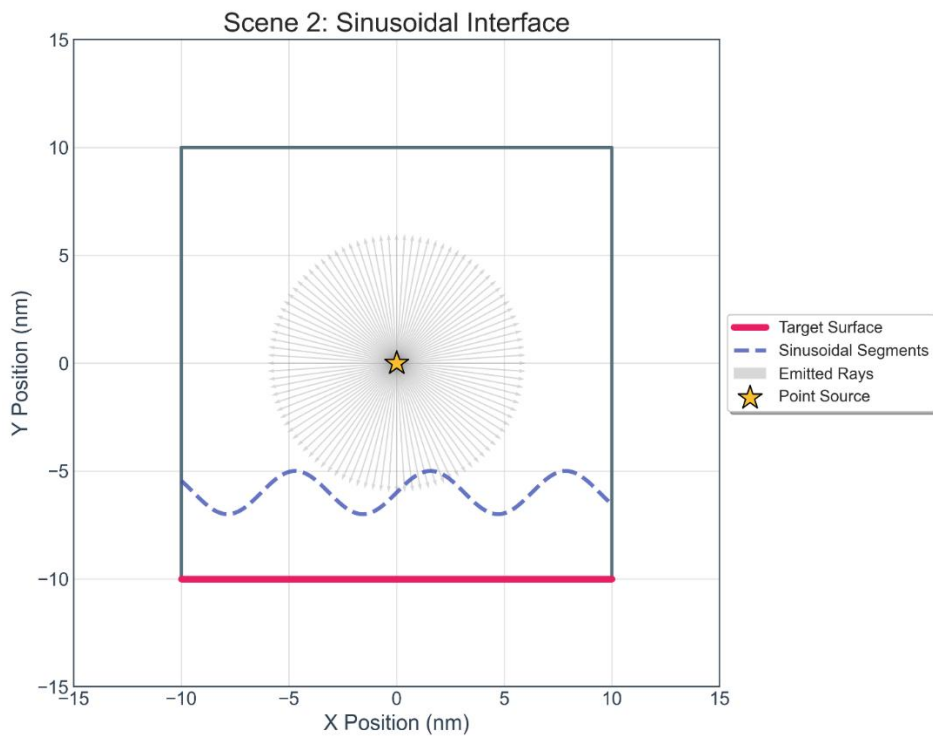


Figure B-4: Python Raytracing Scene 2 testing the Sinusoidal geometric interpretation of a Moth-Eye pattern

B.6 Raytracing Simulation Configurations – C++ Engine

The C++ raytracing engine retains the fundamental optical logic of its Python predecessor, treating light parametrically and evaluating intersections against discrete geometric segments to simulate reflections and refraction. However, the architecture introduces critical enhancements to overcome the severe computational bottlenecks observed in the initial prototype, where benchmarking revealed execution speed was insufficient for large-scale simulations. To achieve the necessary performance, the engine was rewritten in C++, resulting in a 100x increase in processing speed. The most significant difference is the implementation of BVH spatial acceleration structures. Instead of executing a brute-force intersection check against every segment in the same scene, the engine recursively partitions the geometry into a binary tree of bounding boxes. A propagating ray is first tested against these macro-boundaries, if no intersection is detected, the algorithm discards the entire enclosed sub-tree, effectively reducing the time complexity of the intersection search from linear to logarithmic [B-7], [B-8], [B-9].

Furthermore, the compiled C++ environment facilitates robust memory management and enables the integration of complex stochastic procedures, such as Monte-Carlo methods to sample ray parameters and surface normals from Gaussian, solar, or AM1.5G distributions. This high-performance library also supports advanced optical features including the use of Sellmeier equations for dynamic refractive index calculations across various wavelengths. To further reduce computational complexity, the engine was optimized for 2D cross-sectional manipulation, maintaining physical accuracy for the device stack's uniform profile. Each simulation utilized a standardized geometry consisting of a light source, segments representing the device structure, and a target to quantify transmittance. This refined architecture additionally enables future

potential optimizations, such as offloading ray intersection calculations to the GPU via CUDA and dedicated NVIDIA Ray Tracing Cores.

To quantify the optical properties of the device stack, the engine categorizes the final state of each ray based on its interaction with the scene geometry. Rays that intersect with the defined “target” segment are recorded as transmitted light, while rays that exit the simulation volume into the surrounding void are classified as reflected. Conversely, rays that fail to reach the boundary because they have lost 99.5% of their initial power through successive interactions or have exceeded 1000 bounces are considered absorbed by the material.

Initial design verification, executed within the FYDPSims module, focused on establishing a baseline GRIN approximation of the Moth-Eye structure to rapidly prototype the waveguide mechanics. This macroscopic representation defined the refractive index as a function of the structural height utilizing a pre-calculated fifth-degree polynomial as shown in Equation B.12, which is an extrapolated function from the empirical function found in Figure 6 of the paper “Broadband and Crack-Free Antireflection Coatings by Self Assembled Moth Eye Patterns” [B-10]. This polynomial fit generated a smooth optical transition from the air interface to the bulk PDMS substrate, which can be visualized in the approximated refractive index profile shown in Figure B-5.

$$\begin{aligned}
 n(h) &= A - Bh + Ch^2 - Dh^3 + Eh^4 - Fh^5 & (B.12) \\
 A &= 1.41225607 \\
 B &= 0.00122535861 \\
 C &= 0.0000549792342 \\
 D &= 0.000000982578598 \\
 E &= 0.00000000532593744 \\
 F &= 0.0000000000920959868
 \end{aligned}$$

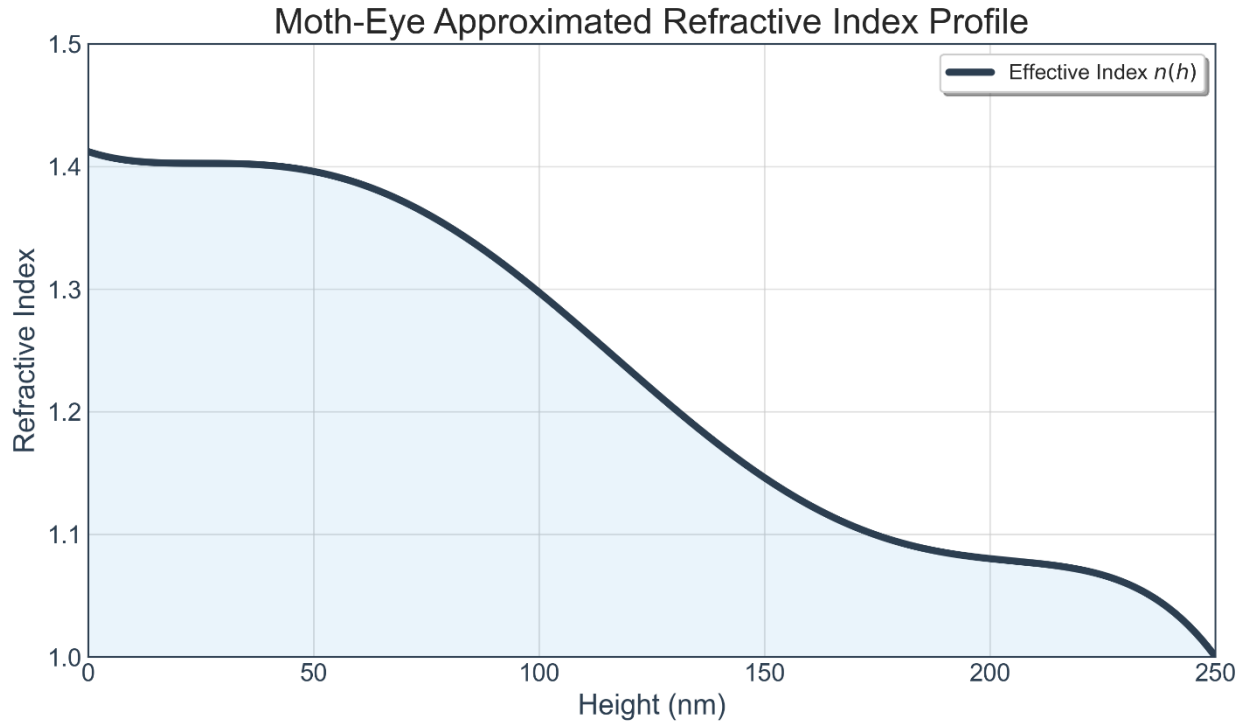


Figure B-5: Extrapolated Empirical Moth-Eye Refractive index profile from the Broadband and Crack-Free Antireflection Coatings by Self-Assembled Moth-Eye Patterns paper

Utilizing this polynomial approximation, the first four simulation categories were configured to validate the light capture and internal trapping capabilities of the device stack. The initial test category evaluated the macroscopic AR performance by sweeping the incident angle and tracking the transmitted power across varying waveguide layer resolutions, validating the relationship between structural discretization and transmittance as shown in Figure E-11. The subsequent three simulation categories characterized TIR efficiency by embedding simulated QDs within the substrate geometry. To roughly map the re-emission profiles, these tests utilized both isotropic point sources and directional cone lights. The simulation environments were alternated between unit cell boundary conditions and isolated large-scale device stacks. These allowed for the precise quantification of absorbed power as a function of the discretized GRIN layer count and the lateral density of the QDs. The configurations for these preliminary verification tests are detailed in Table B-3, visualized in Figure B-6 - Figure B-12, and their results can be found in Raytracing Verification Data Results.

Table B-3: Raytracing Simulation Benchmark parameter sweeps for FYDPSims module

Test Category	QDs Evaluated	Waveguide Layers	Incident Angle (°)	Rays per Source	GRIN Model	Moth-Eye representation
Max Capture Angle	N/A	2 to 250	0 to 90	10 000	Linear and Extrapolated	Waveguide
QD Internal Reflection	1 to 15	2 to 250	N/A	250	Linear and Extrapolated	Waveguide and Sinusoidal
Real Life test	0 to 400	2 to 250	0 & 26	250	Linear and Extrapolated	Waveguide

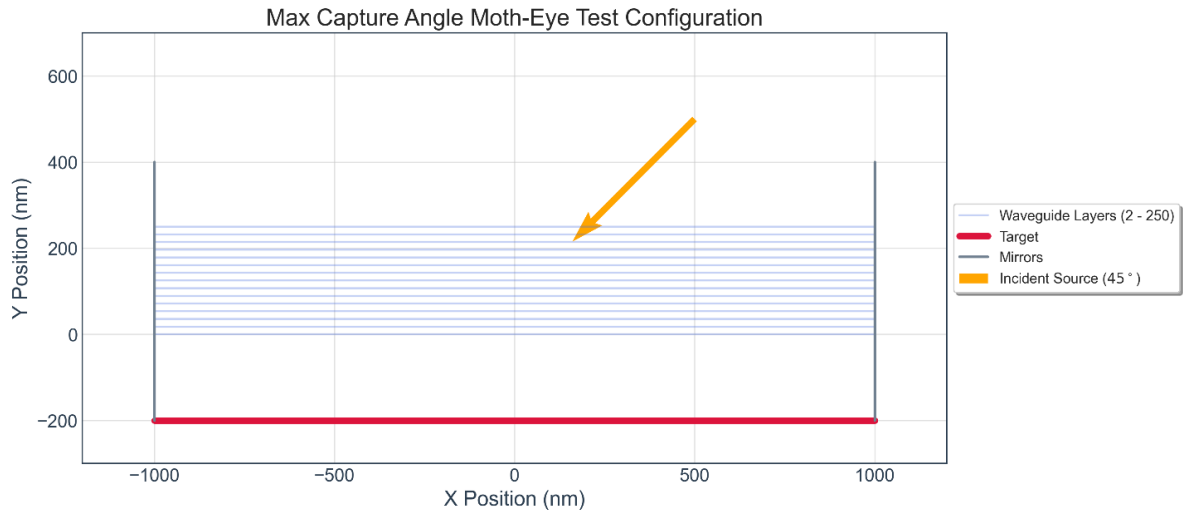


Figure B-6: Max Capture Angle Test Configuration

TIR Sinusoidal Moth-Eye Unit Cell Test Configuration

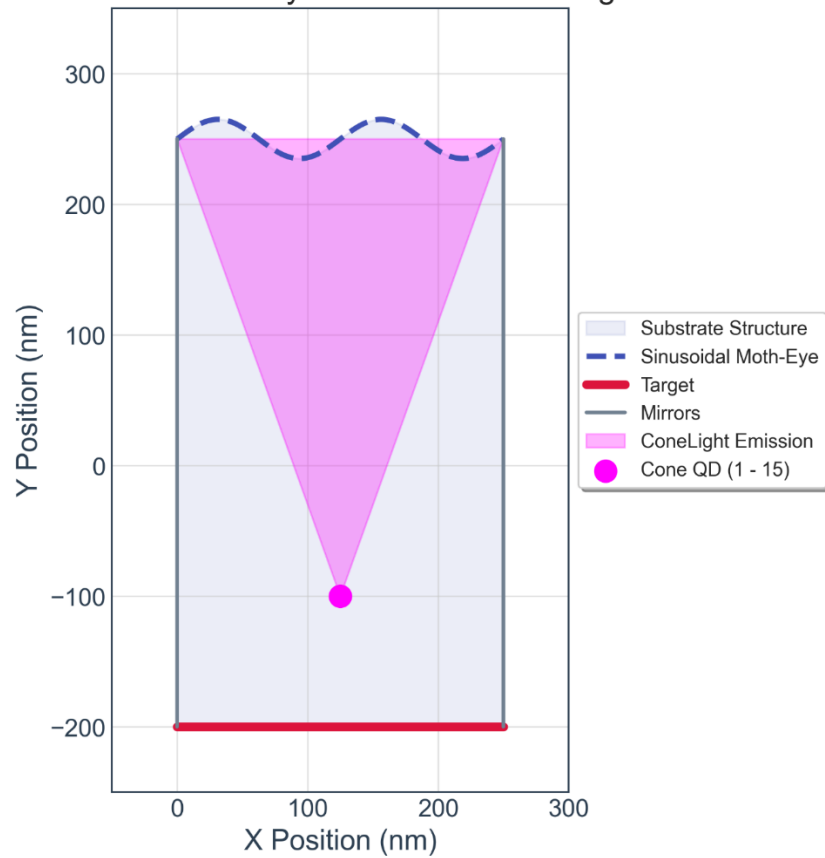


Figure B-7: TIR of Sinusoidal geometric approximation of Moth-Eye layer Test Configuration

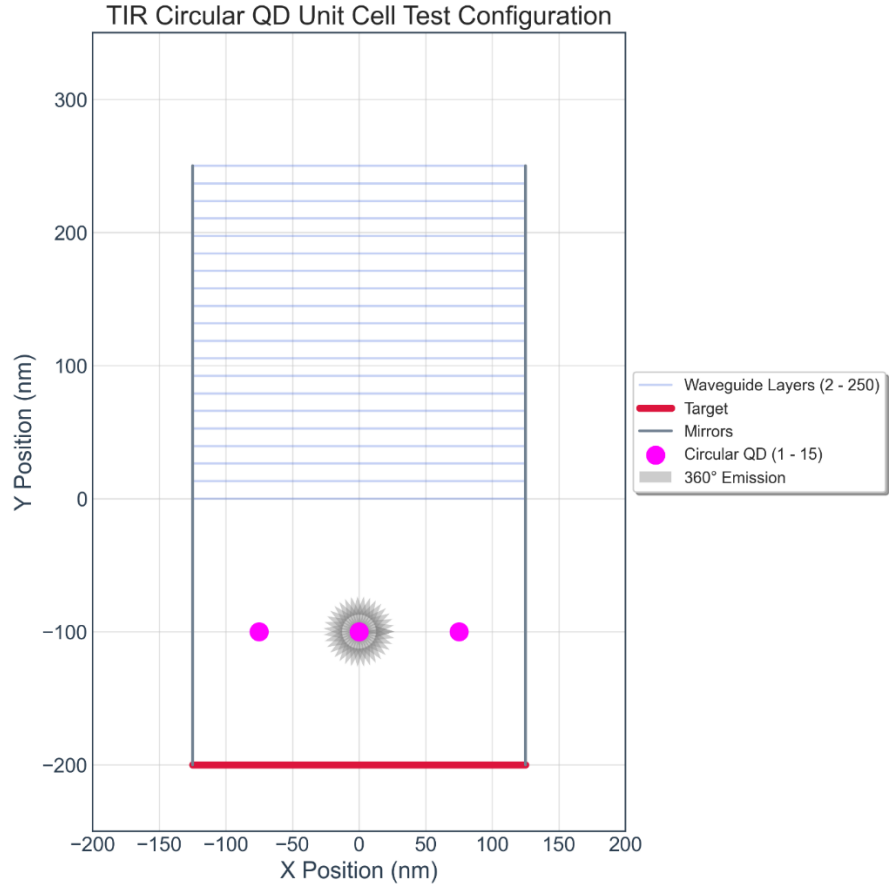


Figure B-8 : TIR of Moth-Eye Waveguide approximation Test Configuration using Circular QDs (Unit Cell)

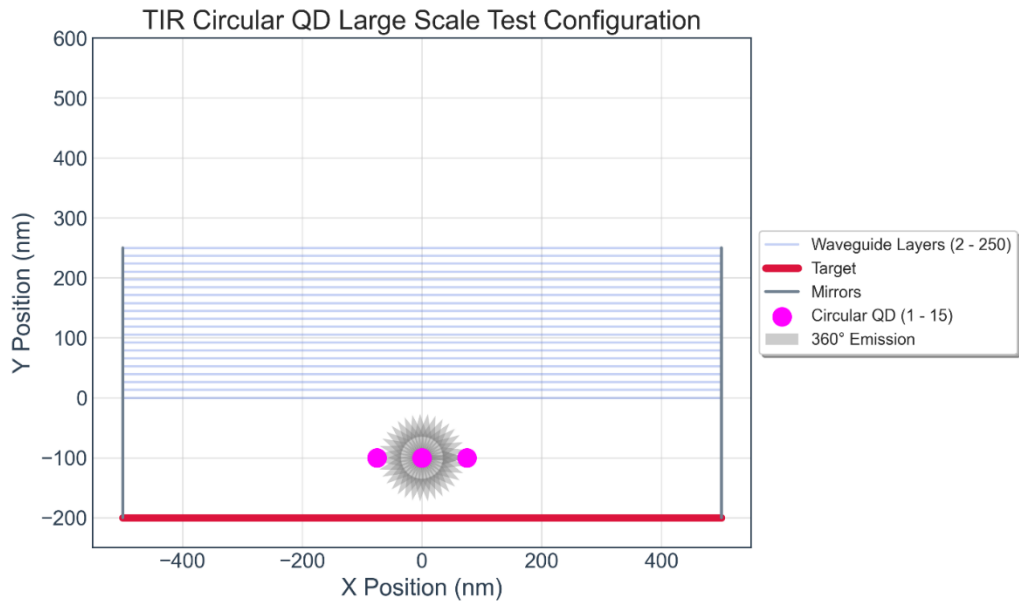


Figure B-9: TIR of Moth-Eye Waveguide approximation Test Configuration using Circular QDs (Large Scale)

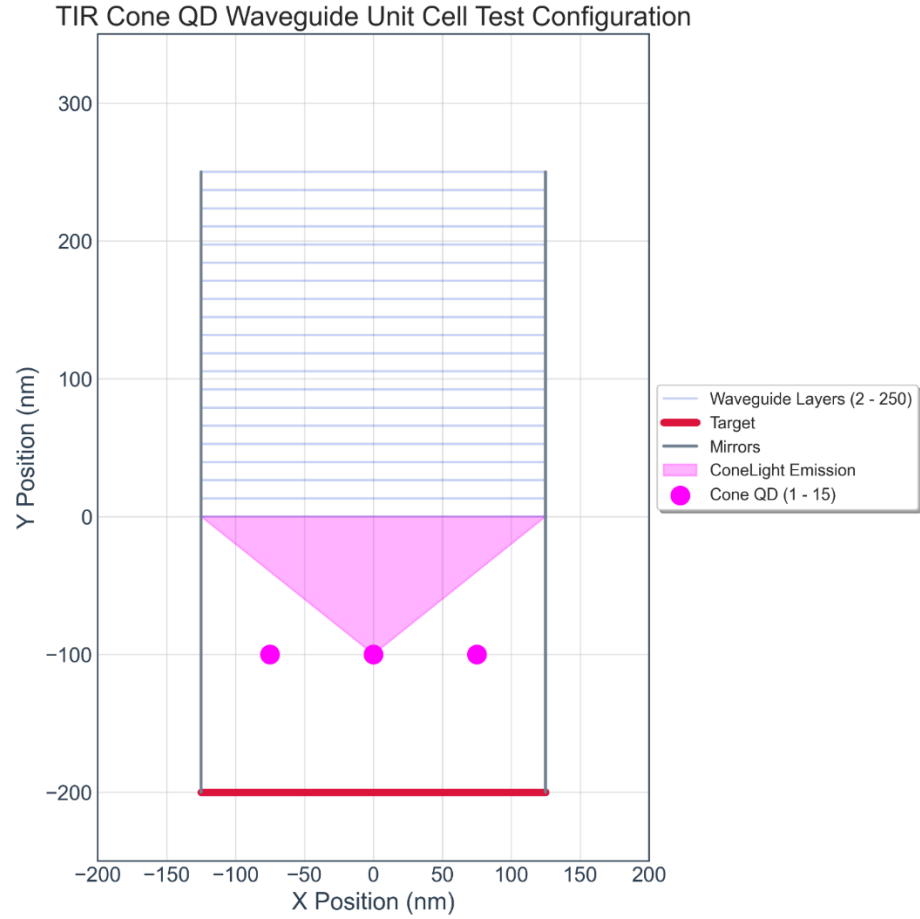


Figure B-10: TIR of Moth-Eye Waveguide approximation Test Configuration using Cone QDs (Unit Cell)

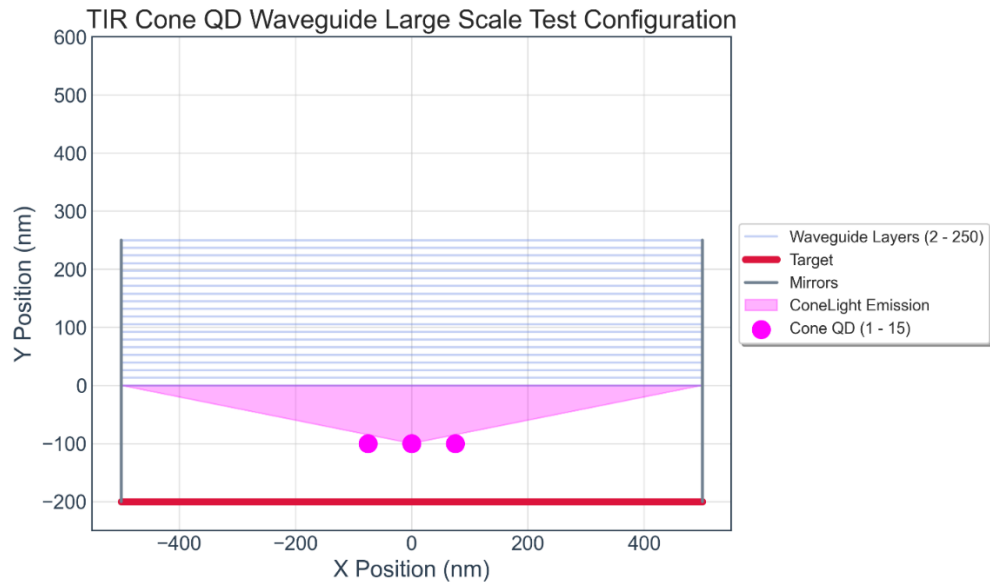


Figure B-11: TIR of Moth-Eye Waveguide approximation Test Configuration using Cone QDs (Large Scale)

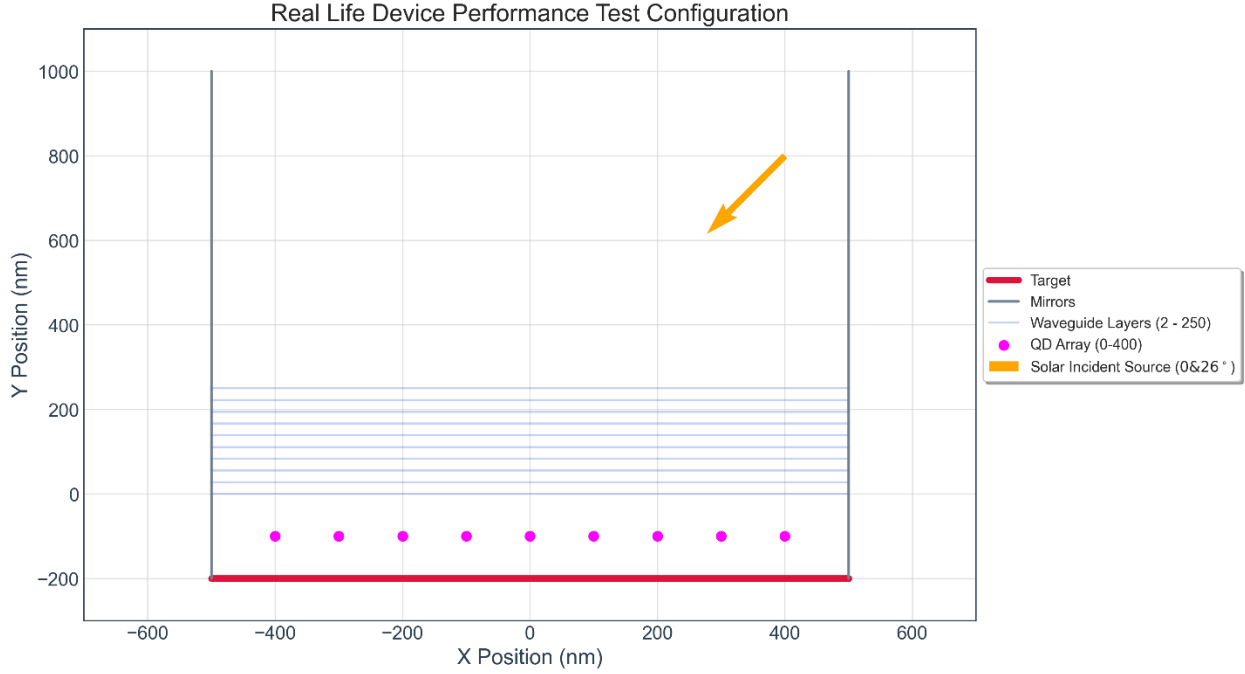


Figure B-12: Real Life Device Test Configuration with Circular Quantum Dots at Incident Angle of 0 and 26°

To generate the spectral and angular response curves required for the Solcore PCE modelling, the simulations transitioned to the SolcoreSims module. This phase necessitated replacing the static polynomial index with a dynamic, wavelength-dependent model capable of spanning the 300 – 1000 nm spectrum. The engine implemented the Sellmeier equation to calculate the bulk refractive indices of the PDMS and Air as shown in Equation B.13 and Equation B.14 [B-11], [B-12].

$$n_{PDMS}(\lambda_{nm}) = \sqrt{1 + \frac{B_1 \lambda^2}{\lambda^2 - C_1}} \quad (B.13)$$

$$B_1 = 1.0093$$

$$C_1 = 13185$$

$$n_{Air}(\lambda_{\mu m}) = 1 + \frac{B_1}{C_1 - \lambda^{-2}} + \frac{B_2}{C_2 - \lambda^{-2}} \quad (B.14)$$

$$B_1 = 0.05792105$$

$$B_2 = 0.00167917$$

$$\begin{aligned} C_1 &= 238.0185 \\ C_2 &= 57.362 \end{aligned}$$

The effective refractive index of the Moth-Eye layer at a specific height fraction h_{frac} was then derived geometrically using a fill factor f . This fill factor computes the ratio of a cross-sectional circles area at a given height relative to the area of its enclosing hexagonal unit cell as shown in Equation B.15. The final index was determined using effective medium blending function as shown in Equation B.16, where the exponent $q = \frac{2}{3}$ dictates the blending regime for the nanoscale structures.

$$f(h_{frac}) = \frac{\pi \sqrt{2rh_{frac} - h_{frac}^2}}{1.5\sqrt{3}s^2} \quad (B.15)$$

$r = 1$ (*Circle Radius*)

$s = \frac{2r}{\sqrt{3}}$ (*Hexagon Side Length*)

$$n_{eff}(\lambda, h_{frac}) = \left(f(h_{frac}) \cdot n_{PDMS}(\lambda)^q + (1 - f(h_{frac})) \cdot n_{Air}(\lambda)^q \right)^{\frac{1}{q}} \quad (B.16)$$

$q = \frac{2}{3}$

Driven by this dynamically calculated fill-factor profile, the final seven simulations categories were executed to provide a comprehensive spectral characterization of the device. Initial exploratory parameters were exceptionally broad to capture the full behavioral envelope of the simulated optical structures. These comprehensive sweeps evaluated incident angles from 0 to 60°, varied structural layer resolutions from 3 to 250 layers, and applied Gaussian normal perturbances ranging from 0 to 20 standard deviations to model manufacturing defects. The configurations for these parameter sweeps are detailed in Table B-4.

Table B-4: Exploratory Parameter sweep bounds for SolcoreSims Module

Parameter	Minimum Bound	Maximum Bound	Step Size
Incident Angle (°)	0	60	5
Waveguide Layers	3	250	1
Normal Perturbance (σ)	0	20	2
Wavelength (nm)	300	1000	1

Based on the convergence stability observed during these wide-ranging tests, the eventual settings were narrowed and refined to narrow and optimize computational resources while maintaining high fidelity. It was determined that the external macroscopic Moth-Eye approximations converged optimally at 3 discrete layers, while TIR calculations required a finer resolution of 40 layers to accurately resolve internal scattering. A nominal normal perturbation of 6 standard deviations was established as the baseline for modelling surface defects. The final configurations utilized to generate the reported spectral datasets present in

Raytracing Verification Data Results are detailed in Table B-5.

Table B-5: Final Parameters for SolcoreSims Module Test Configurations

Index	Evaluation Type	Waveguide Morphology	Layers	Perturbance	Rays/Wavelength	Averaging Runs
1	Transmittance	Flat	3	0	10 000	1
2	Transmittance	Flat	3	6	100 000	5
3	Transmittance	Sinusoidal	3	6	50 000	5
4	TIR	Flat	40	0	1 000	1
5	TIR	Flat	40	6	2 000	5
6	TIR	Sinusoidal	40	6	2 000	5
7	Transmittance (Control)	Flat	1	0	10 000	1

The first three Solcore simulation categories were used to characterize the transmittance of light through the AR layer. To account for structural realities, subsequent variations introduced the perturbances by applying Gaussian-sampled angular shifts to the segment's normal vectors, and

by modelling the layer interfaces as continuous sinusoidal waves rather than flat boundaries. The following three categories inverted the source parameters to perform internal wavelength sweeps using a cone source, specifically measuring the TIR trapping efficiency under the same ideal, perturbed and wavy structural conditions. Finally, a seventh control simulation evaluated a uniform, single-layer PDMS boundary devoid of AR structural features, which was used as control compared to the first 3 simulations. The final configurations for these simulations are visualized in Figure B-13 – Figure B-19.

Due to the massive computational volume required to process millions of intersecting ray paths across the extensive 3 - 250 layer sweeps, the compiled C++ executable was transferred to a remote high-performance server. The execution of these extensive parameter sweeps, and final configurations were managed by a Bash script, which launched the various simulations as parallel background processes. This parallel execution maximized throughput across all available server cores, significantly reducing the total rendering time and efficiently generating the standardized JSON output files required for subsequent Solcore aggregation and analysis.

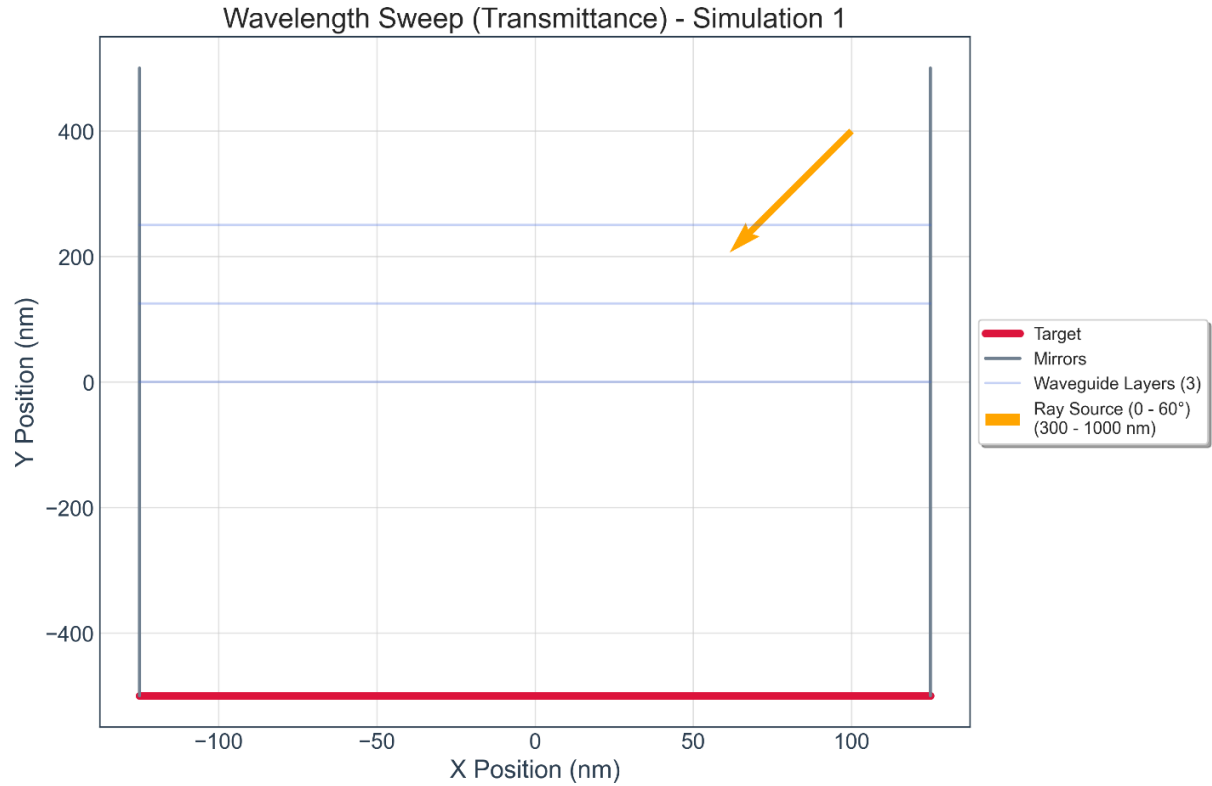


Figure B-13: Solcore Raytracing Simulation 1 Configuration – Transmittance Wavelength Sweep



Figure B-14: Solcore Raytracing Simulation 2 Configuration – Transmittance Wavelength Sweep with Perturbances

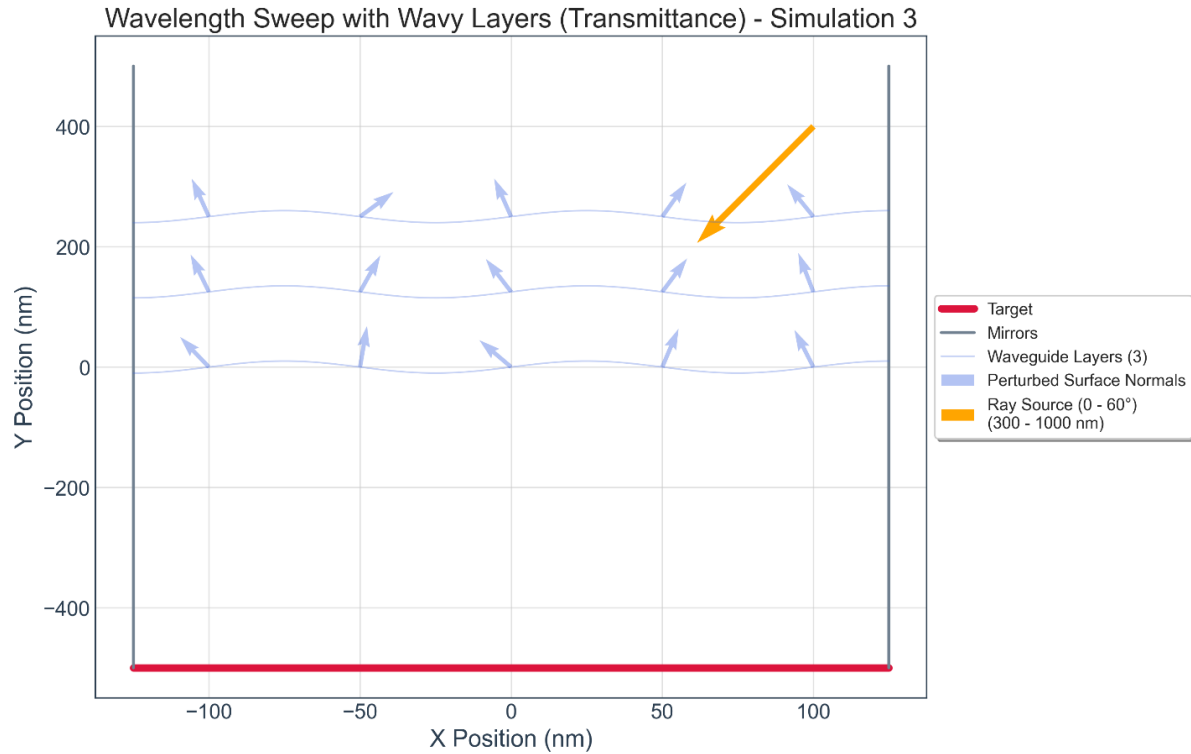


Figure B-15: Solcore Raytracing Simulation 3 Configuration – Transmittance Wavelength Sweep with Wavy layers and Perturbances

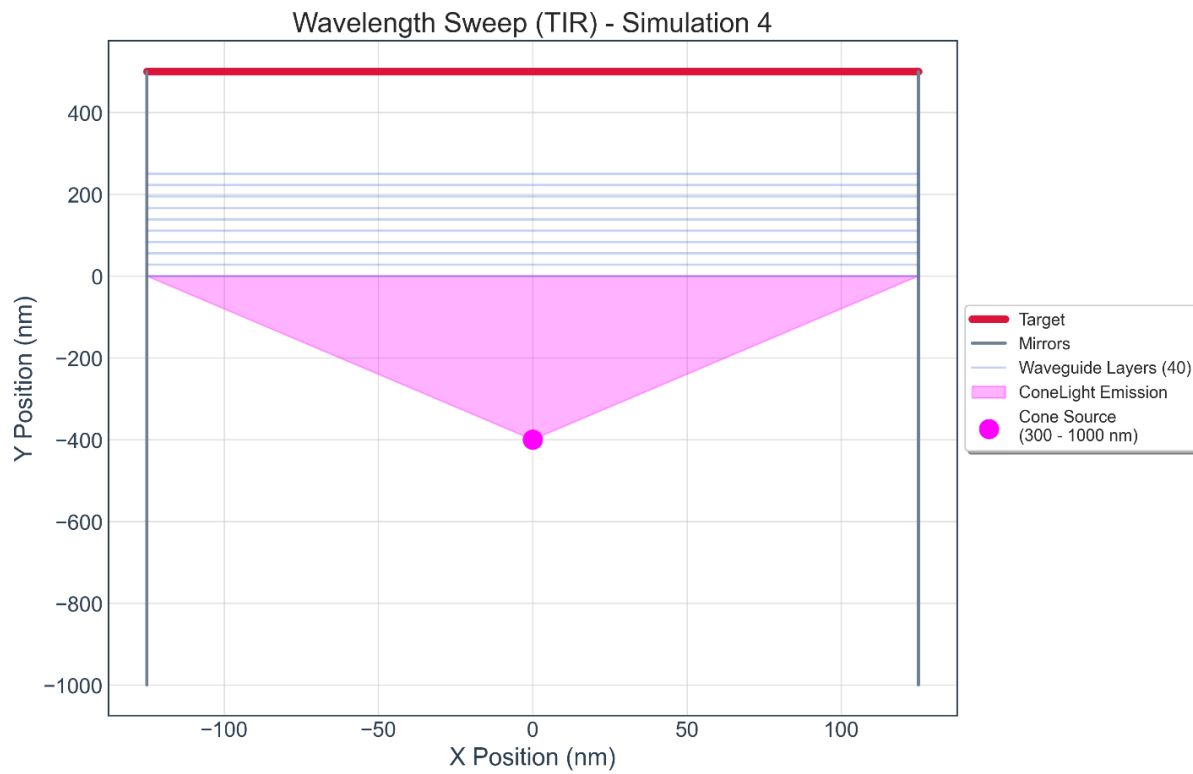


Figure B-16: Solcore Raytracing Simulation 4 Configuration – TIR Wavelength Sweep

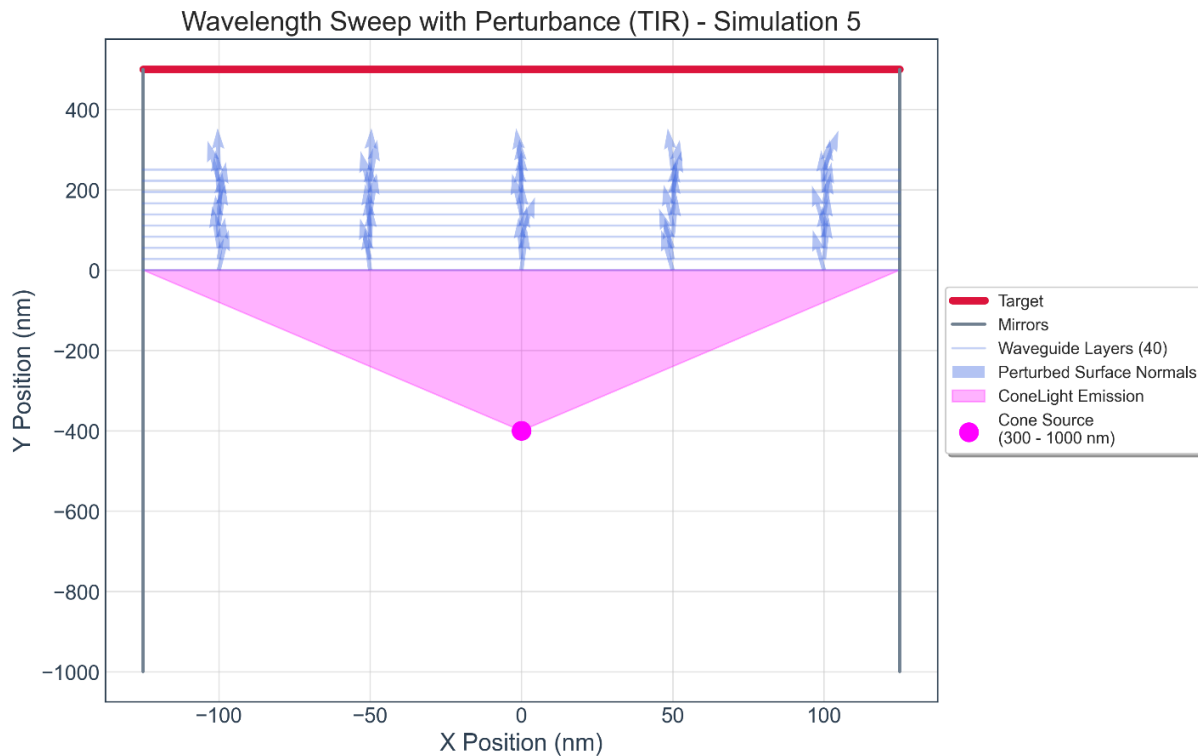


Figure B-17: Solcore Raytracing Simulation 5 Configuration – TIR Wavelength Sweep with Perturbances

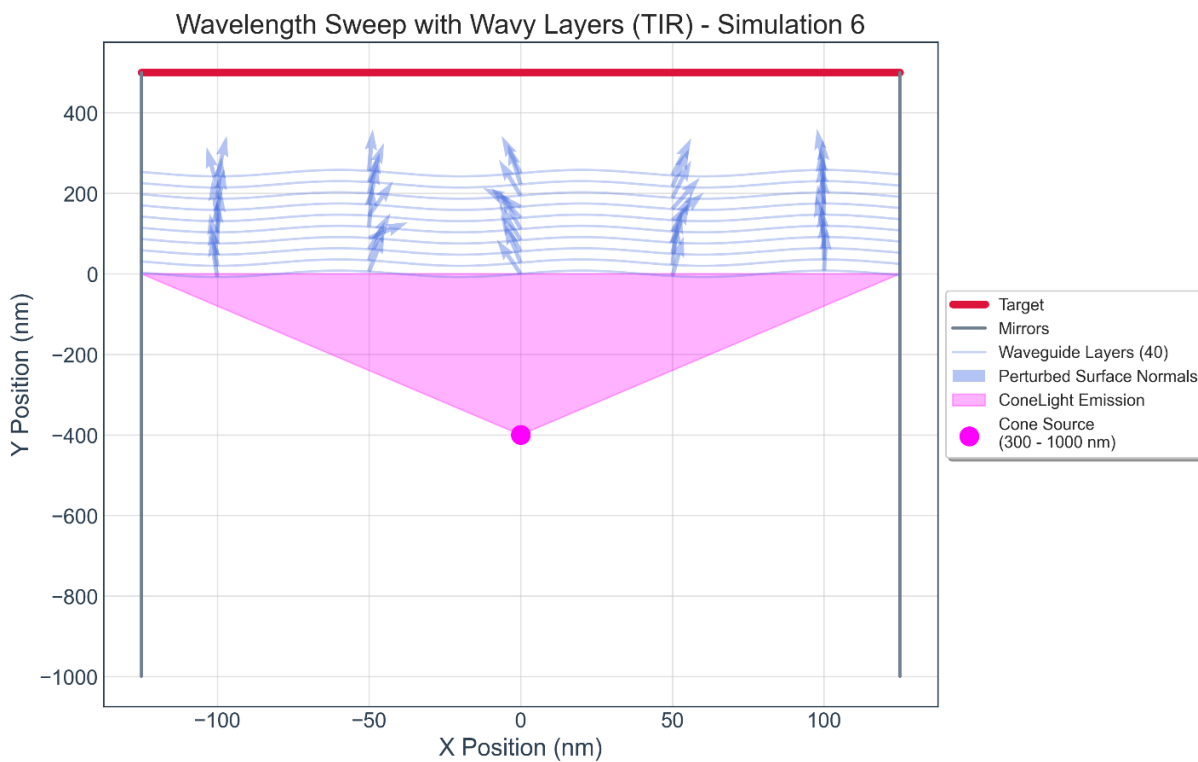


Figure B-18: Solcore Raytracing Simulation 6 Configuration – TIR Wavelength Sweep with Wavy layers and Perturbances



Figure B-19: Solcore Raytracing Simulation 7 Configuration – Control Wavelength Sweep

B.7 Appendix B References

- [B-0] C. Larsen *et al.*, “A tool for identifying green solvents for printed electronics,” *Nat. Commun.*, vol. 12, no. 1, p. 4510, Jul. 2021, doi: 10.1038/s41467-021-24761-x.
- [B-1] “jonschlipf/RigorousCoupledWaveAnalysis.jl: Rigorous Coupled-Wave Analysis (RCWA) for nanophotonics simulations.” Accessed: Apr. 03, 2026. [Online]. Available: <https://github.com/jonschlipf/RigorousCoupledWaveAnalysis.jl>
- [B-2] “Rigorous coupled-wave analysis,” *Wikipedia*. Nov. 30, 2025. Accessed: Apr. 03, 2026. [Online]. Available: https://en.wikipedia.org/w/index.php?title=Rigorous_coupled-wave_analysis&oldid=1324976836
- [B-3] “RCWA Solver Introduction,” Ansys Optics. Accessed: Apr. 03, 2026. [Online]. Available: <https://optics.ansys.com/hc/en-us/articles/4414575008787-RCWA-Solver-Introduction>
- [B-4] “Ray Tracing,” NVIDIA Developer. Accessed: Apr. 03, 2026. [Online]. Available: <https://developer.nvidia.com/discover/ray-tracing>
- [B-5] “Ray tracing (graphics),” *Wikipedia*. Mar. 26, 2026. Accessed: Apr. 03, 2026. [Online]. Available: [https://en.wikipedia.org/w/index.php?title=Ray_tracing_\(graphics\)&oldid=1345468903](https://en.wikipedia.org/w/index.php?title=Ray_tracing_(graphics)&oldid=1345468903)
- [B-6] “Just-in-Time compilation — Numba 0.52.0.dev0+274.g626b40e-py3.7-linux-x86_64.egg documentation.” Accessed: Apr. 05, 2026. [Online]. Available: <https://numba.pydata.org/numba-doc/dev/reference/jit-compilation.html>
- [B-7] “Bounding volume hierarchy,” *Wikipedia*. Mar. 01, 2026. Accessed: Apr. 05, 2026. [Online]. Available: https://en.wikipedia.org/w/index.php?title=Bounding_volume_hierarchy&oldid=1341090674
- [B-8] Sebastian Lague, *Coding Adventure: Ray Tracing*, (Apr. 01, 2023). Accessed: Apr. 05, 2026. [Online Video]. Available: <https://www.youtube.com/watch?v=Qz0KTGYJtUk>
- [B-9] “Coding Adventure: More Ray Tracing! - YouTube.” Accessed: Apr. 03, 2026. [Online]. Available: <https://www.youtube.com/watch?v=C1H4zLiCOaI&t=321s>

Appendix C Test Plan for Constructed Design Prototype

Due to time, equipment availability, and fabrication constraints, not all tests outlined in the original test plan were performed. The experimental work presented in Appendix F primarily focused on system-level optical and electrical characterization, specifically I–V measurements under UV and white light illumination and reflectance testing of both standalone films and film-integrated solar panels. These tests were prioritized because they directly evaluated the core design objectives of improving power output and reducing reflectance. In contrast, several material-level and long-term characterization methods, including PLQY measurements using an integrating sphere, detailed rheological analysis, DLS particle sizing, accelerated weathering tests, and calibrated AM1.5G PCE measurements, were not performed. These tests required specialized instrumentation or extended testing durations that were not feasible within the project timeline. As a result, while the conducted tests were sufficient to validate the primary functional concepts of the prototype, the remaining planned tests represent areas for future work to fully quantify material performance and long-term reliability.

Table C-1: Summary of Primary and Secondary Requirements

Requirement	Primary/Secondary	Functional/ Non-Functional	Minimum Performance Specification
PCE Increase by +1%	Primary	Functional	PCE +1%
Transmit 99% of light	Primary	Functional	99% Transmittance (400 nm -1100 nm)
10-year lifespan	Primary	Non-Functional	10 years
Lower reflectance by 3%	Primary	Functional	-3% Reflectance
Easy to apply	Secondary	Non-Functional	Flexible
Hydrophobic surface	Secondary	Functional	$\geq 100^\circ$ water-film contact angle
Cost efficient/ scalable process	Secondary	Non-Functional	Material cost < energy savings

Winterproof	Secondary	Functional	$\leq 10\%$ mismatch of thermal expansion coefficients
-------------	-----------	------------	--

C.1 Photoluminescent Quantum Yield (*links to 'Increased PCE')

PLQY was the primary metric for determining the quality of the QDs. PLQY measures the ratio of absorbed to emitted photons and provided an indication of how efficient the QDs were.

Ideally, the PLQY of the QDs would be close to 1; however, this was not likely, so a minimum PLQY of 0.5 was determined to be sufficient. There are two methods for measuring PLQY: relative to a standard compound or absolute. The second method was followed, which required the use of an integrating sphere. Access to an integrating sphere was provided by the project consultant, Dr. Sivoththaman.

C.2 QD Absorbance and Emission (*links to 'Invisibility')

QD absorbance and emission spectra were key to ensuring that the film remained invisible to wavelengths already absorbed efficiently by the solar panel. Absorbance was required to be near zero at wavelengths greater than 250 nm. Additionally, the primary emission peak was required to occur in the red range of visible light (~700 nm). These two conditions ensured that the QDs did not absorb light that the solar panel was already able to use and that emission occurred at the optimal wavelength for absorption into the panel. These properties were tested via UV-Vis spectroscopy.

C.3 QD Particle Size and Distribution (*links to 'Increased PCE')

QD particle size and distribution determined the emission wavelength and the width of the emission spectrum. Mean particle size and polydispersity were measured via DLS (dynamic light scattering) and were verified through the emission spectrum obtained from UV-Vis spectroscopy.

C.4 Film Transparency (*links to ‘Invisibility’)

For the film to be manufacturable at scale, several processing parameters required optimization, particularly the rheology of the ink. The viscosity of the ink influenced how the material flowed and leveled during coating, affecting print quality and uniformity. For large-scale processes such as roll-to-roll printing with slot-die coating, the ink required a balance between low viscosity under shear to enable smooth coating and higher viscosity at rest to maintain film consistency. Rheological properties were characterized using a rheometer to obtain viscosity–shear rate profiles. The ink was also evaluated at different levels of crosslinking to assess how printability changed. This allowed determination of the appropriate level of crosslinking or thermal treatment required for scalable processing.

C.5 Print Feasibility (*links to ‘Cost Effectiveness’)

For the film to be manufacturable at scale, several processing parameters required optimization, particularly the rheology of the ink. The viscosity of the ink influenced how the material flowed and leveled during coating, affecting print quality and uniformity. For large-scale processes such as roll-to-roll printing with slot-die coating, the ink required a balance between low viscosity under shear to enable smooth coating and higher viscosity at rest to maintain film consistency. Rheological properties were characterized using a rheometer to obtain viscosity–shear rate profiles. The ink was also evaluated at different levels of crosslinking to assess how printability changed. This allowed determination of the appropriate level of crosslinking or thermal treatment required for scalable processing.

C.6 Anti-Reflectivity (*links to ‘Anti-reflectivity’)

Reflectivity of the moth-eye anti-reflective (AR) polymer layer was measured using a spectrophotometer equipped with an integrating sphere to capture both specular and diffuse

reflected light across the working spectrum (250–1200 nm). This setup ensured accurate quantification of total reflectance regardless of scattering from the surface texture. Measurements were performed on the coated substrate and compared against an uncoated control to isolate the AR effect. The performance target was <3% total reflectance across the relevant spectral range, demonstrating that the moth-eye pattern effectively suppressed surface reflections and enhanced light transmission into the underlying device.

C.7 Device Flexibility (*links to ‘Ease of Application’)

The fully assembled device was required to be wrapped around a 2-inch diameter tube for ease of storage. Each layer was tested separately, as fabrication was not performed simultaneously, and all components were expected to conform to a 2-inch diameter without exterior tearing or irreversible internal wrinkling. These attributes were verified at multiple locations using optical microscopy for macroscopic damage and AFM to confirm preservation of the moth-eye structure.

C.8 Final Stage PCE Characterisation (*links to ‘Increased PCE’)

Power conversion efficiency (PCE) of the full device was tested using a calibrated solar simulator coupled with a source-measure unit to record current–voltage (I–V) characteristics under standard test conditions:

- Illumination: AM1.5G solar spectrum (approximates sunlight at the Earth’s surface at 1.5 air mass, or the sun at $\sim 48.2^\circ$ above the horizon).
- Light intensity: 100 mW/cm² (equivalent to 1 sun).
- Cell temperature: 25 °C (298 K).

Solar cells with and without the full device layers were measured under identical conditions to directly assess performance impact. The target was to demonstrate a measurable PCE improvement of at least 0.5% in coated devices, confirming that the system enhanced light harvesting and overall efficiency.

C.9 Long Term Weathering Resistivity. (*links to ‘Lifespan’)

Weathering was characterized through three primary attributes: yellowing, surface tearing, and transmittance.

Yellowing was quantified using colorimetry before and after accelerated exposure (ISO 4892-2 / ASTM G154 or xenon-arc per ISO 4892-2) using a spectrophotometer to report CIE Lab* values and compute ΔE_{00} (CIEDE2000) and the Yellowness Index (ASTM E313). A 1000-hour accelerated weathering cycle was conducted with measurements at 0, 250, 500, 750, and 1000 hours. Acceptance criteria were $\Delta E_{00} \leq 2$ and $\Delta YI \leq 3$ after 1000 hours.

Surface tearing was evaluated using optical microscopy and SEM to detect microcracks or crazing. Baseline and post-1000-hour measurements were compared, with the requirement that no visible macroscopic cracks or surface rips exceeding 1 mm occurred in active regions

Transmittance was measured using integrating-sphere UV-Vis spectroscopy (300–1200 nm) before and after weathering. Acceptance criteria included $\leq 0.5\%$ absolute loss in total transmittance across 400–1100 nm and $\leq 1\%$ local spectral variation. If thresholds were exceeded, additional I–V characterization was conducted to assess PCE degradation.

Each run with coated devices and uncoated controls should be executed in parallel, test multiple replicates ($n \geq 3$), and pair each optical/mechanical measurement with before/after electrical

testing (I–V) so that any observed optical or mechanical change can be directly correlated to actual PCE impact.

C.10 AFM (*links to ‘Anti-reflectivity’)

To verify if the Breath Figure method was successful in generating consistent pore sizes for the antireflective layer, it will require a physical measurement approach to confirm the sizes due to the nature of the PDMS used. In order to succeed, we need the moth-eye pattern to have pore sizes of around 250 nm; to get these pores, we will use multiple recipes with varying air flow, substrate temperatures and polymers. Each of these variables will affect the consistency and sizes of the pores created. To measure and evaluate the effectiveness of each combination we will scan the patterns surface using an AFM. This will yield a high resolution and high surface area scan of the pattern. Using these scans, AFM image analysis software such as Gwyddian can be used to measure pore diameters, the standard deviation of their diameters and the spacing between them. The most effective Moth Eye pattern should yield pores as close to our target wavelength, that being 250 nm, the lowest Standard Deviation from the ideal diameter, and a consistent spacing between pores of around 250 nm.

Appendix D Prototype Construction Blueprint

As a reminder for the overall process that will be covered, the final prototype consisted of a multilayer film designed for integration onto a solar panel surface. The device stack included, in order from bottom to top: a pre-existing solar cell base, an adhesive-backed PET substrate, a QD layer, and a PDMS encapsulation layer with an imprinted AR surface. This was all fabricated with the intent of roll-to-roll scalability, enabling potential large-area fabrication and application.

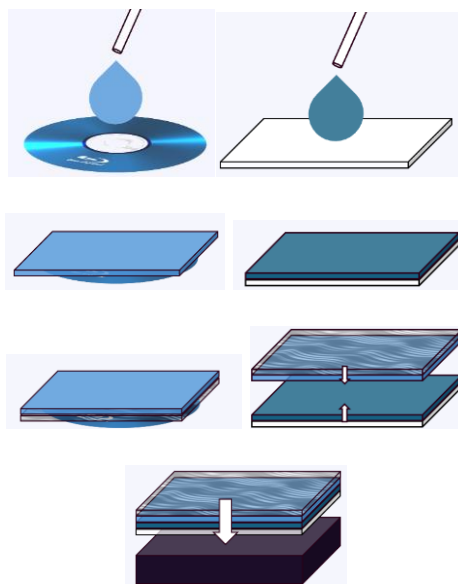
The very first step towards our prototype construction was the PDMS layer. All mixing was done by wt% with respect to the PDMS base, as in if there was 1 g of PDMS base, there would be both 0.1 g of Sylgard 184 and Toluene added to it. First, roughly 2 g of PDMS base was weighed into a tared vial, with the two compounds were dropped in with a pipette until a 0.2 g increase was observed for each. Once the desired weight was achieved, the vial was capped and moved to a vortex mixer, stirred for approximately 10 minutes. Once homogeneity was achieved, the vial was set aside temporarily.

Next, a Teenage Mutant Ninja Turtles Blue-ray disc was first cut in to 1" squares, notably being cut very slowly with very sharp scissors, where sharpening took place between every sample but cutting a large section of aluminum foil repeatedly. To achieve these 1" squares, the disk was cut straight to the middle, then cut into quarters. On one of the straight sides per quarter, it was important to keep moving along the edge from out to in until 1" in height can be achieved, where that position was marked at the base, then measured 1" down from that point to complete the square shape outline. One side of the square would be cut very slowly and controlled with the scissors along the marking's projection, all the way to the end of the disk, repeating the process with the other similar marking, giving one large "rectangle." Lastly, the top part would be cut to form the 1" height. After this and ensuring no cracking was formed, a razor blade was used to pry the

protective plastic lining of the disc, enough to where it could be held and peeled off. This was repeated 4 times to assess and select the best sample. Once the Blue-ray disc was ready, the PDMS vial was then drop-cast onto it, eyeballed to height, allowing it to cure on a hot plate at 120 °C for 10-15 minutes to form the AR surface.

Lastly was the PET layer with the QDs. Due to a divergence in the original plan with the PDMS layer being embedded with the QDs, they were instead drop-casted onto a plain PET film with an adhesive backing (and backing cover). To achieve optimal surface energy matching to avoid aggregation of QDs when drying, the PET underwent Corona treatment. The process introduces polar functional groups onto the otherwise hydrophobic PET surface, allowing better interaction with the toluene-based QD solution. This reduces surface energy mismatch, promotes more uniform spreading during deposition, and improves adhesion of the QD layer [1]. This process was rather unsuccessful and will be mentioned more in the deviations/shortfalls section.

When the two separate layered components were ready to be assembled as the final product, the two stacks were sandwiched together (AR layer facing up, adhesive facing down), placed onto a 1” solar panel, then tape down on each edge, further securing it in place for testing purposes.



*Figure D-1: Entire Prototype Fabrication Process Demonstration;
Left side resembles drop-cast of PDMS mixture onto Blue-Ray disk, allowed to cure on a hot plate. On the right is
the drop-cast of QDs onto PET then dried, where the 2 separate stacks on each side are sandwiched together,
stacked on top of a solar panel*

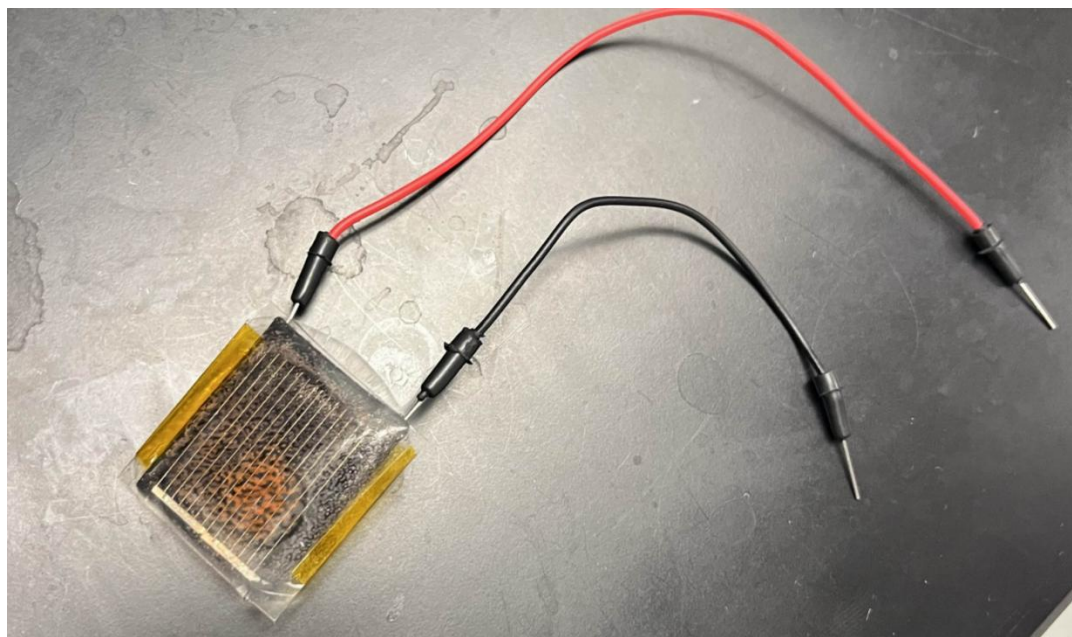


Figure D-2: Final Prototype Assembled Onto a 1" Solar Panel

D.1 Appendix D References

- [1] L. L. Wright, “AFM Studies of Corona Treated Uniaxially Drawn PET Films,” Society of Vacuum Coaters, 2006. [Online]. Available: https://www.svc.org/clientuploads/directory/resource_library/06_673.pdf (accessed Apr. 4, 2026).

Appendix E Verification Data

E.1 RCWA Verification Data Results

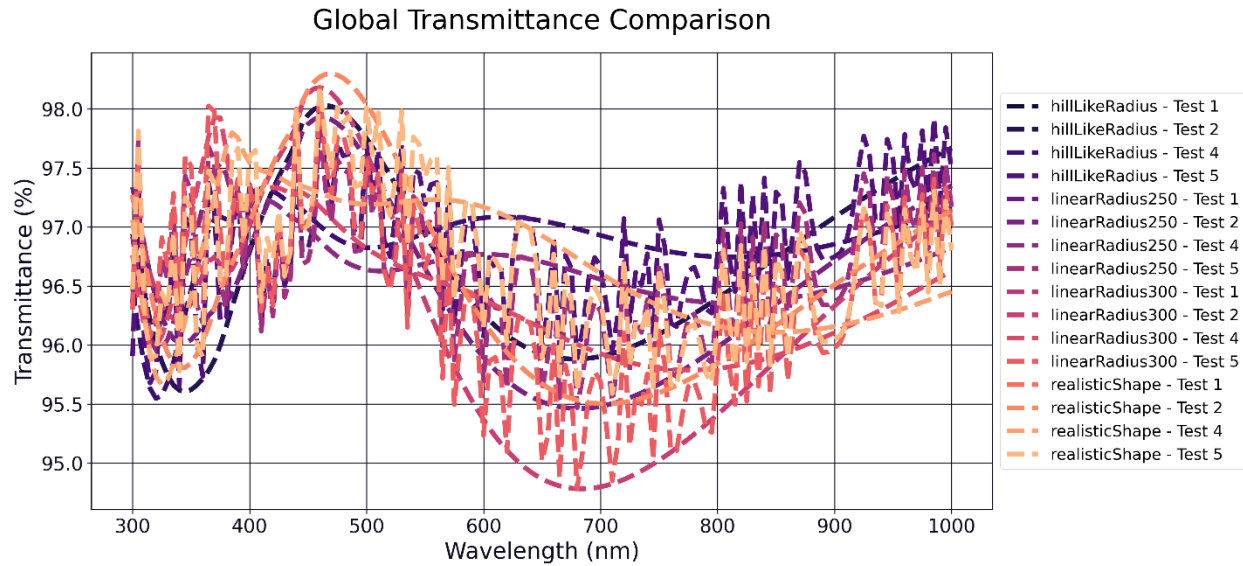


Figure E-1: Comparison of the Transmittance values across a spectral sweep from 300-1000 nm for all RCWA Simulations

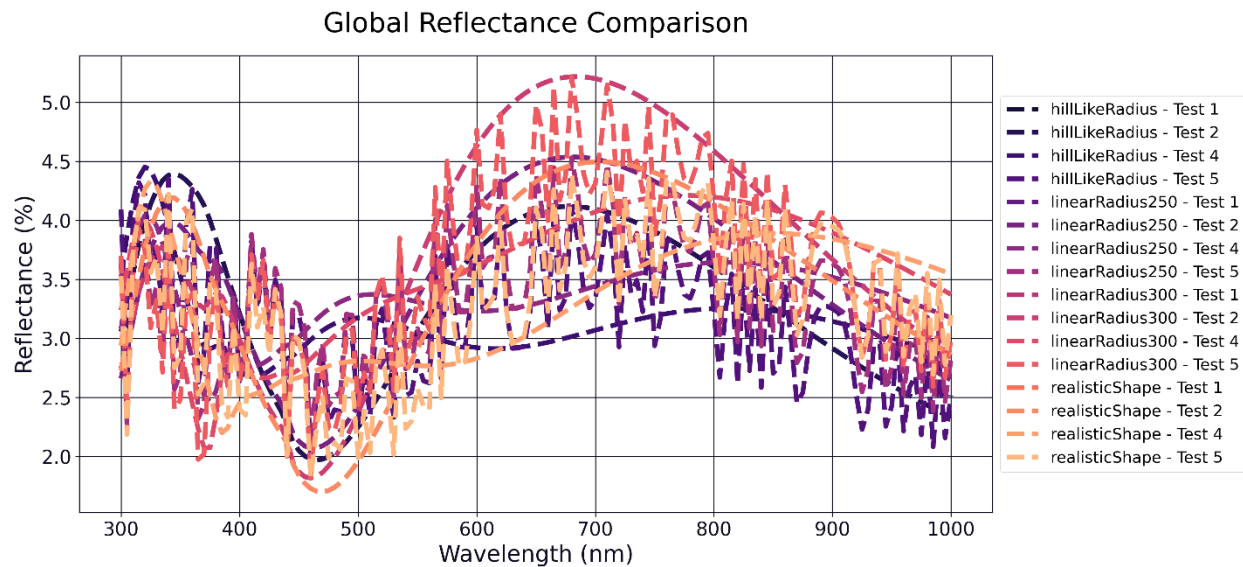


Figure E-2: Comparison of the Reflectance values across a spectral sweep from 300-1000 nm for all RCWA Simulations

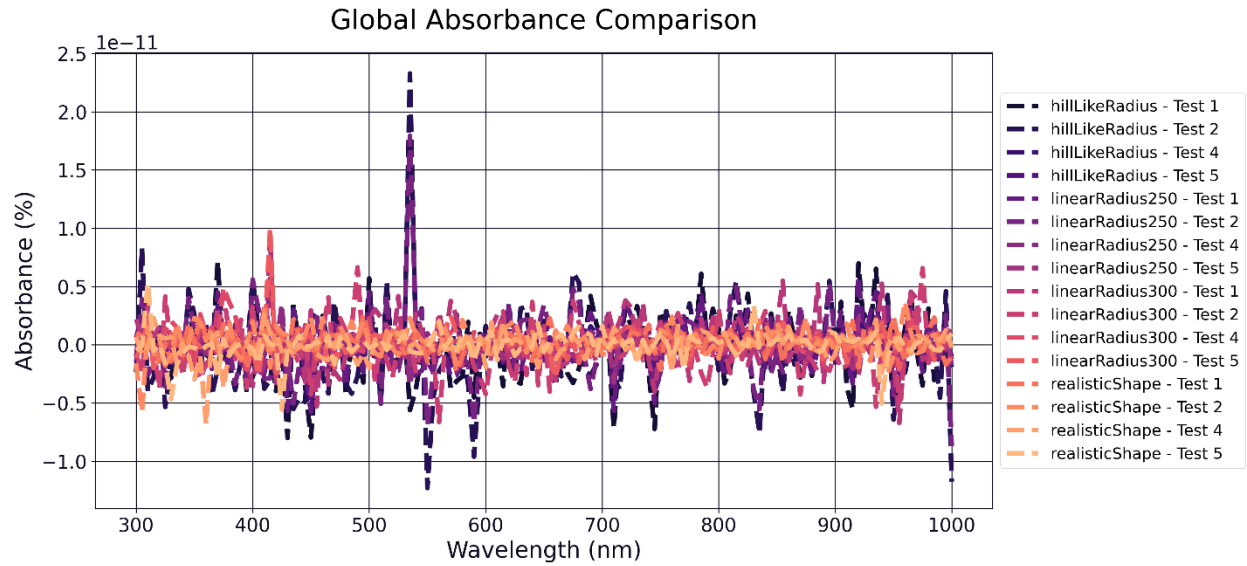


Figure E-3: Comparison of the Absorbance values across a spectral sweep from 300-1000 nm for all RCWA Simulations

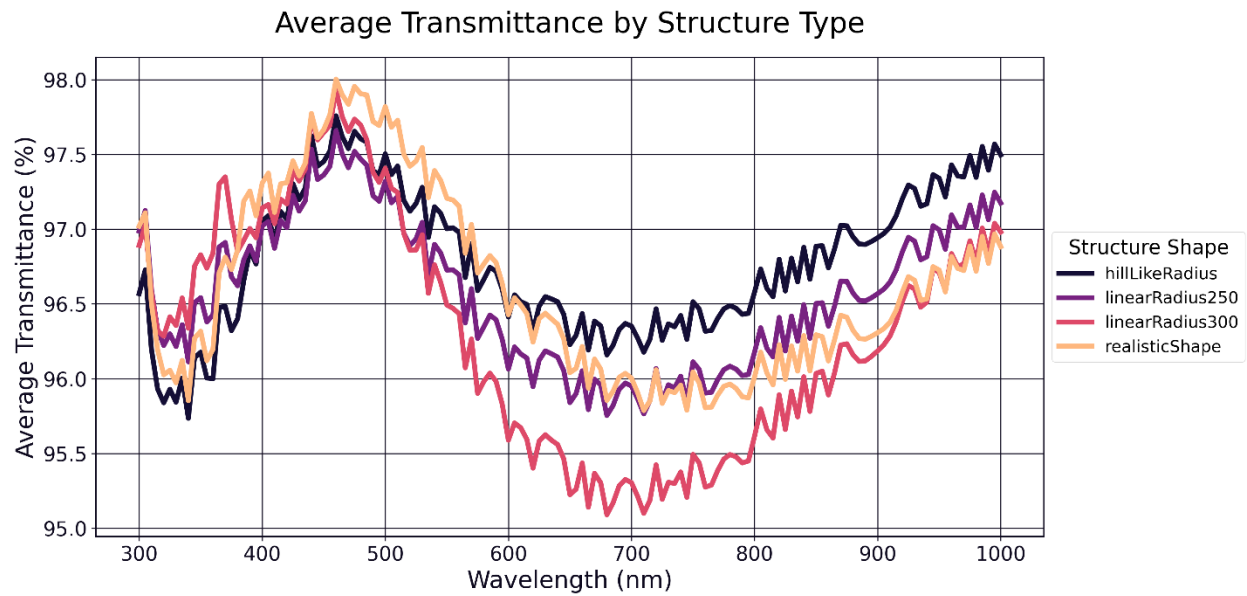


Figure E-4: Average Transmittance values across all RCWA Moth-Eye Cone structures

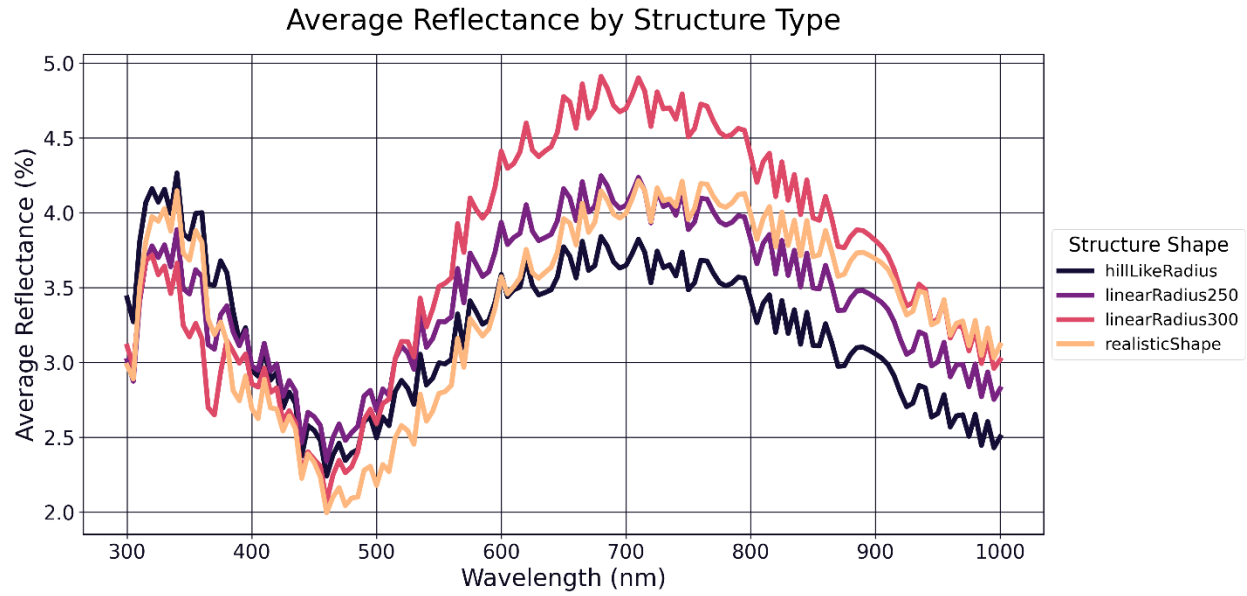


Figure E-5: Average Reflectance values across all RCWA Moth-Eye Cone structures

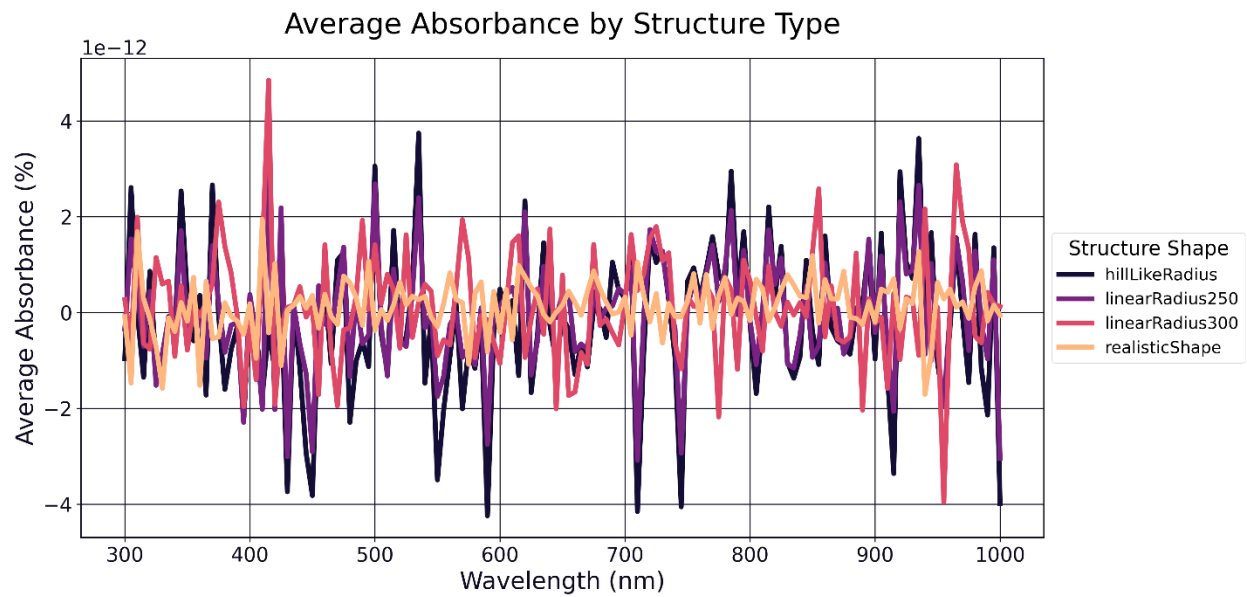


Figure E-6: Average Absorbance values across all RCWA Moth-Eye Cone structures

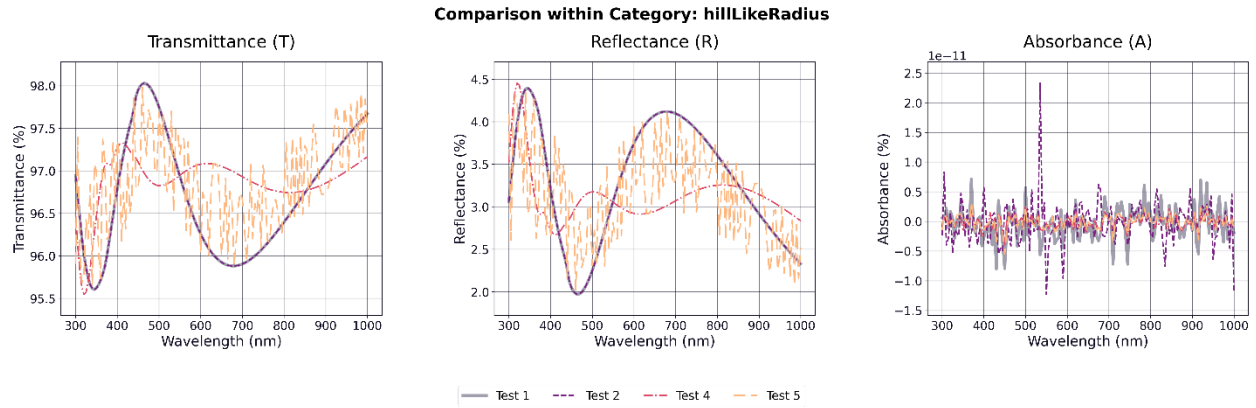


Figure E-7: TRA values across the 4 Completed RCWA tests for the hillLike Moth-Eye cone structure

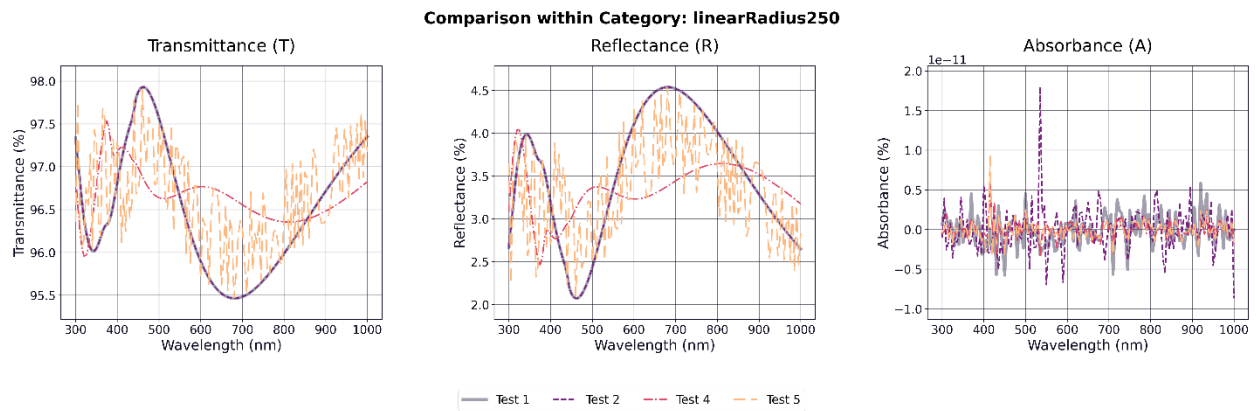


Figure E-8: TRA values across the 4 Completed RCWA tests for the linear 250 Moth-Eye cone structure

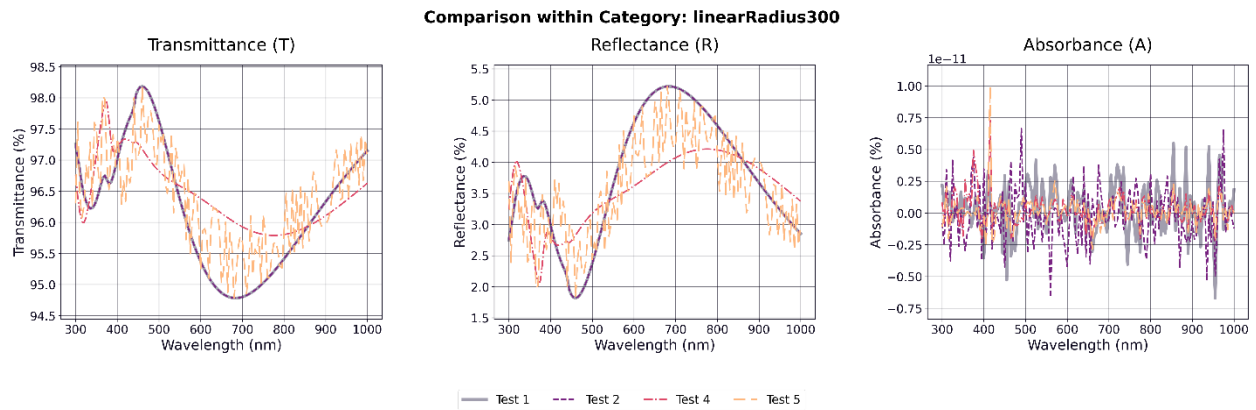


Figure E-9: TRA values across the 4 Completed RCWA tests for the linear 300 Moth-Eye cone structure

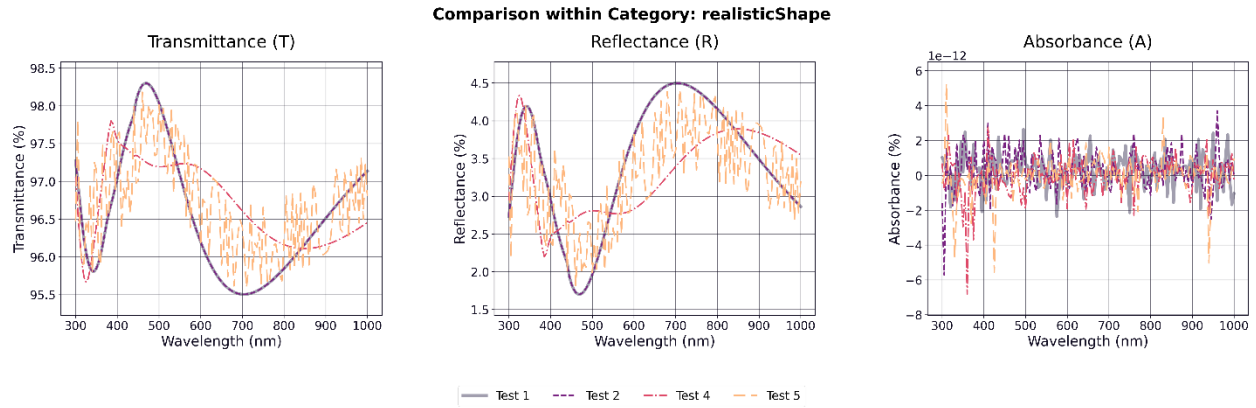


Figure E-10: TRA values across the 4 Completed RCWA tests for the realistic shape Moth-Eye cone structure

E.2 Raytracing Verification Data Results

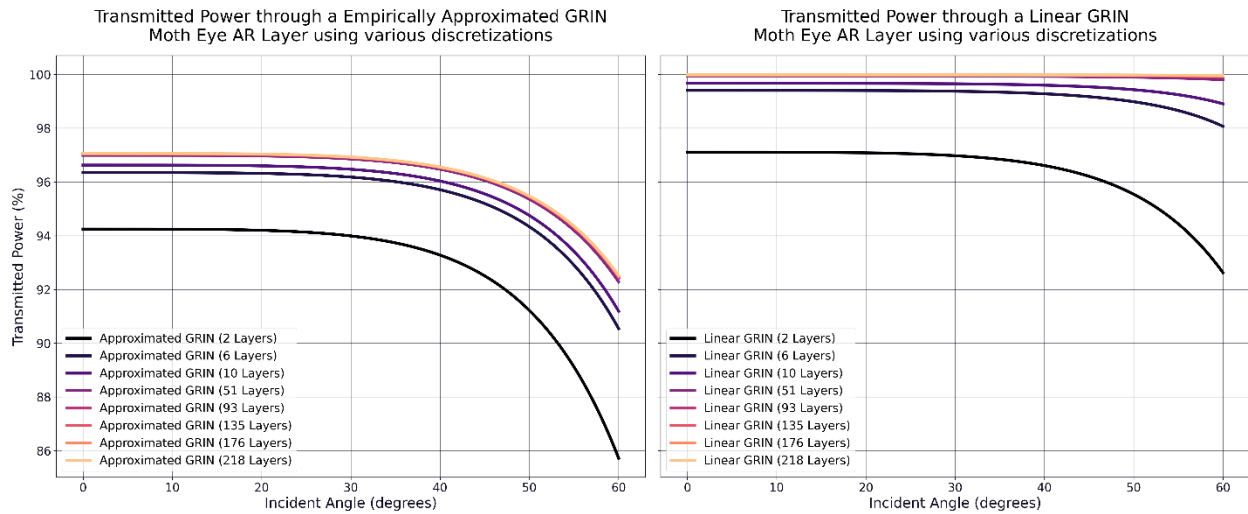


Figure E-11: Comparison of Transmitted power through a Waveguide interpretation of a Moth-Eye AR layer using Empirically Approximated and Linear GRIN

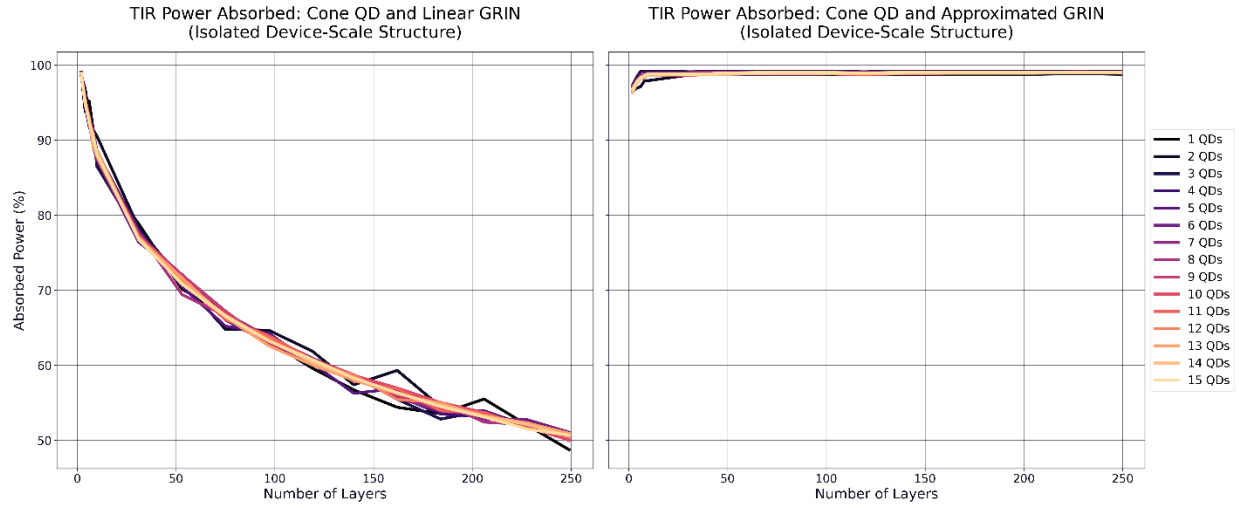


Figure E-12: TIR Comparison of Linear and Empirically Approximated GRIN using a Cone QD (Large Scale)

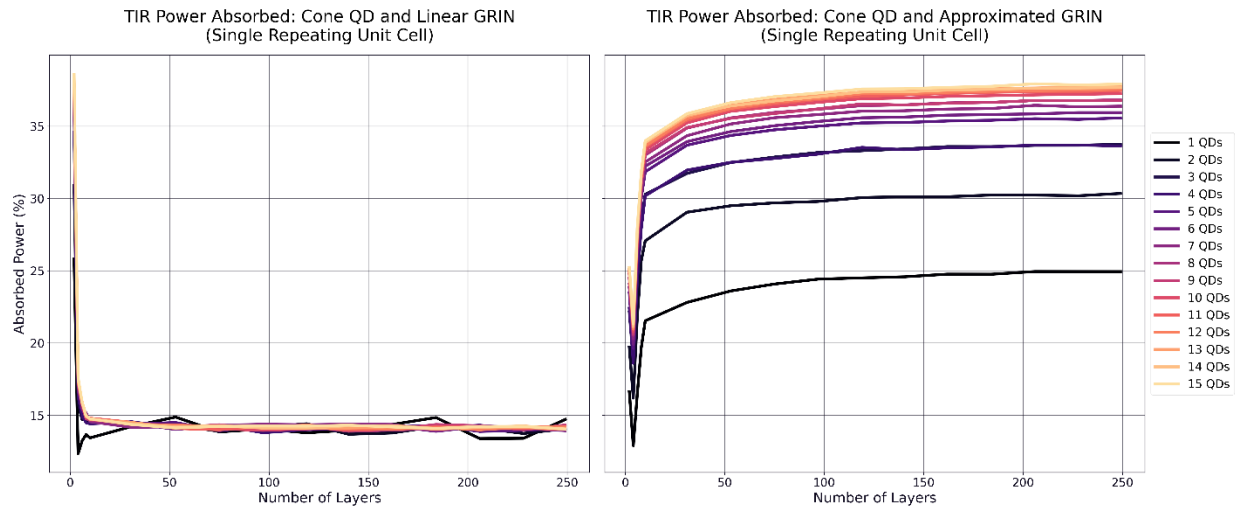


Figure E-13: TIR Comparison of Linear and Empirically Approximated GRIN using Cone QD (Unit Cell)

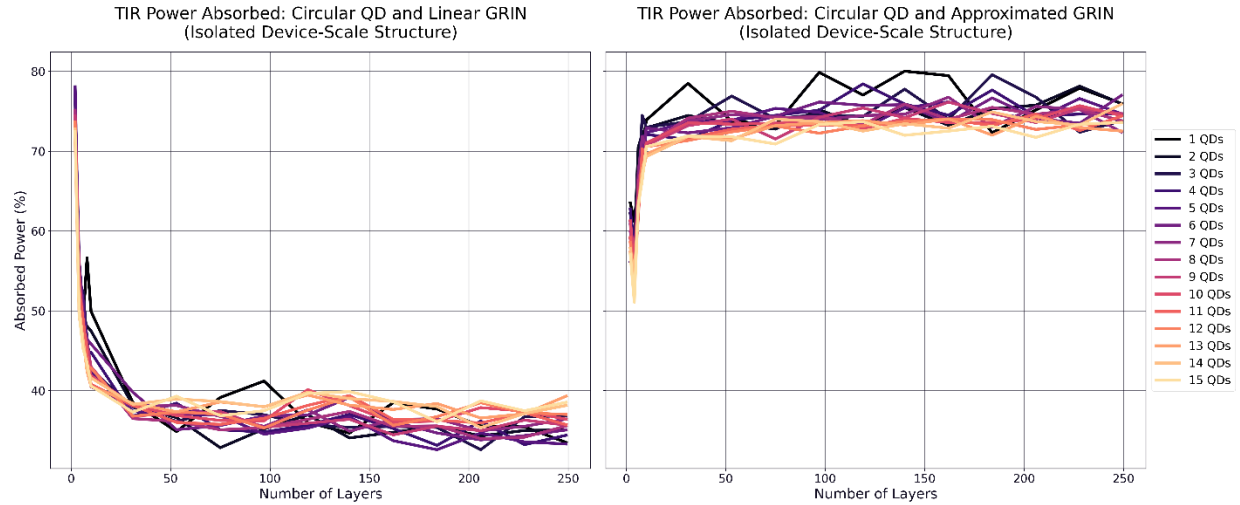


Figure E-14: TIR Comparison of Linear and Empirically Approximated GRIN using a Circular QD (Large Scale)

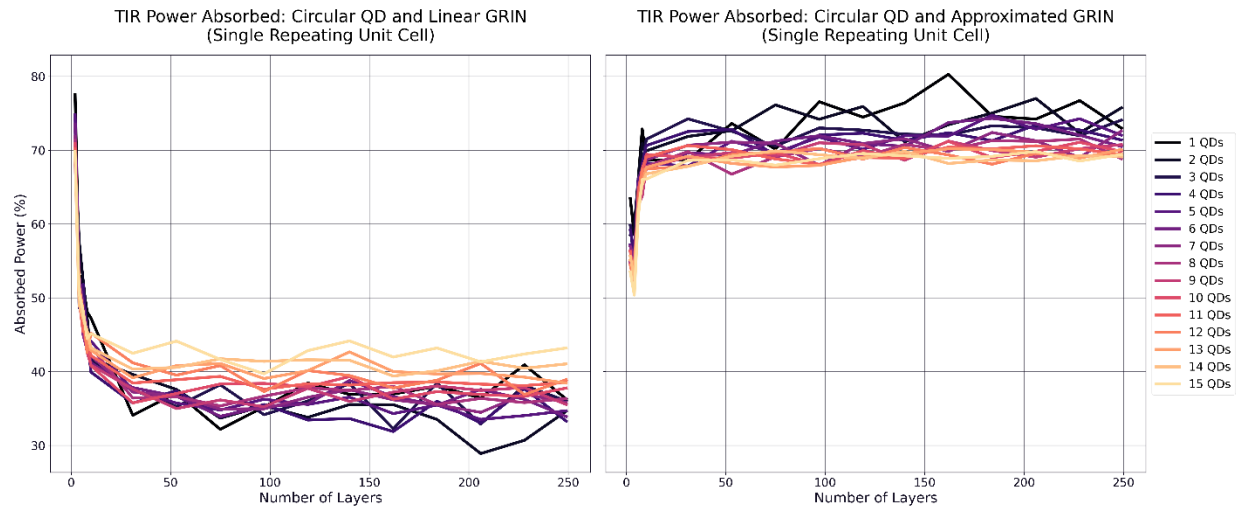


Figure E-15: TIR Comparison of Linear and Empirically Approximated GRIN using Circular QDs (Unit Cell)

TIR Power Absorbed from QD Emission using Moth Eye Sinusoidal Geometric Representation

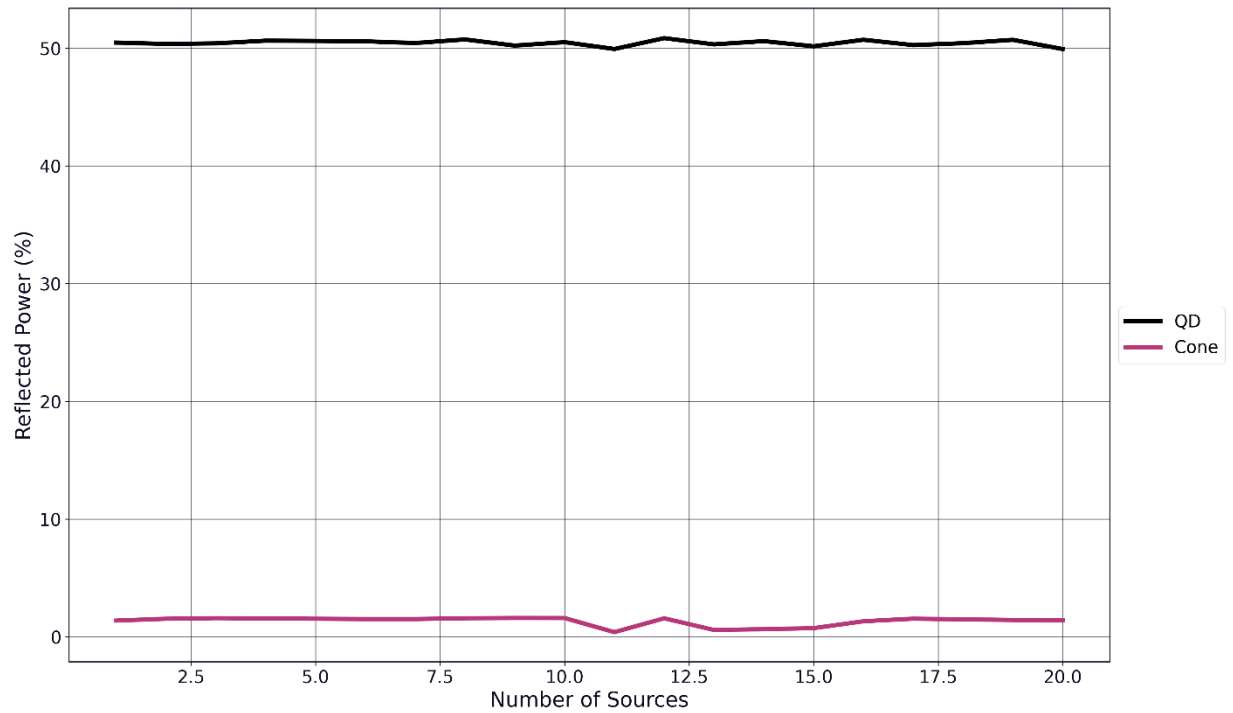


Figure E-16: TIR power absorbed using a Circular and Cone QD's using a Sinusoidal geometric representation of a Moth-Eye pattern

Real Life Scenario: Transmitted Power vs Layers

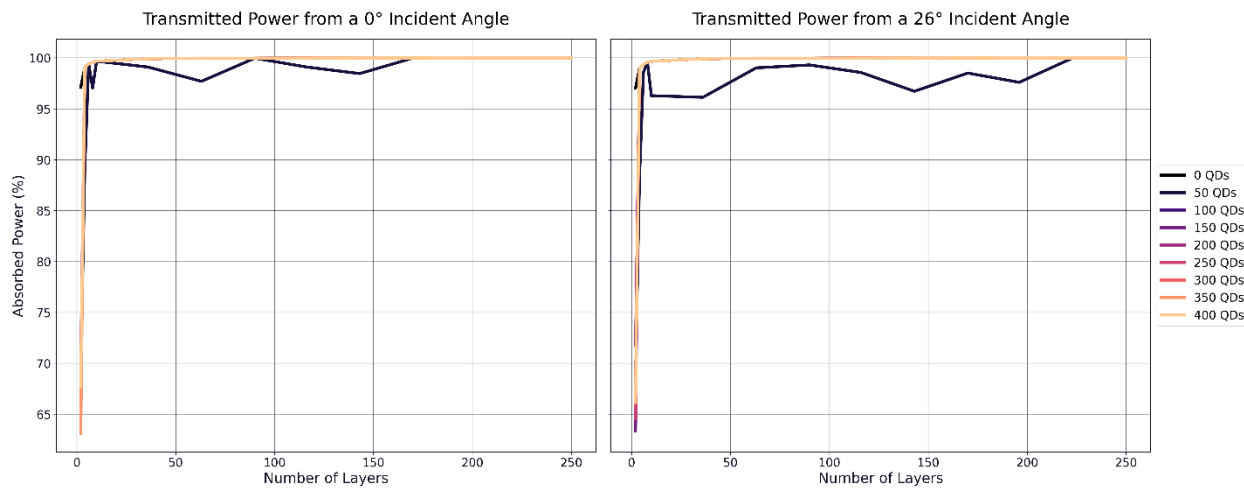


Figure E-17: Transmitted Power through a Waveguide approximation of the Film Device Stack and Circular QD's with an Incident Angle of 0 and 26°

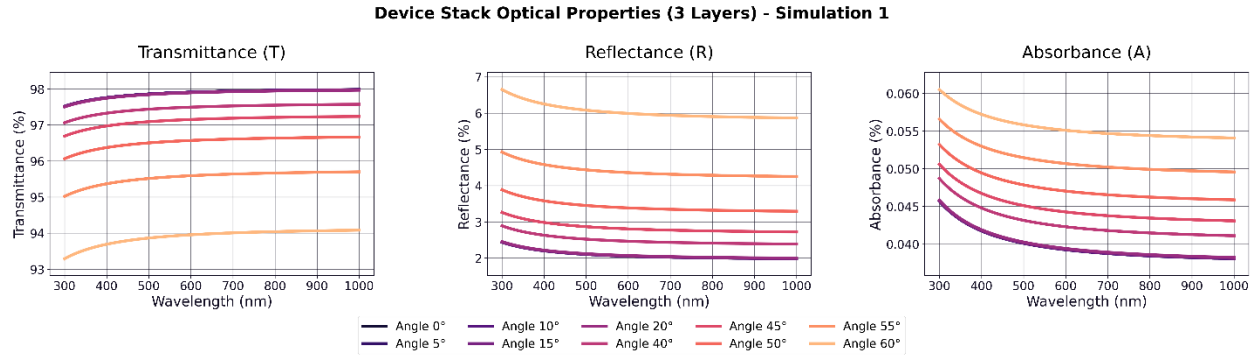


Figure E-18: TRA AR spectral responses from C++ Raytracing Moth-Eye layer (Simulation 1)

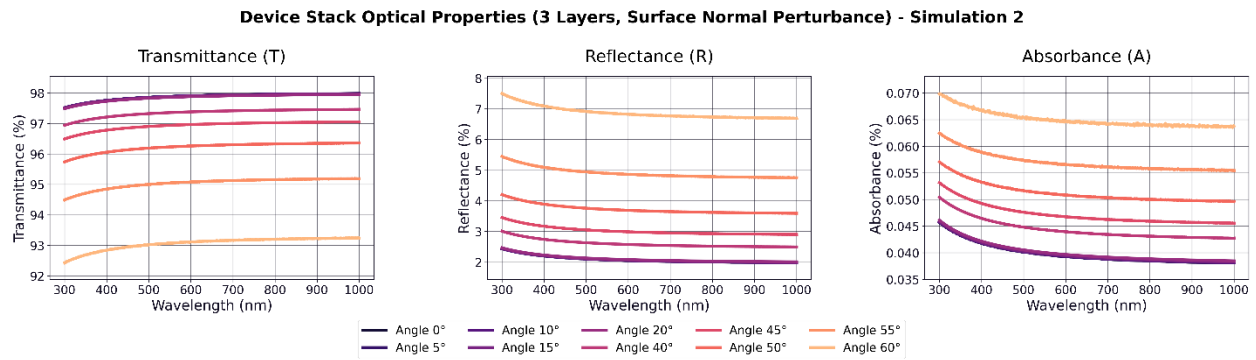


Figure E-19: TRA AR spectral responses from C++ Raytracing Moth-Eye layer (Simulation 2)

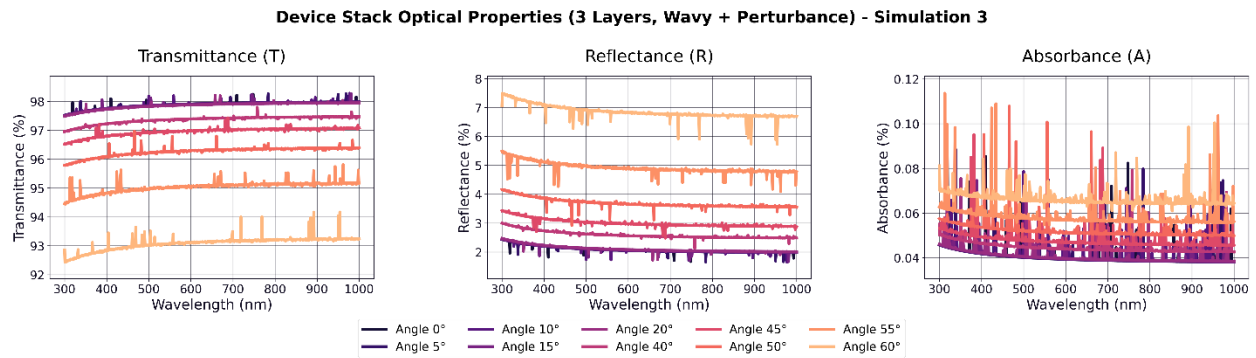


Figure E-20: TRA AR spectral responses from C++ Raytracing Moth-Eye layer (Simulation 3)

Device Stack TIR Optical Properties (40 Layers) - Simulation 4


Figure E-21: TRA TIR spectral responses from C++ Raytracing Moth-Eye layer (Simulation 4)

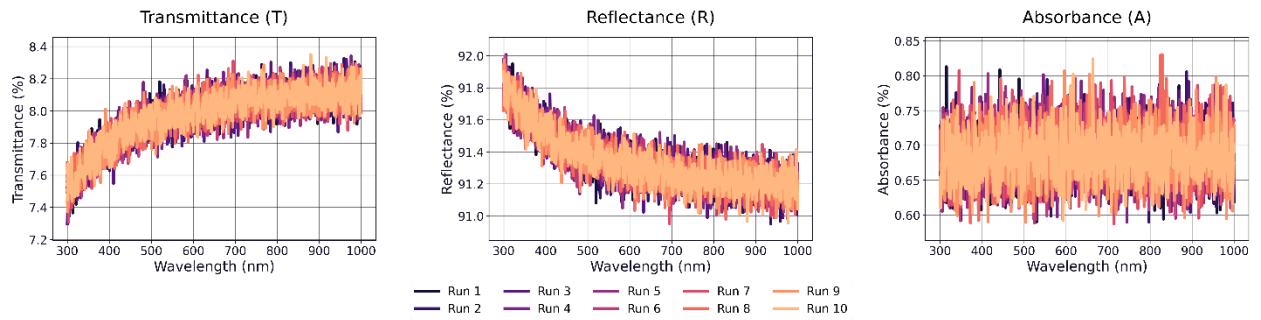
Device Stack TIR Optical Properties (40 Layers, Surface Normal Perturbance) - Simulation 5


Figure E-22: TRA TIR spectral responses from C++ Raytracing Moth-Eye layer (Simulation 5)

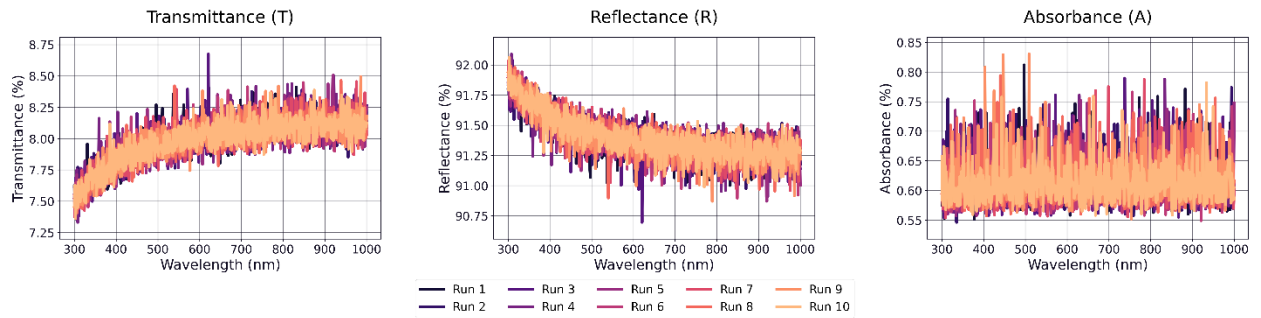
Device Stack TIR Optical Properties (40 Layers, Wavy + Perturbance) - Simulation 6


Figure E-23: TRA TIR spectral responses from C++ Raytracing Moth-Eye layer (Simulation 6)

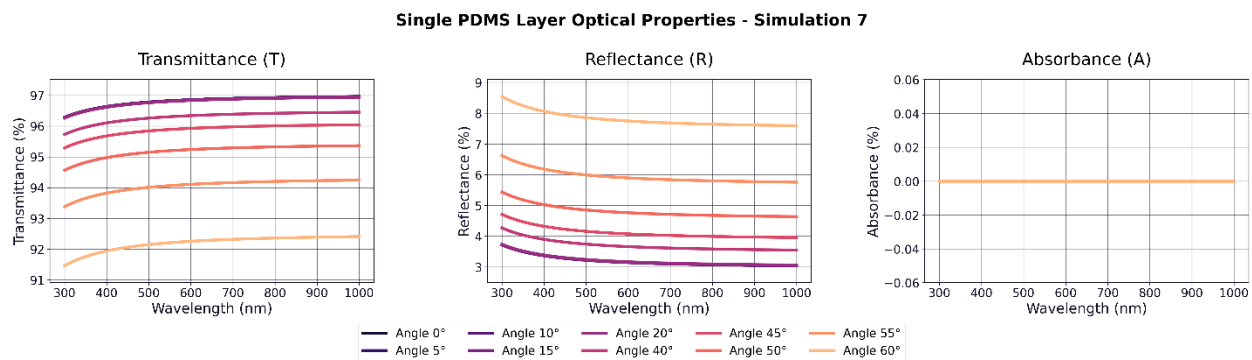


Figure E-24: TRA AR spectral responses from C++ Raytracing Single PDMS layer (Simulation 7)

E.3 Cost-Savings Analysis

The cost savings analysis was done using a model domestic system in Kitchener, Ontario. This model system has 29 m² of panels (5 kW system) installed on their roof at an angle of 35°. These installation conditions give the system an average yearly irradiance of 1556.92 kWh/m² [E.1]. The model system has panels with an average efficiency of 15%. The cost-savings analysis was done using the Ontario Ultra-Low Overnight pricing scheme which has four time-divisions [E.2]. The total power generation of the panels was split into a rough cost estimate based on these time-divisions as can be seen in Table E-11. The solar energy output was calculated using Equation E.1, where A is the system area, r is the solar panel efficiency, H is the average yearly irradiance, and PR is the performance ratio which accommodates losses within the system (default = 75%) [E.3]. The presented model system would provide the household energy cost savings of \$881.65 per year.

$$E = A * r * H * PR \quad (E.1)$$

Table E-1: Annual Cost-Savings with 15% Efficient Solar Panel System

	Generation Times	Cost Savings
On-Peak	725.64 kWh/yr	283.72 \$/yr
Mid-Peak	2902.54 kWh/yr	455.70 \$/yr
Weekend Off-Peak	1451.27 kWh/yr	142.22 \$/yr
Ultra-Low Overnight	0.00 kWh/yr	0.00 \$/yr
Total Power:	5079.45 kWh/yr	881.65 \$/yr

Upgrading the model system to 16% efficient panels to symbolize the use of the SunDOG film results in annual cost savings of \$940.42 as seen in Table E-22: an increase of \$58.78. Expanding this over the target product lifespan of 10 years results in total additional savings of \$587.77. This represents significant savings for the end-user and provides pricing flexibility when considering additional cost saving measures from scaled-up manufacturing.

Table E-2: Annual Cost-Savings with 16% Efficient Solar Panel System

	Generation Times	Cost Savings
On-Peak	774.01 kWh/yr	302.64 \$/yr
Mid-Peak	3096.05 kWh/yr	486.08 \$/yr
Weekend Off-Peak	1548.02 kWh/yr	151.71 \$/yr
Ultra-Low Overnight	0.00 kWh/yr	0.00 \$/yr
Total Power:	5418.08 kWh/yr	940.42 \$/yr

When comparing to SunDOG production costs to the lifetime cost-savings, it was found that using commercial QDs and small-scale Sylgard 184 is not economically viable with a cost to cover the model system of at least \$650 for a 20 μm thick film. This shows how critical improving the manufacturing process of the system would be for future applications. In-house synthesis of QDs and industrial-scale purchasing of Sylgard 184 would both serve to significantly lower production costs and make SunDOG more viable as a product.

E.4 Appendix E References

- [E.1] “Solar Panels Power Calculator for Kitchener, Ontario Canada.” Accessed: Apr. 04, 2026. [Online]. Available: <https://solarcalculator.ca/report/Ontario/Kitchener/>
- [E.2] “Electricity rates | Ontario Energy Board.” Accessed: Apr. 04, 2026. [Online]. Available: <https://www.oeb.ca/consumer-information-and-protection/electricity-rates>
- [E.3] “How to calculate the annual solar energy output of a photovoltaic system?” Accessed: Apr. 04, 2026. [Online]. Available: <https://photovoltaic-software.com/principle-ressources/how-calculate-solar-energy-power-pv-systems>

Appendix F Prototype Test/Measurement Data

F.1 Measurement Conditions

- Illumination sources:
 - UV source: Cole-Parmer UV Lamp 365 nm UV.
 - White light source: Garpsen Grow Lights for Indoor Plants
- Reflectance measurement:
 - Incident angle: 8°
 - Wavelength range: 300–800 nm
- Electrical measurements:
 - Voltage sweep range: 0–1.2 V
 - Measurement device: Keysight B2900 Series Source Measure Unit

F.2 Extracted Electrical Parameters

Table F-1 Electrical Parameters Extracted from I-V Testing to get Power Increase

Condition	Sample	V _{oc} (V)	I _{sc} (mA)	P _{max} (mW)
UV	Control	~1.0	~0.40	0.18
UV	Coated	~1.0	~0.45	0.21
White	Control	~1.1	~9.8	7.74
White	Coated	~1.1	~9.5	7.48

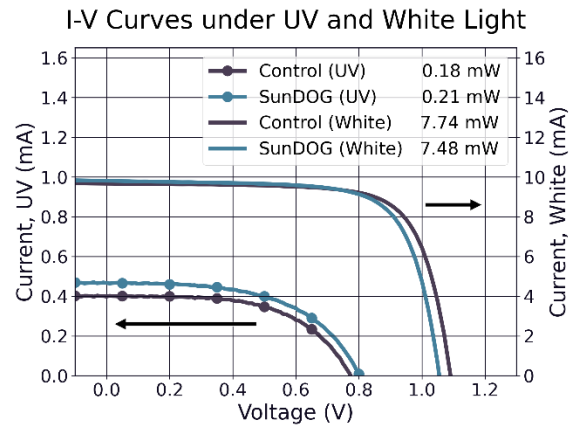


Figure F-1: I-V Testing with a Control Solar Panel and One with Device Stack with Different Illumination Conditions

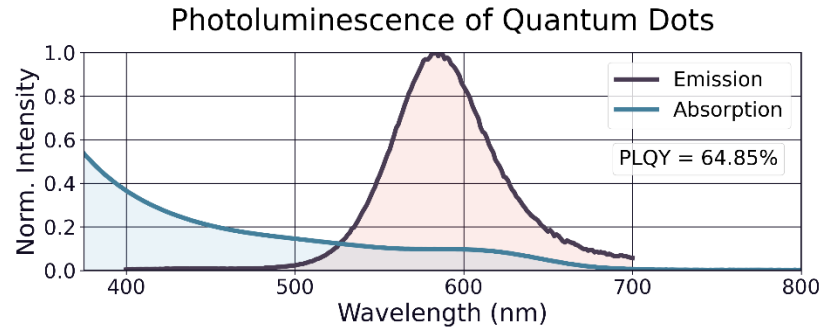


Figure F-2: PLE of Quantum Dots. Emission measured with excitation $\lambda=380\text{nm}$. Absorption from UV-Vis

F.3 Reflectance Data Summary

Table F-2: Table of Reflectance Data Summarized

Sample	Avg Reflectance (%)	Notes
Flat PDMS	~10–12	baseline
Textured PDMS	~9–11	AR effect
Solar panel	~7–9	baseline
Panel + film	~10–13	integration losses

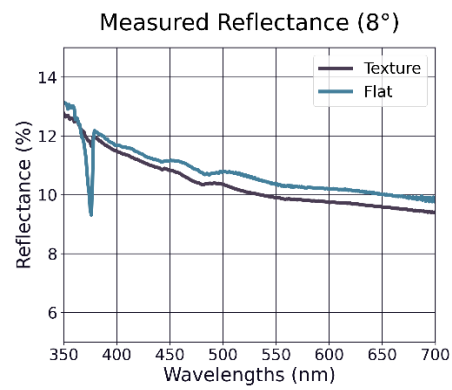


Figure F-3: Measured Reflectance with Incoming Light at 8° Under /h Conditions of a Blue-ray Imprinted AR PDMS Surface Film Vs. A Flat PDMS Sample

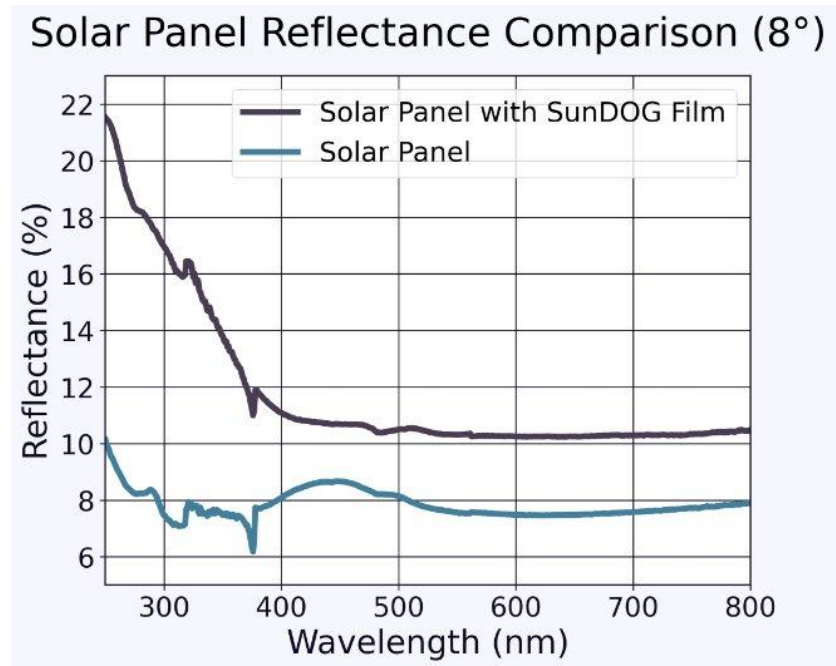


Figure F-4: Measured Reflectance with Incoming Light at 8° Under /h Conditionsof Plain Solar Panel Vs. One With the Device Stack

F.4 Requirements Summary

Table F-3: Copy of Level of Success for Each Requirement

Requirement	Test Conducted	Level of Success
PCE Increase by +1%	IV-Testing	Semi-successful
Transmit 99% of light	UV-Vis	Semi-successful
10-year lifespan	Not tested	N/A
Lower reflectance by 3%	UV-Vis	Semi-successful
Easy to apply	Not tested	N/A
Hydrophobic surface	Not tested	N/A
Cost efficient/scalable process	Calculated cost vs power generated per area, Designed to be scalable	Semi-successful
Winter Proof	Not tested	N/A

Appendix G FYDP Resources

G.1 Budget

Table G-1: NE_2026_20 Final Budget

Item	Cost	Currency	Supplier
SYLGARD 184	\$ 145.46	CAD	Newark
InP/ZnSe/ZnS QDs	\$ 475.00	USD	NanoOptical Materials
Tetrakis(triphenylphosphine) palladium	\$ 15.00	USD	A2B Chem LLC
Phenylboronic acid pinacol ester	\$ 8.00	USD	A2B Chem LLC
2,7-Dibromofluorene	\$ 8.00	USD	A2B Chem LLC
2-(6-Bromohexyloxy)tetrahydropyran	\$ 94.00	USD	A2B Chem LLC
9,9-Dioctylfluorene-2,7-bis(boronic acid pinacol ester)	\$ 24.00	USD	A2B Chem LLC
Bromobenzene	\$ 11.00	USD	Oakwood Chemical
Clear Moisture-Resistant Polyester Film	\$ 4.00	CAD	McMaster Carr
Clear Moisture-Resistant Polyester Film	\$ 13.94	CAD	McMaster Carr
Clear Static Cling PVC Film	\$ 7.02	CAD	McMaster Carr
Metal Plates	\$ 18.62	CAD	Home Depot
Solar Panels	\$ 25.00	CAD	Amazon
Lab Consumables	\$ 30.00	CAD	UWaterloo
AFM Lab Time	\$ 10.00	CAD	UWaterloo
UV-Vis Lab Time	\$ 10.00	CAD	UWaterloo
Total	\$ 1,149.61	CAD	-

G.2 Lab Equipment Usage

Table G-2: NE_2026_20 Lab Equipment Usage

Equipment	Amount	Source
Gloves	1 box	NE UG Lab
Kimwipes	1 box	NE UG Lab
Glass Vials	20	NE UG Lab
Plastic Pipette	25	NE UG Lab
Parafilm	6"	NE UG Lab
Glass Pipette	10	NE UG Lab
TLC Plates	6" x 6"	NE UG Lab
Capillary Tubes	3	NE UG Lab
Needles 18 Gauge 1.5"	1	NE UG Lab
AFM	<1 hour	NE UG Lab
UV-Vis	<1 hour	NE UG Lab
Hot Plate + Stir Bar	1	NE UG Lab
Glassware	-	NE UG Lab
UV-Vis	-	Consultant Lab
Photoluminescence Excitation Spectrometer	-	Consultant Lab
4-point Probe	-	Consultant Lab
IV Probe Station	-	Consultant Lab
Mayer Rod	1	Trusscore

G.3 Activity Log

Table G-3 Lab Equipment Usage per Group Member with Associated Usage Time

Name	Activity	Date	Hrs	Signature
Ryan Becze	Solar panel and PCB soldering	02/26/2026	2	<i>Ryan Becze</i>
	IV Characterization of Solar Cell + Device	03/09/2026	3	
	Solar Cell + Device Reflectance Test	03/11/2026	2	
	Final I-V Testing with UV light	03/12/2026	2	
	Capstone Poster	03/09-13/2026	20	
	Seminar Presentation	03/13-18/2026	10	
	Final Report	04/03-06/2026	33	
Ryan Miller	SD-RM-09: Retest QD PL	02/25/2026	3	<i>Ryan Miller</i>
	SD-RM-10: UV-Vis of QDs	02/25/2026	3	
	SD-RM-11: QD in PDMS	02/26/2026	2	
	SD-RM-12: Repeat RM-11	02/27/2026	4	
	SD-RM-13: Repeat RM-11	03/02/2026	2	
	SD-RM-14: Poisoned Catalyst	03/03/2026	3	
	SD-RM-15: QD Thin Film	03/05/2026	2	
	SD-RM-16: Device fabrication	03/06/2026	2	
	SD-RM-17: Device fabrication	03/09/2026	2	
	SD-RM-18: IV Testing day 1	03/10/2026	4	
	Trusscore Visit, SD-RM-19	03/11/2026	3	
	SD-RM-20: IV Testing day 2	03/12/2026	5	
	Capstone Poster + Data Plotting	03/09-13/2026	20	
	Seminar Presentation	03/13-18/2026	10	
	Final Report	04/03-05/2026	32	
Alexandre Dufresne-Nappert	Design PDMS Well	02/24/2026	2	<i>Alexandre Dufresne-Nappert</i>
	RaspberryPi Multimeter Research	02/27/2026	3	
	RaspberryPi Multimeter Reading	02/28-01/2026	9	
	PCE Power Calculation + Reflectance Graphing	03/10/2026	2	
	Solar Cell + Device Reflectance	03/11/2026	2	
	Capstone Poster	03/13-18/2026	2	
	Seminar Presentation	03/13-18/2026	10	
	Final Report	04/03-06/2026	26	
Nathan Frey	SD-NF-04: Stripping Epoxy off Solar Cell	02/27-28/2026	5	<i>Nathan Frey</i>
	SD-NF-05: Resistivity Measurement	03/03/2026	2	
	SD-NF-06: Solar Cell Characterization	03/06/2026	6	
	SD-NF-07: SunDOG IV Characterization	03/09/2026	3	
	SD-NF-08: Solar Cell + Device Reflectance Test	03/11/2026	2	
	Capstone Poster	03/10-13/2026	12	
	Seminar Presentation	03/13-18/2026	10	
	Final Report	04/03-06/2026	32	
Total Hours	Ryan B: 72	Ryan M: 97	Alex: 56	Nathan: 72

